MASON EVANS

Geo-Environmental Consultants

CAUSEWAYSIDE, EDINBURGH

REPORT ON

SITE INVESTIGATIONS

DATE

JULY 2023

CLIENT

BALFOUR BEATTY INVESTMENTS LIMITED C/O ETIVE CONSULTING ENGINEERS LIMITED

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ETIVE CONSULTING ENGINEERS LIMITED

CAUSEWAYSIDE,

EDINBURGH

REPORT ON SITE INVESTIGATIONS

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CONTENTS

		PAGE NO
	EXECUTIVE SUMMARY	
1.0	INTRODUCTION	1
1.1	Commission	1
1.2	Investigation Proposals	1
1.3	Limitations	1
1.4	Access	4
2.0	SUMMARY OF DESK STUDY INFORMATION	5
2.1	The Site	5
2.2	Site History	6
2.3	Published Geological Information	8
2.4	Preliminary Conceptual Site Model	12
2.5	Receptor Characterisation	12
2.6	Source Characterisation	12
2.7	Pathway Characterisation (Pollutant Linkages)	13
3.0	SITE INVESTIGATIONS	17
3.1	Objectives	17
3.2	Scope and Methods of Investigations	17
3.3	Summary of Ground Investigation Data	18
3.4	Investigation Rationale	20
3.5	Analytical Procedures	23
4.0	INVESTIGATION RESULTS	24
4.1	Ground Conditions	24
4.2	Rockhead	26
4.3	Obstructions	26
4.4	Groundwater	26
4.5	Visual/Olfactory Evidence of Contamination	28
5.0	CONTAMINATION RISK ASSESSMENT	29
5.1	Human Health and Groundwater Risk Assessment Screening Criteria	29
5.2	Statistical Analysis of Data	30
5.3	Ground Gas Assessment	30
5.4	Building Materials Assessment	30

6.0	HUMAN HEALTH RISK ASSESSMENT	31
6.1	Contaminants in Soils	31
6.2	Volatile Organic Compounds (VOCs) and Semi-Volatile Organic Compounds (SVOCs) Testing	33
7.0	WATER ENVIRONMENT RISK ASSESSMENT	34
7.1	Water Environment Vulnerability	34
7.2	Surface Water and Groundwater Assessments	34
7.3	Conclusions	45
8.0	GROUND GAS EMISSIONS	46
8.1	General	46
8.2	Ground Gas – Results	46
8.3	Ground Gas – Assessment	47
8.4	Radon	51
8.5	Conclusions	51
9.0	RISKS TO CONSTRUCTED DEVELOPMENT	52
9.1	Sulphate Attack on Construction Materials	52
9.2	Water Supply Pipes	52
9.3	Phytotoxicity	52
10.0	REVISED CONCEPTUAL SITE MODEL	54
10.1	Contamination Sources	54
10.2	Pollutant Linkage Assessment	54
10.3	Mitigation Measures	56
10.4	Waste Management Legislation	57
10.5	Waste Acceptance Criteria (WAC)	59
10.6	Rate of Scottish Landfill Text	59
10.7	Contingent Liabilities	60
11.0	GEOTECHNICAL ASSESSMENT	61
11.1	General	61
11.2	Atterberg Limits	61
11.3	Triaxial/Undrained Shear Strength Testing	61
11.4	Sedimentary Bedrock	62
11.5	Point Load and Unconfined Compressive Strength (UCS) Testing	62

12.0	FOUNDATION RECOMMENDATIONS	64
12.1	Details of the Development	64
12.2	Foundations (relative to existing site levels)	64
12.3	Excavations	64
12.4	Obstructions	66
13.0	ROAD CONSTRUCTION	67
13.1	General	67
13.2	Ground Conditions	67
13.3	Chemical Contamination	67
13.4	Gas Emissions	67
13.5	Mining and Mineral Stability	67
14.0	INVASIVE PLANT SURVEY	68
14.1	General	68
15.0	CONCLUSIONS AND RECOMMENDATIONS	69
15.1	General	69
15.2	Contamination and Gas Emissions	69
15.3	The Built Environment	70
15.4	Fuel Tanks	70
15.5	Foundation Recommendations (relative to existing site levels)	70
15.6	Mining	70
15.7	Invasive Plants	70
15.8	Additional Investigations	71
15.9	Consultations with Public Authorities	71

LIST OF DRAWINGS

		PAGE NO
Drawing No P23/075/SI/R/F/01	Site Location Plan	2
Drawing No P23/075/SI/R/F/02	Study Area	3
Drawing No P23/075/SI/R/F/03	Extract from Published Geological Survey Map (Drift Lithology)	9
Drawing No P23/075/SI/R/F/04	Extract from Published Geological Survey Map (Solid Geology)	10
Drawing No P23/075/SI/R/F/05	Preliminary Conceptual Site Model	15
Drawing No P23/075/SI/R/F/06	Location of Exploratory Holes	21
Drawing No P23/075/SI/R/F/07	Recorded Made Ground Thickness	25
Drawing No P23/075/SI/R/F/08	Recorded Depth to Definite Rockhead	27
Drawing No P23/075/SI/R/F/09	Ground Gas Emissions Survey	48
Drawing No P23/075/SI/R/F/10	Revised Conceptual Site Model	55
Drawing No P23/075/SI/R/F/11	Conjectured Depth to a Potential Foundation Bearing Horizon	65

LIST OF APPENDICES

Appendix 1	Site Walkover Survey (March 2023)
Appendix 2	Envirocheck Report
Appendix 3	SEPA Databases
Appendix 4	The Coal Authority Interactive Viewer Map
Appendix 5	BGS Borehole Scans and Data
Appendix 6	Records of Exploratory Holes
Appendix 7	Gas and Groundwater Monitoring Results
Appendix 8	Chemical Laboratory Analysis Results
Appendix 9	RTM Groundwater Modelling Worksheets
Appendix 10	Waste Classification Report (HazWaste Online, May 2023)
Appendix 11	Geotechnical Laboratory Analysis Results

This Phase 2 Site Investigation Report is prepared on behalf of Balfour Beatty Investments (BBI) Ltd ('the Applicant') who is seeking detailed planning permission for the demolition of the existing building and the erection of purpose-built student accommodation ('PBSA') with associated works at 140 Causewayside, Edinburgh, EH9 1PR ('the application site').
The description of development for the planning application is:

"*Demolition of existing building and erection of purpose-built student accommodation with landscaping, infrastructure, access, and associated works."* ('the application').

This Phase 2 Site Investigation Report forms part of a suite of supporting documents submitted with the planning application as outlined below. These supporting documents are in addition to the formal planning application documents comprising the planning application form and certificates and the accompanying plans, sections, and elevations. The suite of supporting documents comprises:

- Planning Statement
- Pre-application Consultation Report
- Design & Access Statement
- Landscape Drawings & Strategy
- Heritage Statement
- Transport Statement & Travel Plan
- Flood Risk Assessment and Drainage Strategy including Surface Water Management Plan and SUDs
- Noise Impact Assessment
- Sustainability Statement including S1 Form
- Daylight and Sunlight Assessment
- Archaeology Desk-based Assessment
- Ecology BREEAM Report including Preliminary Ecological Assessment, Bat Survey and Biodiversity Enhancement Statement
- Tree Survey and Arboricultural Report
- Arboricultural Impact Assessment and Tree Protection Plan
- **Site Investigations Phase 1 and Phase 2 Report**
- Student Accommodation Demand and Supply Report
- PBSA Management Plan
- Verified Visualisations

EXECUTIVE SUMMARY

Client	**Balfour Beatty Investments Limited c/o Etive Consulting Engineers Limited**
Site	Causewayside, Edinburgh
Project Objectives	▪ To investigate the possible presence of ground contamination and any potential associated risks. ▪ To investigate the ground conditions and provide recommendations on foundation and infrastructure design. ▪ To provide recommendations (if any) for additional works/remediation required.
Assessment of Risks to Human Health & the Water Environment	The asbestos encountered in BH02 (at 0.50 mbgl) has been subject to analysis that has indicated the quantity of asbestos to be less than 0.1% vol. of the sample. Therefore, we would recommend that this *'hotspot'* is delineated and removed or relocated prior to development. It is anticipated that these works would comprise the removal of a limited volume of material around the affected area and would require to be supervised and validated by a suitably qualified geo-environmental scientist. The excavated soils would then need to be disposed of off-site, or alternatively, the soils could be retained on site and placed beneath less sensitive areas, such as beneath roads and/or areas of hardstanding/building cover. Note that we would not recommend that the asbestos contaminated soils are placed beneath roads due to the potential risk to future maintenance personnel. Conversely, the asbestos identified in BH03 (at 0.50 mbgl) was recorded to contain visible asbestos, described as *'visible insulation board debris'* (at 0.50 mbgl). Therefore, this *'hotspot'* will require to be delineated in the first instance to confirm its size and extent and then removed by a suitably licensed contractor, with the excavated soils being removed from site so a specialist *'hazardous'* waste facility site. The asbestos identified in BH03 at 1.00 mbgl was below the 0.1% vol. threshold and was recorded to contain no visible ACM's (asbestos containing materials). Therefore, once the ACM at 0.50 mbgl has been removed, the deeper lying asbestos (at 1.00 m depth) can be retained onsite beneath a less sensitive area or disposed offsite to a suitably licensed waste facility site, as per the *'hotspot'* identified in BH02. Chemical analysis and groundwater modelling indicates that the proposed development site was not considered to pose a significant risk to the Water Environment (surface water and groundwater).
Assessment of the Built Environment	We would recommend that a UKWIR assessment is undertaken along the proposed water supply route, once the site layout and levels have been finalised. Note that this assessment would inform the likely water supply pipe selection material. Concrete class DS-1, AC-1 is considered sufficient to protect buried concrete from pH and sulphate levels in the soils and groundwater.

EXECUTIVE SUMMARY - ***CONT***

Assessment of Ground Gas	Based on the results of the ground gas monitoring undertaken to date and subsequent risk assessment, gas protection measures are not considered necessary within the future development. Confirmatory gas monitoring is continuing at the site and the results will be reported as a separate addendum letter report upon completion. However, it is assumed that gas protection measures are unlikely to be required.
Foundation Construction	Given the nature of the proposed development, which is understood to be for 5 to 6 No. storey student accommodation, and therefore the associated loadings, piling to the underlying competent rock strata will likely be required. Rockhead was definitively encountered beneath the site at varying depths of between 15.50 m and 16.00 m, below current ground level, and should provide a a presumed bearing value of 3.0 MPa. However, the design of piles and method of piling should be developed in consultation with a specialist piling contractor, who will be able to devise the most economical and practical measures to be adopted.
Mining	Mining instability constraints are not considered to pose a significant risk to the proposed development site.
Radon	Radon protection measures are not considered necessary within the proposed development given that the site is located within a Lower Probability Radon Area (less than 1% of homes are estimated to be at or above the Action Level).
Invasive Plants	No obvious evidence of *'Commonly Known Species Controlled by Legislation'* (i.e. Giant Hogweed, Japanese Knotweed of Himalayan Balsam) or *'Problematic Non-Legislated Species'* (i.e. Horsetail) were identified during our recent 2022 site walkover survey (included in Appendix 1). However, we would recommend that an invasive plat survey is undertaken at the site, preferably during the new growing season, by a specialist contractor. Kleerkut, or another invasive plant specialist, would be able to assist in the matter.
Further Investigations	At the time of our investigation, the Scottish Law Commission building (with associated outbuildings) occupied at least half of the site. Therefore, some further limited confirmatory site investigation works, including chemical testing, will be required following demolition of the onsite building.

1.0 INTRODUCTION

1.1 Commission

1.1.1 Mason Evans (ME) were commissioned by Etive Consulting Engineers Limited (the Client's Engineer), on behalf of Balfour Beatty Investments Limited (the Client), to undertake ground investigations at a site located off of Causewayside, Edinburgh in connection with the proposed redevelopment of the site (Drawing Nos. P23/075/SI/R/01 and 02). The development proposals were not available at the time of reporting but were understood to be for 5 to 6 No. storey student accommodation, with areas of soft landscaping and associated road infrastructure.

1.1.2 We had previous knowledge of the site, having undertaken a Phase 1 Desk Study Report in advance of the Phase 2 intrusive site investigations works. The report referred to was entitled: *'Causewayside, Edinburgh – Phase 1 Desk Study Report, dated April 2023'* (MEP Ref: P23-075).

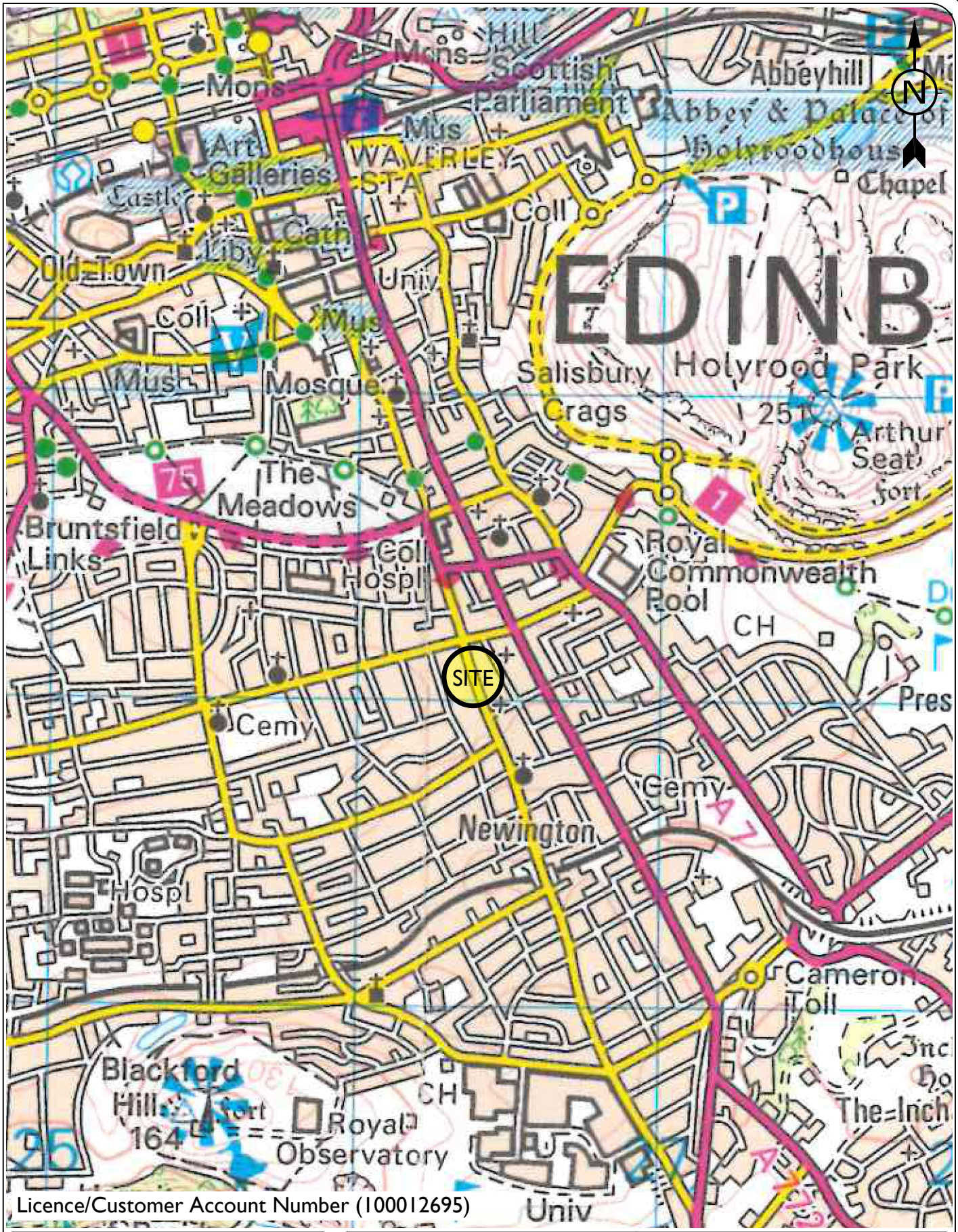
1.2 Investigation Proposals

1.2.1 The investigation proposals were provided in our correspondence to the Client, dated April 2023. The intention of the investigations was to provide further information on the following:

- Soil profile beneath proposed development areas of the site
- Chemical contamination conditions
- Gas emissions
- Geotechnical characteristics of the materials
- Foundation bearing characteristics
- Possible existence of invasive plants
- Mining Constraints

1.3 Limitations

1.3.1 Our interpretations of the ground conditions are based on the information retrieved from the exploratory bores sunk at the site during site investigations. While we have carried out some interpretation of the ground conditions between the exploratory locations, it should be recognised that soil and groundwater conditions can vary from point to point. As such, ground conditions at variance with those indicated by the exploratory pits/bores may exist in areas not investigated.



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project title:	drawing title:
CAUSEWAYSIDE EDINBURGH	SITE LOCATION PLAN

project no:	drawing no:	revision:	date:	drawn by:	approved by:	scale:
P23/075	P23/075/SI/R/F/01		21.07.23	TR	LR	Not to Scale

A4 border: 277 x 190

NOTES

Approximate site boundary

REV	DATE	DETAILS

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PROJECT TITLE

CAUSEWAYSIDE EDINBURGH

DRAWING TITLE

STUDY AREA

DRAWN BY	CHK'D BY	APP'D BY	DATE	SCALES
TR	LR	LR	21.07.23	1:250 @ A3

PROJECT No.	DRAWING No.	REVISION
P23/075	P23/075/SI/R/F/02	

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A3 border: 400 X 273

1.3.2 It should be recognised that this report is prepared in accordance with current recommended practice and existing legislation. It is written in the context of the proposed student accommodation development previously described. Should there be any alternative end-use, it would be prudent to consult us further to ensure the continued pertinence of the recommendations advised.

1.4 Access

1.4.1 Note that a partially occupied building, with associated outbuildings, was still present onsite at the time of our investigations. Therefore, we would recommend that some limited additional investigations are undertaken following demolition of the onsite buildings.

2.0 SUMMARY OF DESK STUDY INFORMATION

2.1 The Site

2.1.1 A summary of the current site conditions as understood from the supplied survey information and site reconnaissance is included in Table 1. A site walkover was undertaken in late March 2023 (included in Appendix 1), an Envirocheck report was procured (Appendix 2) and a review of selected publicly available databases was undertaken (Appendices 3 and 4). A summary of the findings of these researches are included in the table below:

TABLE 1 - Site Details and Review of Public Records

Site Name	Causewayside, Edinburgh
National Grid Ref	326330, 672120 (approximately)
Site Area	0.21 Ha (approximately)
Topography	Generally flat lying.
Current Usage	At the time of the investigation, the site was occupied by the Scottish Law Commission building with associated car parking and outbuildings at the rear.
Proposed Use	The development proposals were understood to be for multi-storey student accommodation (5 to 6 No. storey) with areas of soft landscaping and associated road infrastructure.
Surface Water Bodies	The nearest surface water body to the proposed development site was considered to be the Pow Burn/Braid Burn, located approximately 1.00 km to the south-east (at its closest point), flowing generally towards the north-east towards the coast. SEPA's Water Classification Hub, dated 2020, indicated the Braid Burn (Upstream Dreghorn Barracks to Portobello (ID: 3500)) to have an overall status of *'moderate ecological potential'. H*owever, this water body has also been designated as *'an artificial water body on account of physical alterations that cannot be addressed without a significant impact on navigation'.* Duddingston Loch was recorded approximately 1.70 km to the north-east of the proposed development site (at its closest point). However, SEPA have no further information on this water body. The Union Canal was recorded approximately 1.85 km to the north-west of the proposed development site (at its closest point). SEPA's Water Classification Hub, dated 2020, indicated the Union Canal (Murray Burn to Fountainbridge (ID: 9) to have an overall status of '*good ecological potential/'*. Similarly, this water body has also been designated as *'an artificial water body on account of physical alterations that cannot be addressed without a significant impact on navigation'.* Please see Appendix 3 for further details.
Groundwater	SEPA's Water Classification Hub indicated there to be no classified groundwater body within the superficial soils beneath the proposed development site. However, the deeper groundwater within the bedrock aquifer was indicated by the SEPA Water Classification Hub to be named 'Edinburgh' (ID: 150591). The most recent SEPA classification sheet, dated 2020, indicated this deeper groundwater body to have an overall status of *'Good.* Please see Appendix 3 for further details.

TABLE 1 *CONT* - Site Details and Review of Public Records

Flooding	The SEPA Flood Maps indicate that the proposed development lies outwith the surface water, river and coastal water flood maps. A review of the future flood maps was also undertaken and the proposed development site is not at risk from future river or coastal flooding as a result of climate change. However, these comments are preliminary only and for a detailed assessment of these risks, specialist flood advice should be sought. Please see Appendix 3 for further details.
Public Register Information	No Registered Radioactive Substances, BGS Recorded Mineral Sites, Areas of Adopted Green Belt, Sites of Special Scientific Interest (SSSI) or World Heritage Sites present within 250 m of the proposed development site. 1 No. Local Authority Pollution Prevention and Control was recorded 204 m to the south of the proposed development site at Kennet Ltd (2-40 Ratcliffe Terrace, Edinburgh, WH9 1SS) for an air pollution control (PG1/14 petrol filling station). The status of this control was *'not supplied'*. 22 No. Contemporary Trade Directory Entries (14 No. recorded as *'inactive'*) were recorded within 250 m of the proposed development site for; stairlifts – manufacturers and installers, painting and decorating suppliers, bed and mattress manufacturers, foam products – rubber and plastics, furniture – repairing and restoring, catering equipment, jewellery manufacturers and repairers, printers, cinema equipment, dry cleaners, garage services, petrol filling stations, joinery manufacturers and electricity generating and distributing equipment. 2 No. Fuel Station Entries were recorded within 250 m of the proposed development site at; 'Causewayside Service Station' (35 Causewayside, Edinburgh, EH9 1QF), approximately 207 m to the north and at 'Mfg Ratcliffe' (2-4 Ratcliffe Terrace, Edinburgh, EH9 1SS), approximately 207 m to the site. The fuel station at 35 Causewayside was indicated as *'obsolete'*, while the fuel station at Ratcliffe Terrace was indicated as *'open'*.
Mining	The Coal Authority Interactive Map (Appendix 4) indicates that the proposed development site does not lie within a *'Coal Mining Reporting Area'* or *'Development High Risk Area'*, as defined by the Coal Authority. Therefore, mining instability was not considered to pose a potential constraint to the proposed development. In addition, no mine entries were recorded to be present within the immediate vicinity of the proposed development site. An extract from the Coal Authority Interactive Map is attached in Appendix 4 for reference.
Radon	The site is located within a Lower Probability Radon Area (less than 1% of homes are estimated to be at or above the Action Level) and therefore radon protection measures are not considered necessary within the proposed development.

2.2 Site History

2.2.1 Information on the site's historical use was obtained through an inspection of available Ordnance Surveys maps (copy of maps included in Appendix 2) dating from 1854 to present day (2023). A summary of the information is presented below:

TABLE 2 - Site History

OS Map	Description
1855	The proposed development area was indicated to be occupied by a building, of an unspecified usage, within the 'Newington' area of Edinburgh. A road, later named 'Causewayside', with tramway was recorded to extend parallel to the eastern site boundary. The surrounding area was noted to be a mixture of residential and commercial properties.
1896 - 1897	The building onsite was noted to be a school. Within the surrounding area, 'Grange Works' was recorded adjacent to the southern site boundary and a hospital (later named 'Longmore Hospital') had been developed approximately 250 m to the north-east of the site. The tramway along Causewayside was no longer noted, although a further tramway was now recorded to extend along 'Grange Road', approximately 25 m to the north.
1908 – 1909	No significant changes were recorded onsite. Within the surrounding area, a biscuit factory had been developed approximately 20 m to the north-east of the proposed development site and a smithy, 'Paul's (printing) Works' and a timber works had been developed approximately 135 m to the south-south-east. Within the wider site area, 'St Katherine's (engineering) Works' and a distillery (with associated chimney) had been developed approximately 250 m to the north-north-west.
1914	No significant changes were recorded onsite or within the immediate surrounding area.
1920 – 1932	No significant changes were recorded onsite or within the immediate surrounding area.
1947 - 1949	The building onsite was now indicated as a 'school meals centre'. Within the surrounding area, the works to the immediate south was now indicated as a printing works, a confectionary works (with associated chimney) had been developed approximately 130 m to the south and several garages had been developed approximately 235 m to the south. A chimney was also now noted at the biscuit factory to the north-east and the tramway along Grange Road had been dismantled. Within the surrounding area, the distillery and chimney to the north-north-west were no longer noted.
1951 – 1967	No significant changes were recorded onsite or within the immediate surrounding area.
1967 - 1973	Sometime after 1967, the onsite school meals centre had been demolished and the site redeveloped with a building, similar to that of todays configuration.
1984 - 1985	No significant changes were recorded onsite. Within the surrounding area, the biscuit factory had closed and the area was redeveloped as the 'National Library of Scotland'. The works to the immediate south had also closed and the building redeveloped as government offices.
1989	By 1989, the National Library of Scotland building had expanded northwards and St Katherine's Work to the north-north-west had closed and the area redeveloped with residential properties.
1993	An electrical substation was recorded to be present within the far southern site area and the confectionary works to the south was no longer noted.
1994 - 1999	No significant changes were recorded onsite or within the immediate surrounding area.

TABLE 2 *CONT* - Site History

OS Map	Description
Present Day (2023)	Our site walkover survey (included in Appendix 1) indicated the proposed development site to be occupied by the Scottish Law Commission building, with the electrical substation still indicated to be present in the southern site area. The surrounding area was noted to be a mixture of residential and commercial properties.

2.3 Published Geological Information

Superficial Deposits

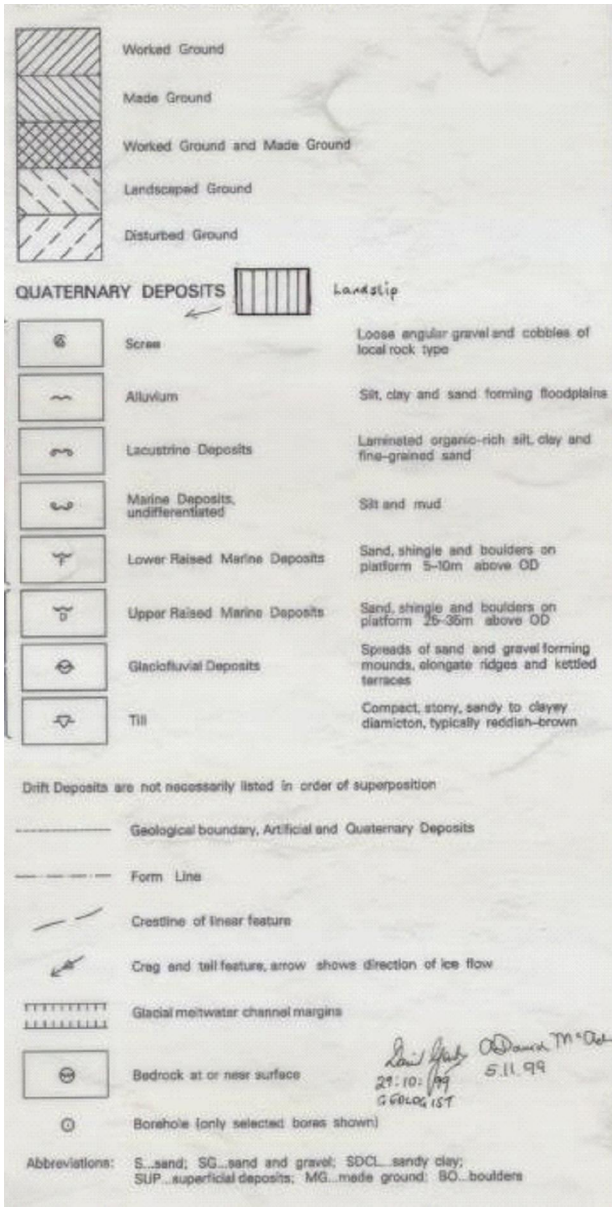
2.3.1 The British Geological Survey (BGS) indicated that the soils across the site were likely to comprise of glacial till (Drawing No. P23/075/SI/R/F/03). Made ground deposits, of a limited thickness, could also be anticipated within the area of former building. Historical boreholes (Appendix 5) sunk within the immediate surrounding area indicated made ground of up to 2.10 m in thickness, underlain by stiff glacial soils, which extended to rockhead at varying depths of between 10.70 mbgl and 16.30 mbgl.

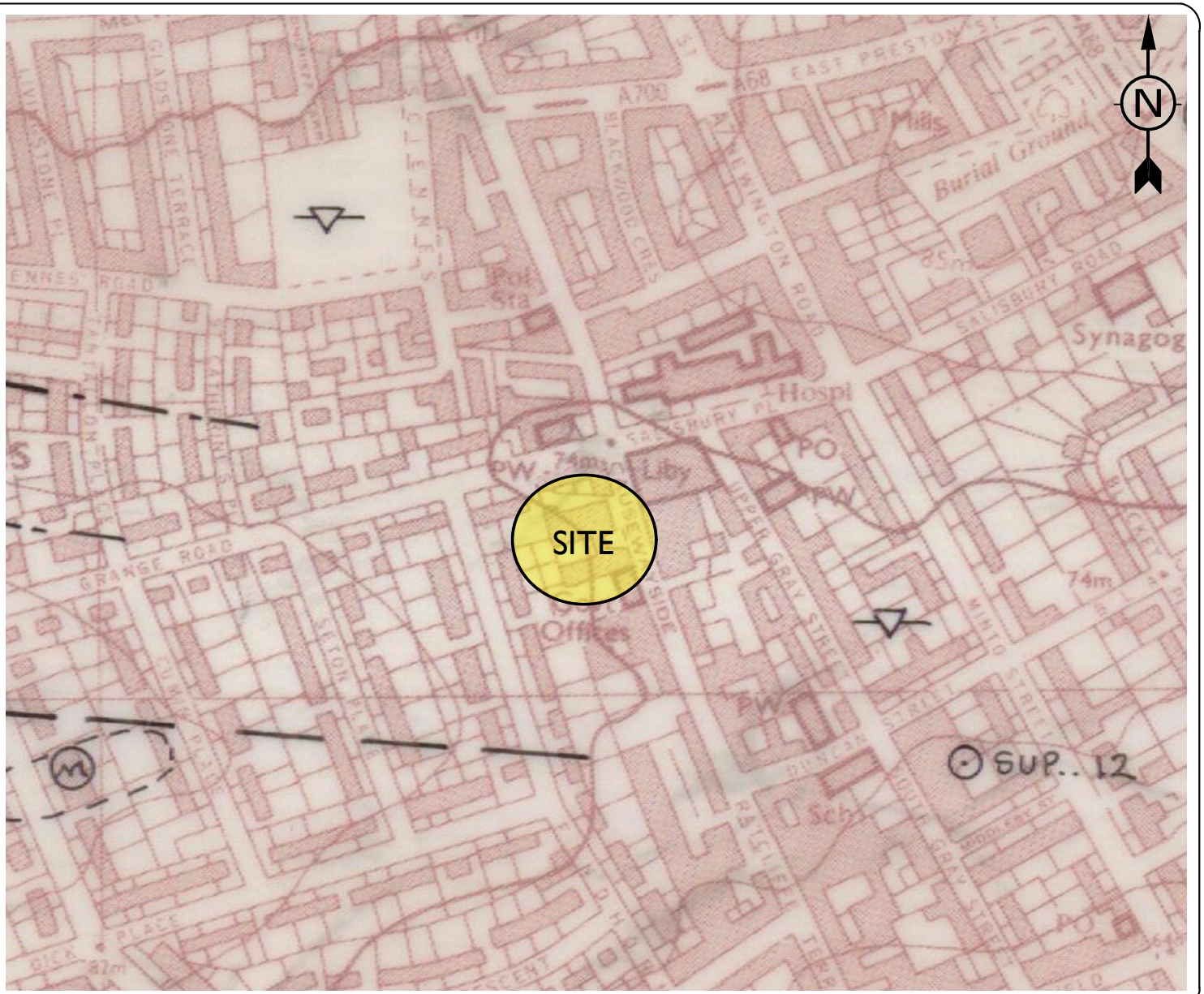
2.3.2 The underlying solid strata were indicated to belong to the Ballagan Formation of the early Carboniferous period, which typically comprises of sandstones, siltstones and mudstones with occasional beds of cementstones (Drawing No. P23/075/SI/R/F/04). Historical boreholes (Appendix 5) sunk within the immediate surrounding area indicated rockhead to be present at varying depths of between 10.70 mbgl and 16.30 mbgl and was generally described as, '*very weak to weak highly weathered brown becomes red clayey SILTSTONE*' (i.e. weathered rockhead), underlain by *'strong to very strong fresh grey fine grained SANDSTONE'* and *'moderately strong moderately weathered grey clayey SILTSTONE'*. A geological fault, trending north-east to south-west, was recorded approximately 300 m to the south-east of the proposed development site, and downthrowing the strata towards the north-west.

2.3.3 No coal seams or seam of economic importance were recorded to outcrop within the vicinity of the site. A review of the Coal Authority Interactive Viewer map (Appendix 4) also indicated that the site does not lie within a *'Coal Mining Reporting Area'* or *'Development High Risk Area'*, as defined by the Coal Authority. Consequently, no further investigations were considered necessary relative to this aspect of the ground conditions.

2.3.4 No recorded mine entries were indicated to be present within, or within 20 m of the proposed development site.

2.3.5 Interpretation of the site hydrogeology required consideration of the general geological conditions. In this instance, the available information indicated the site to be potentially comprised of three geological units: made ground (of a limited thickness), glacial till and sedimentary bedrock. The typical permeabilities of each of these strata are recorded in Table 3 below.





NOTES

REV	DATE	DETAILS

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PROJECT TITLE

CAUSEWAYSIDE
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DRAWING TITLE

EXTRACT FROM PUBLISHED GEOLOGICAL SURVEY MAP (DRIFT LITHOLOGY)

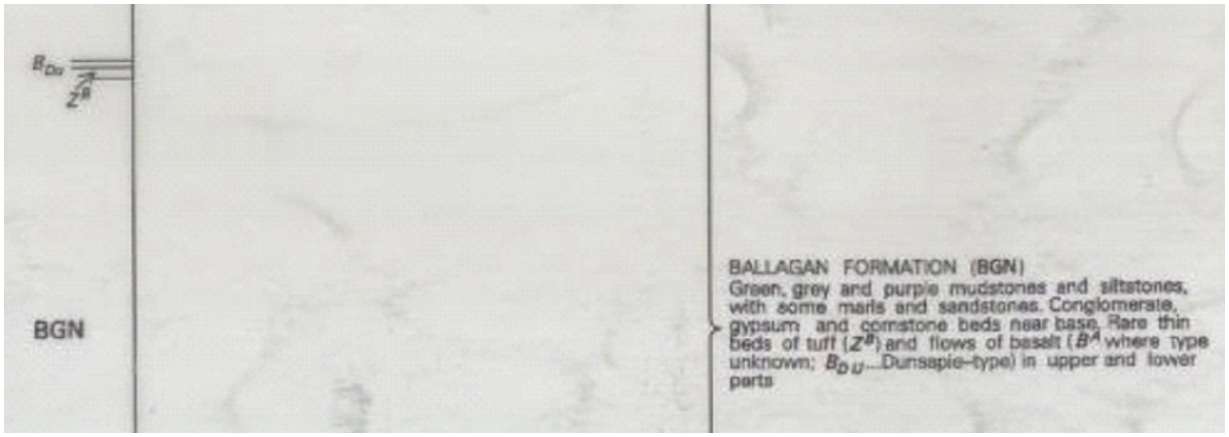
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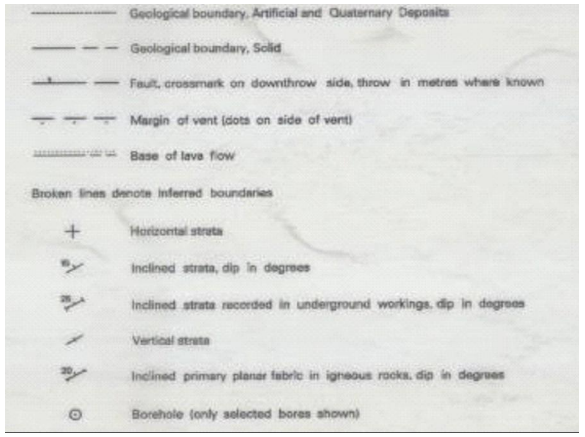
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NOTES

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PROJECT TITLE

CAUSEWAYSIDE
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DRAWING TITLE

EXTRACT FROM PUBLISHED GEOLOGICAL SURVEY MAP (SOLID GEOLOGY)

DRAWN BY	CHK'D BY	APP'D BY	DATE	SCALES
TR	LR	LR	21.07.23	Not to Scale

PROJECT No.	DRAWING No.	REVISION
P23/075	P23/075/SI/R/F/04	

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TABLE 3 –Typical Material Permeability

Material	Typical Permeability Range*
Made Ground (of a limited thickness)	Variable
Glacial Till	Very Low to Low
Sedimentary Bedrock	Low to High

* Sourced from the BGS National Environmental Research Council Open Report CR/06/160 N

2.3.6 At present, surface run-off would be relatively low given that the site is surfaced in building cover and tarmac/concrete hardstanding (i.e. impermeable surfaces). Infiltration of surface water was therefore considered to be high.

2.3.7 As the shallow soils could potentially allow limited infiltration, it was considered possible that a shallow, perched and discontinuous groundwater table could existing within the made ground deposits (if encountered), within the underlying cohesive glacial till deposits acting as a relatively impermeable barrier to downward migration of groundwater. The potential for this and deeper groundwater below rockhead to interact was, therefore, considered to be low, although lateral migration of shallow groundwater was considered possible.

2.3.8 It was considered likely that shallow groundwater could exist either within any made ground or natural soils. The Scottish Environmental Protection Agency (SEPA) provides guidance in document WAT-PS-10-01 'Assigning Groundwater Assessment Criteria for Pollutant Inputs' (March 2012) for assessing contamination risks to groundwater and the water environment. It was considered unlikely that groundwater within the superficial soils beneath the site could meet the minimum criteria to be classified as a water body i.e. an abstraction could achieve 10 m^3 per day. However, it was considered prudent to regard groundwater (with the superficial soils) as a potential sensitive receptor at this stage. It should be noted that SEPA's Water Classification Hub did not indicate there to be a classified groundwater body within the superficial soils beneath the proposed development site.

2.3.9 The Envirocheck Report (Appendix 2) indicated the underlying sedimentary strata to be a '*minor or moderately permeable aquifer – fractured or potentially fractured rocks which do not have a high primary permeability or other formations of variable permeability*' and therefore, it is considered unlikely that extraction from the bedrock aquifer could meet the minimum criteria to be classified as a water body i.e. abstraction could achieve 10 m^3 per day. Furthermore, we are not aware of any records that suggest that extraction from the deeper groundwater (bedrock aquifer) is occurring within the immediate vicinity of the proposed development site for private use (Appendix 5) and groundwater abstraction is not being proposed as part of the proposed development. However, it was considered prudent to also regard the deeper groundwater (bedrock aquifer) as a potential receptor at this stage. SEPA's Water Classification Hub indicated the groundwater, within the bedrock aquifer, beneath the site area to be named 'Edinburgh'. The most recent SEPA classification sheet, dated 2020, indicated this deeper groundwater body to have an overall status of *'Good'*.

2.4 Preliminary Conceptual Site Model

2.4.1 In order to fully evaluate the potential presence and impact of contamination at the site, the area must be considered in an environmental context taking account of its geology, topography and past and present land-use. Science Report SC050021/SR3, published by the Environment Agency in January 2009, supersedes the previous Contaminated Land Reports (CLR7 to CLR10 and briefing notes) series and provides standard guidance for the assessment of sites that may be contaminated. This essentially highlights the importance of developing a robust *Conceptual Site Model*. The model then forms an integral part of the contamination assessment for the proposed development site, looking at conventional source-pathway-receptor linkages.

2.4.2 Statutory guidance sets the definition of contaminated land within the context of the "suitable for use" approach. It is based on the principles of risk assessment, including the concept of a **pollutant linkage** between a **source** contaminant and a **receptor**, by means of a **pathway.** The presence of all three elements identifies a plausible pollutant linkage. An assessment of the potential sources, pathways and receptors constitutes a conceptual mode for the site. This concept is considered further below. We would highlight that the approach, while perhaps rendering the site suitable for its current use, may provide inappropriate to a change in site designation or specific land use, arising from existing site conditions.

2.5 Receptor Characterisation

2.5.1 Potential receptors at the site are defined on the basis of the site proposals for multi-storey student accommodation, with areas of soft landscaping and associated road infrastructure. The following receptors are therefore considered relevant to this project:

- Humans – site end users and construction work (outdoor),
- Humans – site end users (indoor),
- Buildings and services (including water supply pipes),
- Vegetation (plants in areas of soft landscaping (if any);
- Water Environment (groundwater (within the superficial soils and bedrock aquifer) and surface water).

2.6 Source Characterisation

2.6.1 The potential on-site sources of contamination identified by this desk study are:

- Possible deposition of contaminated fill materials associated with the previous development of the site as a school and then the Scottish Law Commission
- Potential spillages/leakages of fuels/oils associated with onsite vehicles, given that part of the site is currently occupied by a carpark (although indicated to be of a limited size)
- Generation and accumulation of ground gases sourced from any made ground deposits encountered.

2.6.2 Note that we recognise the presence of the electrical substation within the far southern site area. However, the substation was constructed in approximately 1993 i.e. after polychlorinated biphenyls (PCBs) were banned in the UK (PCB's were banned in the UK after 1981). Therefore, spillages/leakages of PCBs, associated within substation, are not considered to pose a significant risk to the proposed development.

2.6.3 The typical processes involved and associated Contaminants of Concern (COC) are discussed and summarised in Table 4 below:

TABLE 4: Contaminants of Concern

THE SITE	**Industrial Activity/ Site Use**	**Potential Pathways**	**Associated Potential Contaminants***
CURRENT AND PREVIOUS	• Possible deposition of contaminated fill materials associated with the development of the site as a school and then the Scottish Law Commission • Potential spillages/leakages of fuels/oils associated with onsite vehicles, given that part of the site is currently utilised as a carpark • Generation and accumulation of ground gases sourced from any made ground deposits encountered	• Deposition of waste materials • Generation and accumulation of ground gasses • Leaching of contaminants to groundwater • Migration of gases and vapours • Spillage/leakage of fuels and chemicals	Metals: As, Cd, Cr, Ni, Zn, Cu, Hg, Pb, Organics: Fuel oils, PAHs, Miscellaneous: Asbestos, Cyanide, VOCs and SVOCs Ground Gasses: CO_2, CH_4
IMMEDIATE SURROUNDING AREA	**Industrial Activity/ Site Use**	• **Potential Pathways**	**Associated Potential Contaminants**
CURRENT AND PREVIOUS	• Possible deposition of contaminated fill materials associated with the residential, commercial and infrastructural development • Former industry including; biscuit factory, engineering works, distillery and printing works etc • Processes associated with former 'Longmore Hospital' • Spillages/leakages of fuels/oils associated with the former tramways and from the current garages	• Deposition of waste materials • Generation and accumulation of ground/mine gasses • Leaching of contaminants to groundwater • Migration of gases and vapours. • Leakage/spillage of hydrocarbon products	Leachates (metals, semi-metals and non-metals) Ground Gasses: CO_2, CH_4 Fuel oils, PAHs, phenol.

** It should be noted that the potential contaminants listed above are intended as a guide to the possible contaminant species which may be encountered in such sites, and that other contaminants could potentially be present. The species will, however, be considered in the site assessment and evolution of appropriate chemical testing regime based upon the finding of the investigations and recommendations of the industry profiles will be carried out.*

2.7 Pathway Characterisation (Pollutant Linkages)

2.7.1 The pathways by which sensitive receptors may be exposed to potential sources of contamination, as determined by the proposed end use for the site are as follows:

1. Humans – site end users and construction workers (outdoor)

- Dermal (skin) contact with contaminated soil, fugitive dust and the absorption of any contaminants through the skin into the body.
- Inhalation of fugitive soil dust or vapour.
- Ingestion of soil by hand to mouth activity.
-

2. Humans – site end users (indoor)

- Inhalation of any ground gas migrating into the buildings.
- Inhalation of soil derived dust.

3. Buildings

- Potential soil gas generated in the ground vertically migrating and pooling within the structure.
- Contact with aggressive or acidic soils will affect the concrete design of the foundations.

4. Services including the domestic water supply

- Direct contact with contaminated soil or groundwater.
- Leaching of contaminants through the soil.
- Service trenches acting as preferential migration pathways for contamination.
- Permeation of plastic water supply pipes.

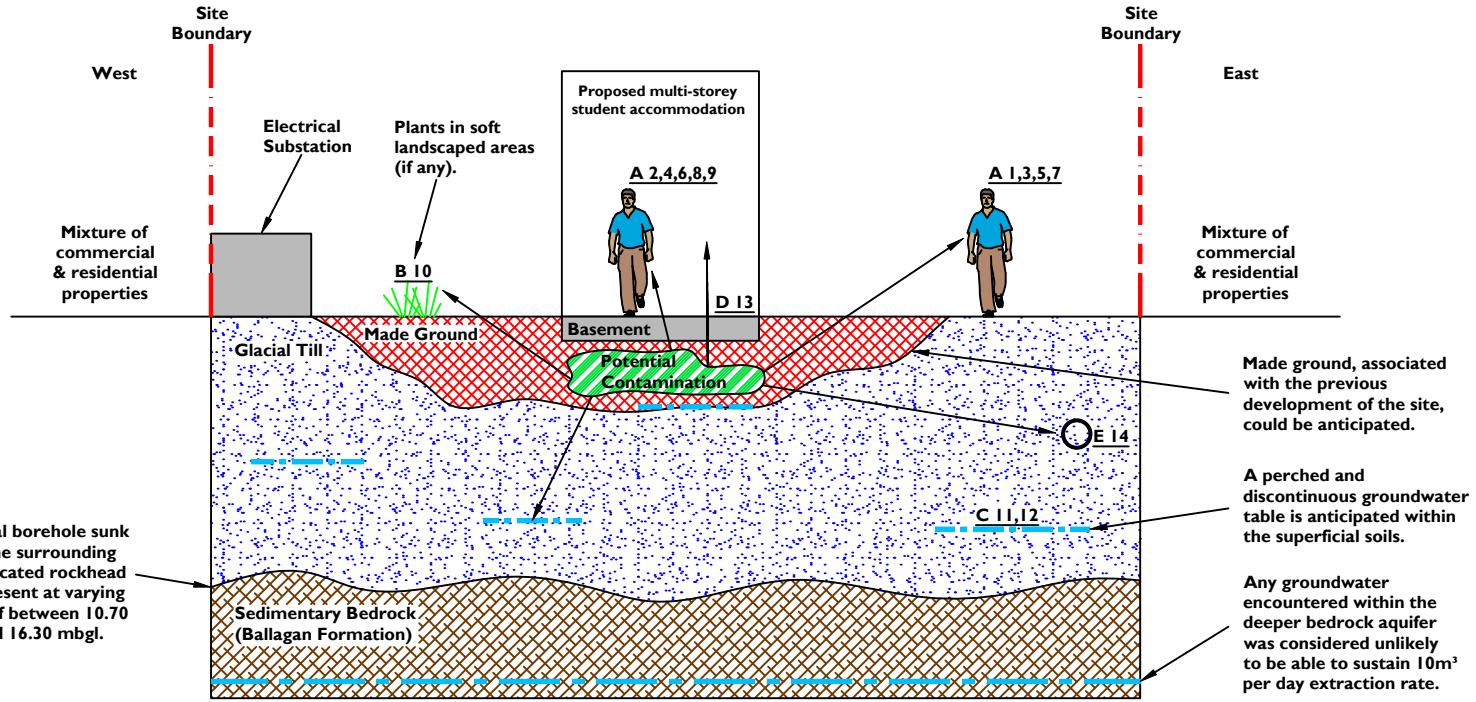
5. Vegetation (plants in areas of soft landscaping (if any)).

- Direct contact with contaminated soils and groundwater.
- Uptake of contaminants from the soil or groundwater into the plant.

6. Water Environment (groundwater and surface water)

- Leaching of contaminants from the soil to groundwater.
- Contaminant migration offsite in the groundwater.
- Contaminant uptake as base flow within surface watercourse *(considered unlikely given that the nearest surface water to the proposed development site i.e. Pow Burn/Braid Burn is located approximately 1.00 km to the south-east of the site (at its closest point), but should be considered at this stage).*
- Direct entry of contaminants (e.g. spillage or via outfall pipes) into surface water *(considered unlikely given that the nearest surface water to the proposed development site i.e. Pow Burn/Braid Burn is located approximately 1.00 km to the south-east of the site (at its closest point), but should be considered at this stage).*

2.7.2 The potential source-receptor-pathway linkages identified for the site are illustrated within our Preliminary Conceptual Site Model (Drawing No. P23/075/SI/R/F/05) and on Tables 5A and 5B. Site investigations were required to confirm or otherwise the existence of such linkages in addition to providing further confirmation of the geological and geotechnical conditions.



Potential Source

- Potentially contaminated made ground.
- Ground gasses potentially sourced from any made ground deposits encountered.

Potential Exposure Pathways

1. Outdoor ingestion of dust.
2. Indoor ingestion of dust.
3. Skin contact with outdoor soil.
4. Skin contact with indoor dust.
5. Outdoor inhalation of dust.
6. Indoor inhalation of dust.
7. Outdoor inhalation of soil vapour.
8. Indoor inhalation of soil vapour.
9. Inhalation of ground gases.
10. Contaminant uptake by vegetation.
11. Leaching of contaminants to the groundwater.
12. Contaminant migration in the groundwater.
13. Detrimental effects on buried concrete.
14. Permeation of plastic water supply pipes.

Potential Receptors

A. Site users / construction personnel.
B. Vegetation / fauna.
C. Groundwater.
D. Buried concrete (Service and foundations)
E. Plastic water supply pipes.

NOTES

REV	DATE	DETAILS

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PROJECT TITLE

CAUSEWAYSIDE
EDINBURGH

DRAWING TITLE

PRELIMINARY CONCEPTUAL SITE MODEL

DRAWN BY	CHK'D BY	APP'D BY	DATE	SCALES
TR	LR	LR	21.07.23	Not to Scale

PROJECT No.	DRAWING No.	REVISION
P23/075	P23/075/SI/R/F/05	

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TABLE 5A: Preliminary Qualitative Risk Assessment – On Site

Source	COCs	Pathway	Receptors (s)	Assessment	Further Investigation Required
1. Possible deposition of contaminated fill materials associated with the development of the site as a school and then the Scottish Law Commission 2. Potential spillages/leakages of fuels/oils associated with onsite vehicles, given that part of the site is currently utilised as a car park 3. Generation and accumulation of ground gases sourced from any made ground deposits encountered	Metals, semi-metals and non-metals: As, Cd, Cr, Ni, Zn, Cu, Hg, Pb, Organics: Phenols Hydrocarbons and PAH's VOCs and SVOCs Anions: Cyanide, Sulphate Ground gasses: CO_2, CH_4 Asbestos	Dermal contact, ingestion, inhalation	Human – site workers Humans – end users (outdoor)	Spillage/leakage of contaminants impacting near surface soils. Contaminated materials may have been deposited within the site.	Yes
		Leaching through soil or direct migration	The water environment - groundwater	Contaminants may be leached and potentially mobilised from the soil by percolation and/or shallow groundwater movement.	Yes
		Direct contact, leaching through soil, groundwater migration	Buildings and services	Potential for aggressive chemical environments for concrete due to sulphate and acidic conditions. Presence of contaminants in soil that may permeate water supply pipes.	Yes
		Gas/vapour inhalation, vertical/lateral migration	Buildings and services Humans – end users (indoor)	Contamination may include gas/vapour producing materials or compounds that could vertically migrate into overlying buildings producing a potentially asphyxiating or explosive environment.	Yes
		Direct contact, uptake	Plants	Direct contact or uptake of contamination from the soil or groundwater could adversely affect any plants grown.	Yes
		Migration in the groundwater	Groundwater	Contaminants could impact the groundwater and migrate offsite.	Yes
		Point source discharge	Surface water	Direct entry of contaminants into surface water via accidental spillage/leakage or from discharge pipework.	Yes
		Diffuse source	Surface Water	Contaminants could migrate in the groundwater and act as base flow for surface water recharge.	Yes

TABLE 5B: Preliminary Qualitative Risk Assessment – Off-Site

Source	COCs	Pathway	Receptors (s)	Assessment	Further Investigation Required
1. Possible deposition of contaminated fill materials associated with the residential, commercial and infrastructural development 2. Former industry including; biscuit factory, engineering works, distillery and printing works etc 3. Processes associated with the former 'Longmore Hospital' 4. Spillages/leakages of fuels/oils associated with the former tramways and from the current garages	Metals, non-metals and semi-metals: (Leachates) As, Cd, Cr, Ni, Zn, Cu, Hg, Pb, Organics: Phenols Hydrocarbons, PAHs Anions: Cyanide, Sulphate Ground gasses: CO_2, CH_4 Asbestos	Dermal contact, ingestion, inhalation	Human – site workers Humans – end users (outdoor)	Contaminants likely to migrate into the site via groundwater or via pore spaces in granular soils and therefore unlikely to be exposed to site users via direct contact, ingestion of inhalation pathways.	No
		Leaching through soil or direct migration	The water environment - groundwater	Contaminants may be leached and potentially mobilised by shallow groundwater movement.	Yes
		Direct contact, leaching through soil, groundwater migration	Buildings and services	Potential for aggressive chemical environments for concrete due to sulphate and acidic conditions. Presence of contaminants in soil that may permeate water supply pipes.	No
		Gas/vapour inhalation, vertical/lateral migration	Buildings and services Humans – end users (indoor)	Contamination may include gas/vapour producing materials or compounds that could vertically migrate into overlying buildings producing a potentially asphyxiating or explosive environment.	Yes
		Direct contact, uptake	Plants	Uptake of contamination from the soil or groundwater could adversely affect plant growth.	No
		Migration in the groundwater	Groundwater	Leachates may migrate either into the site (from the offsite source) or offsite (via an onsite source).	Yes
			Surface water	Contaminants may migrate in the groundwater and act as base flow for surface water recharge, allowing contaminants to migrate significant distance in the surface watercourse.	Yes

3.0 SITE INVESTIGATIONS

3.1 Objectives

3.1.1 The investigations were designed relative to the preliminary conceptual site model and in recognition of the nature of the proposed redevelopment of the as multi-storey student accommodation. The objectives of the investigation included the determination of:

a) The conjectured distribution and composition of (any) made ground and natural soils (including depth to rockhead)
b) The geological context.
c) The groundwater regime.
d) Chemical contamination.
e) Potential foundation solutions.

3.2 Scope and Methods of Investigations

3.2.1 The scope and method of investigation to fulfil objectives (a) to (e) is summarised in Table 6 below.

TABLE 6 - Site Investigations Based on Objectives

Objective		Site Investigation
a)	*The conjectured distribution and composition of (any) made ground and natural soils (including depth to rockhead)*	3 No. cable-percussive boreholes (BH01 to BH03), with rotary follow-on (RBH01 to RBH03), sunk by Phoenix Drilling Limited, on behalf of Mason Evans. *Note that hard obstructions were encountered at shallow depths in BH01 and BH02. Therefore, these bores were relocated and redrilled as BH01A, BH01B and BH02A, respectively.* *The rotary follow-on element was carried out within boreholes BH01B, BH02A and BH03.*
b)	*The geological context*	3 No. terrier boreholes (BH04 to BH06) sunk by Phoenix Drilling Limited, on behalf of Mason Evans. *Again, hard obstructions were encountered at shallow depths within BH04 and BH05. As such, these bores were relocated and redrilled as BH04A., BH04B and BH05A, respectively.*
c)	*The groundwater regime*	Cable-percussive/rotary follow-on boreholes with gas/groundwater monitoring wells i.e. 3 out of 3 boreholes (BH01B, BH02A and BH03).

TABLE 6 *CONT* - Site Investigations Based on Objectives

Objective		Site Investigation
d)	*Chemical contamination*	**Soil Contamination** Cable-percussive boreholes with soils analysis (including an asbestos screen, plus quantification (where applicable)) i.e. 8 samples (6 No. made ground and 2 No. natural samples) recovered from BH01 to BH03. Cable-percussive boreholes with specific pH and sulphate testing i.e. 1 sample (1 No. natural sample) recovered from BH01. Cable-percussive boreholes with semi-volatile organic compounds (SVOCs) and volatile organic compounds (VOC) testing i.e. 1 sample (1 No. made ground sample) recovered from BH03. Cable-percussive boreholes with Waste Acceptance Criteria (WAC) testing i.e. 3 samples (3 No. made ground samples) recovered from BH01, BH02 and BH03. Terrier boreholes with soils analysis (including an asbestos screen, plus quantification (where applicable)) i.e. 3 samples (2 No. made ground and 1 No. natural sample) recovered from BH05 and BH06. Terrier borehole with Waste Acceptance Criteria (WAC) testing i.e. 1 sample (1 No. made ground sample) recovered from BH06.
d)	*Chemical contamination*	**Water Contamination** Cable-percussive boreholes with leachate analysis i.e. 5 samples (4 No. made ground and 1 No, natural sample) recovered from BH01 to BH03. Terrier boreholes with leachate analysis i.e. 2 samples (2 No. made ground samples) recovered from BH05 and BH06 Groundwater samples obtained from BH01B/RBH01B on 3 No. occasions and from BH02A/RBH02A on 2 No. occasions.
e)	*Potential foundation solutions*	All terrier and cable-percussive with rotary follow-on bores sunk across the site.

3.3 Summary of Ground Investigation Data

3.3.1 The scope and location of the works was determined by Mason Evans, where access permitted. The sampling was generally targeted across the site, where access allowed, in relation to geo-environmental matters as dictated by the potential limited onsite contamination risks identified in the Preliminary Conceptual Site Model. Site works were implemented generally in accordance with BS10175:2011 + A2: 2017 and were designed to achieve an approximate 35 m spacing across the site, where access permitted. The sample spacing was considered appropriate given the history of the site as a school and then more recently as the Scottish Law Commission, and also given the development proposals for multi-storey study accommodation.

3.3.2 Note that at the time of our investigation, the Scottish Law Commission building (with associated outbuildings) occupied at least half of the site. Therefore, we would recommend that some limited confirmatory site investigations works, including chemical testing, are carried out beneath the building footprint following demolition. Any supplementary SI, carried out post-demolition, will be reported in a separate addendum letter report upon completion.

TABLE 7 –Site Investigations

Cable-Percussive Borehole with Rotary Follow-on	3 No. cable-percussive boreholes (BH01 to BH03) sunk by Phoenix Drilling, on behalf of Mason Evans, to depths of up to 4.00 mbgl. Following termination of each cable-percussive borehole, the dual sampling rig was utilised to confirm the depth rockhead at each of the 3 No. locations (RBH01 to RBH03). Once rockhead was confirmed, it was then cored to recover at least 9.00 of intact core. The rotary follow-on element was carried out within boreholes BH01B, BH02A and BH03. *Note that hard obstructions were encountered at shallow depths in BH01 and BH02. Therefore, these bores were relocated and redrilled as BH01A, BH01B and BH02A, respectively.* Standpipe installations were installed in BH01B, BH02A and BH03 to allow for a period of gas and groundwater monitoring within the superficial soils.
Terrier Boreholes	3 No. terrier boreholes (BH04 to BH06), sunk by Phoenix Drilling, on behalf of Mason Evans, to depths of up to 1.90 mbgl. *Hard obstructions were encountered at shallow depths within BH04 and BH05. As such, these bores were relocated and redrilled as BH04A., BH04B and BH05A, respectively.*
Chemical Testing	11 No. soil samples were tested for a comprehensive range of potential contaminants (including an asbestos screen (plus quantification (where applicable)), 1 No. soil samples was tested for VOCs and SVOCs, 1 No. soil sample was tested specifically for pH and sulphate, 4 No. soil samples were tested for Waste Acceptance Criteria (WAC), 7 No. samples were tested for leachate analysis and 5 No. groundwater samples were tested for a comprehensive suite of chemical analysis, including TPHs and PAHs.
Geotechnical Testing	In-situ SPT tests were undertaken in all of the boreholes (terrier, cable-percussive and rotary follow-on). In addition, moisture content analysis, Atterberg Limits and triaxial testing was undertaken on a selection of samples. Samples of the underlying rock strata, obtained from the rotary follow-on boreholes, were also tested for unconfined compressive strength (UCS) and point load testing.

3.3.3 The ***cable-percussive boreholes*** were intended to provide coverage across the proposed development site (where access allowed), in order to define the nature of the shallow soils, allow a selection of representative samples for a comprehensive suite of chemical analysis and to provide geotechnical and hydrogeological data associated with the proposed development. Given the proposed end-use, the purpose of the sampling and testing was to identify potential risks to site users and the Water Environment. Gas and groundwater standpipe installations were also installed in each of the 3 No. cable-percussive boreholes to allow for a period of ground gas monitoring within the superficial soils and to monitor fluctuations in the groundwater levels beneath the site.

3.3.4 The ***rotary-follow element,*** carried out within the cable-percussive boreholes (BH01B, BH02A and BH03) was intended to confirm the depth to rockhead and to obtain further information on the nature of the underlying rock strata, inorder to aid possible piling designs.

3.3.5 The ***terrier borehole***s were intended to increase coverage across the proposed development site and to allow a selection of additional soil samples to be obtained for chemical and geotechnical analysis.

3.3.6 ***Waste Acceptance Criteria (WAC)*** testing was undertaken on a selection of made ground samples to determine the waste classification for the receiving waste disposal facility site, should any the made ground materials require to be removed from site.

3.3.7 As previously discussed, in accordance with BS10175:2011 + A2:2017, the intrusive investigation locations were designed to achieve an approximately 35 m spacing across the proposed development site. This was considered appropriate given the proposed end-use for student accommodation.

3.3.8 Representative samples of the made ground and underlying natural soils were obtained during the investigations and tested for a comprehensive suite of analysis, included total petroleum hydrocarbons (TPHs), polycyclic aromatic hydrocarbons (PAHs) and an asbestos screen. VOCs (volatile organic compounds) and SVOCs (semi-volatile organic compounds) were also tested on a soil sample obtained from BH03, to assess the for the potential presence of spillages/leakages of fuels/oils, associated with on-site vehicles as the front of the building is currently being utilised as a car park (although for a limited number of cars). The results of the analyses were utilised in a site-specific risk assessment in accordance with the current UK technical guidance for human health and SEPA guidance for the Water Environment. A selection of soil samples were also tested for leachable metals, as a precautionary measure, encase shallow groundwater samples could not be obtained during the later groundwater monitoring period, given the cohesive nature of the indicated superficial soils.

3.3.9 All soil samples recovered for chemical analysis were contained in sealed plastic tubs and glass amber jars, labelled and stored onsite in cool boxes to maintain natural temperature. The glass amber jars were collected at every sample location in order to prevent chemical breakdown as a result of exposure to light and limit absorption of the contaminant to the sample container. This procedure is designed to maintain sample integrity and ensure that the chemical analysis is as representative of the site conditions as possible.

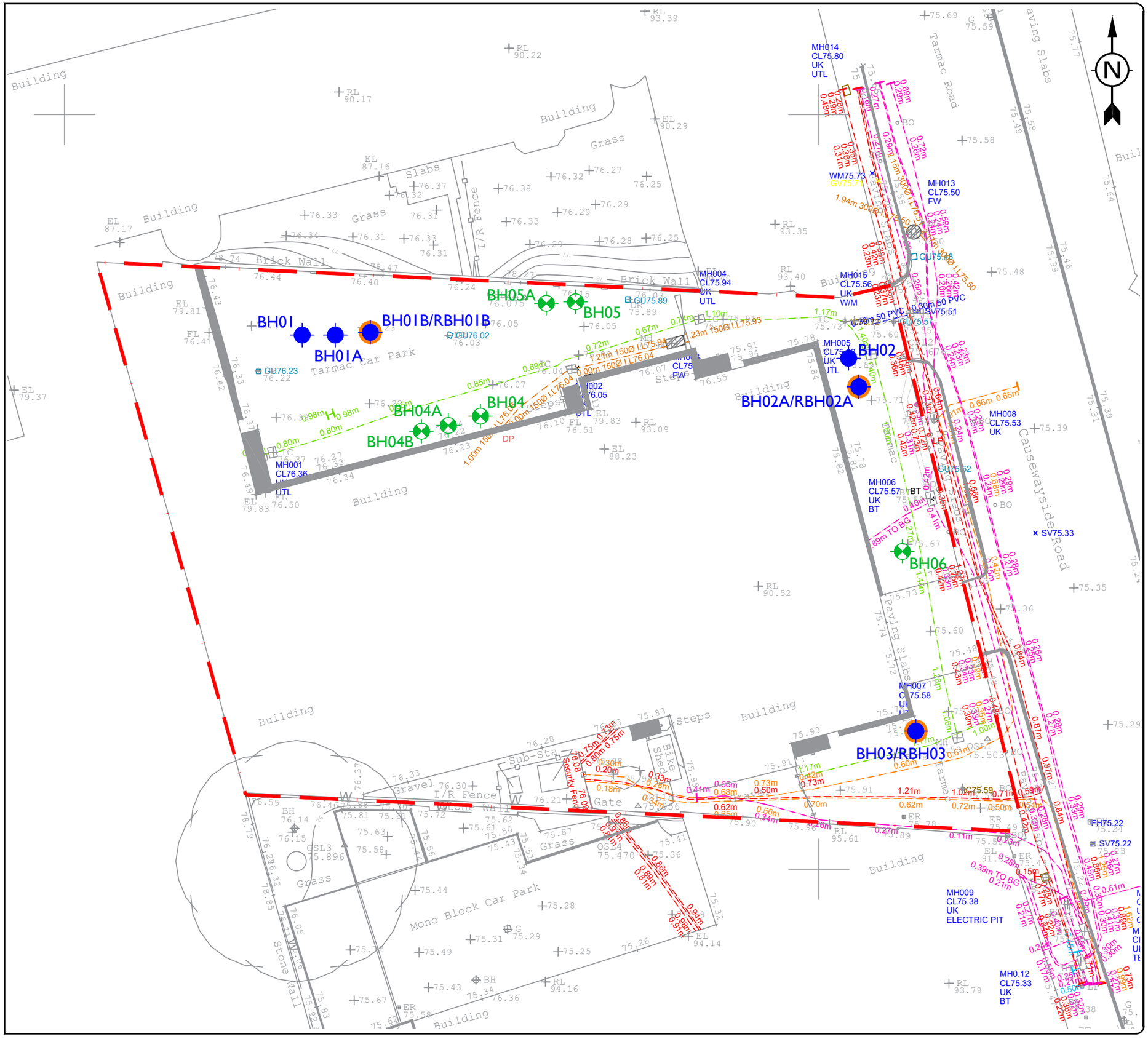
3.3.10 Soil samples were collected and dispatched to the laboratory for immediate testing. The scope of the chemical testing of soil samples recovered during the investigation are discussed in detail in section 6.0 of this report.

3.3.11 Properties recorded during logging of the shallow soils included the general composition, strength, material, description, colour, density, state of weathering and any other notable features. These were generally described in accordance with the guidelines provided by the Code of Practice for Site Investigations BS5930:2015+ A1:2020.

3.3.12 The locations of the investigative works are indicated on Drawing No. P23/075/SI/R/F/06 and records of the exploratory holes are included in Appendix 6.

3.4 Investigation Rationale

3.4.1 The findings of our preliminary CSM indicated the potential presence of contaminated fill materials associated with previous development of the site as a school and then as the Scottish Law Commission, as well as potential spillages/leakages of fuels/oils associated with onsite vehicles, given that the site is partially occupied by a carpark (although of a limited size and extent). The generation and accumulation of ground gases potentially sourced from any made ground deposits encountered was also considered. Consequently, the sampling strategy for the investigations undertaken was largely targeted around the existing building footprint, where access allowed, to try and give as much coverage of the site as possible.



NOTES

- Approximate site boundary
- BH01 to BH03 — Cable-percussive boreholes sunk by Phoenix Drilling Ltd, on behalf of Mason Evans (April 2023)
  - Rotary follow-on-element, also carried out by Phoenix Drilling Ltd undertaken in boreholes BH01B, BH02A and BH03 (referenced as RBH01B, RBH02A and RBH03)
- BH04 to BH06 — Terrier boreholes sunk by Phoenix Drilling Ltd, on behalf of Mason Evans (April 2023)
- Gas/groundwater monitoring well installation

REV	DATE	DETAILS

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PROJECT TITLE

CAUSEWAYSIDE
EDINBURGH

DRAWING TITLE

LOCATION OF
EXPLORATORY HOLES

DRAWN BY	CHK'D BY	APP'D BY	DATE	SCALES
TR	LR	LR	21.07.23	1:250 @ A3

PROJECT No.	DRAWING No.	REVISION
P23/075	P23/075/SI/R/F/06	

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3.4.2 The scope and location of exploratory holes was determined by ourselves, where access permitted. The chemical analysis involved the sampling of any made ground and natural soils at regular depth intervals, to allow for an assessment of the risk to human health as well as to evaluate the risk to potential receptors including vegetation/fauna, groundwater, concrete structures/foundations and/or buried services (i.e. water supply pipes). The analytic schedule was then based on the interpreted origin of the soils and their description, which is consistent with best practice under current contaminated land guidance BS:10175:2011 *'Code of Practice for the Investigation of Contaminated Land'*. The soil samples submitted to the laboratory for chemical analysis were considered representative of the made ground and natural soils encountered across the site.

3.4.3 In recognition of the above, the following site practices were implemented:

- The drilling works have been undertaken by a suitably accredited sub-contractor;
- The geological succession at each borehole location has been logged by an experienced field specialist and samples taken for laboratory analysis. A visual assessment was made of the geological character and potential contamination, if present. Soil samples have predominantly been taken at approximately 0.5 m intervals, or at a change in lithology, or where evidence of potential contamination impact was observed.
- In selecting the appropriate samples for testing, we have taken cognisance of a number of factors, including the proposed site use. Sampling rationale has been determined in accordance with R&D Technical Report P5-066/TR Secondary Model Procedure for the Development of Appropriate Soil Sampling Strategies for Land Contamination, as indicated in the table below.
- The scope of the testing implemented considered the interpreted origin of the materials in association with their description. This is consistent with best practice under current contaminated land guidance. The chemical composition of these materials was assessed for a wide spectrum of potential contaminants, comprising a broad range of common organic and inorganic substances primarily of a toxic or phytotoxic nature, and appropriate to the past usage of the site.
- During sample collection, relevant information such as notes of field observations has been logged before transferring the samples to laboratory-prepared sample bottles of appropriate type. Care was also taken to minimise the aeration of samples during transfer to the bottles.

TABLE 8 – Sampling Strategy

Rationale for Sampling at Different Depths	
Depth Range	**Rationale**
Ground Level – 1.0 m	To assess: • Human/ animal intake arising from ingestion and dermal contact. • Potential for wind entrainment leading to inhalation (of contaminated soils and dusts) or deposition onto neighbouring land. • Surface water run-off (e.g. due to flash flooding). • Uptake by shallow rooting plants (e.g. crops, ornamental and wild species). • Surface leaching to groundwater.
>1.0 m in made or natural ground	To assess: • Intake via ingestion/ inhalation/ dermal contact arising from 'abnormal' (or unpredicted) excavation (e.g. children digging dens) or for other purposes such as swimming pools, ponds, house extensions. • Uptake by deep rooting shrubs or trees. • Intake by or arising from the activities of burrowing animals. • Intake arising from construction/ maintenance of buildings and services, for example: a. Foundations (usually within 2.0 m of final formation level). b. Water supply pipes, telecommunications, gas and power (0.5 m to 1.0 m of final formation levels). c. Sewers (from 0.5 m to >1.0 m of final formation level). • To locate perched water of groundwater. • To confirm depth of made ground. • To locate possible lateral pathways for gas or vapour migration in made ground. • To establish the extent of any leaching of soluble constituents from superficial soils. • To detect 'deep' contamination (e.g. gas generating materials, leachable materials, dense solvents located above an impermeable stratum). • To obtain information on 'background' soil properties. • To locate 'natural' lateral migration pathways.

3.4.4 The samples scheduled for chemical analysis were selected by ourselves and were intended to provide site wide analysis of the soils encountered and also provide representative coverage of the proposed development site. Samples were generally collected at regular intervals throughout the made ground (where encountered), as well as from the underlying natural soils, inorder to provide a comprehensive assessment of all soils encountered onsite.

3.5 Analytical Procedures

3.5.1 Analytical procedures adopted during the chemical analyses, carried out on behalf of the consultant, by Derwentside Environmental Testing Services (DETS), conformed to recognised practices, allowing the award of UKAS accreditation (unless indicated otherwise).

4.0 INVESTIGATION RESULTS

4.1 Ground Conditions

4.1.1 The ground conditions encountered during the investigation were generally consistent with the anticipated sequence of strata indicated by the desk study information. The soils across the site were noted to comprise of made ground, underlain by stiff to very stiff glacial till. Rockhead was definitively encountered in the rotary-follow-on bores at varying depths of between 15.50 mbgl and 16.00 mbgl.

4.1.2 The generalised soil sequence encountered was as follows:

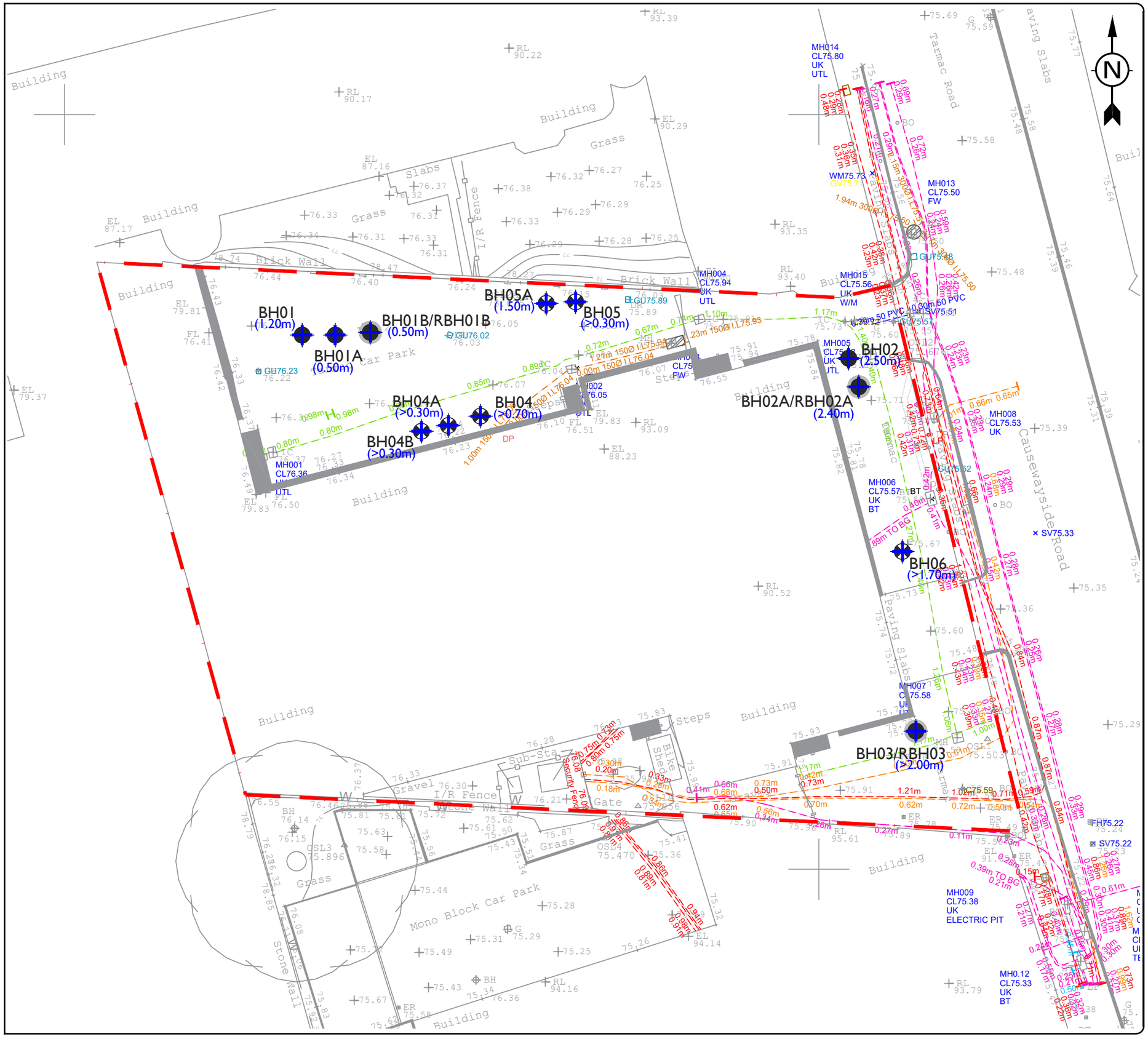
TABLE 9 – Summary of Ground Conditions

Soil Type	Depth Range (mbgl)
Made Ground (surface coverings)	0.00 – 0.10
Made Ground	0.10 – 2.50
Glacial Till	0.50 – 16.00
Rockhead	15.50 – 16.00

- **Made Ground (surface coverings)** – encountered across the site from the ground surface to a maximum recorded depth of 0.10 mbgl. Described as; 'tarmacadam'.

- **Made Ground** – encountered underlying the surface coverings across the site and extending to a maximum recorded depth of 2.50 mbgl (BH02). Described as; black/brown slightly clayey very gravelly sand with cobbles of brick (i.e. sub-base); grey/brown slightly sandy angular cobbles of igneous rock (i.e. sub-base); *'stone fill'* (drillers description); *'orange and brown angular cobbles of brick and igneous rock'* (drillers description); *'brick, ash and stone fill'* (drillers description); *'brown clay stone and concrete fill'* (drillers description); grey/brown slightly clayey gravelly sand; grey/brown slightly silty slightly sandy slightly gravelly clay (i.e. reworked natural soils); or stiff becoming very stiff or hard dark brown sandy gravelly clay (i.e. reworked natural soils), with varying cobble content and extraneous materials including; sandstone, brick and concrete. The recorded made ground thickness (including made ground surface coverings) is indicated on Drawing No. P23/075/SI/R/F/07.

- **Glacial Till** – the underlying natural soils, which were considered to be representative of the indicated glacial till deposits, were encountered across the site at varying depths of between 0.50 mbgl and 2.50 mbgl, extending to a maximum recorded depth of 16.00 mbgl (BH03/RBH03). These deposits were generally described as; '*brown sandy clay'* (drillers description); *'brown sandy gravelly CLAY'* (drillers description); stiff and very stiff grey/brown slightly silty slightly sandy slightly gravelly CLAY; or very stiff dark brown sandy very gravelly CLAY, all with varying cobble and boulder content.



NOTES

- Approximate site boundary
- BH01 to BH03 — Cable-percussive boreholes sunk by Phoenix Drilling Ltd, on behalf of Mason Evans (April 2023)
  - Rotary follow-on-element, also carried out by Phoenix Drilling Ltd undertaken in boreholes BH01B, BH02A and BH03 (referenced as RBH01B, RBH02A and RBH03)
- BH04 to BH06 — Terrier boreholes sunk by Phoenix Drilling Ltd, on behalf of Mason Evans (April 2023)
- Gas/groundwater monitoring well installation
- (1.00m) — Recorded made ground thickness (m)

REV	DATE	DETAILS

BALFOUR BEATTY INVESTMENTS Ltd
3rd FLOOR, FOCUS POINT
21 CALEDONIAN ROAD
LONDON, N1 9GB

PROJECT TITLE

CAUSEWAYSIDE EDINBURGH

DRAWING TITLE

RECORDED MADE GROUND THICKNESS

DRAWN BY	CHK'D BY	APP'D BY	DATE	SCALES
TR	LR	LR	21.07.23	1:250 @ A3

PROJECT No.	DRAWING No.	REVISION
P23/075	P23/075/SI/R/F/07	

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A3 border: 400 X 273

4.2 Rockhead

4.2.1 Rockhead was definitively encountered in each of the rotary follow-on boreholes at varying depths of between 15.50 mbgl and 16.00 mbgl. Rockhead was described as; *'very weak, locally extremely weak light grey thin to thickly bedded micaceous MUDSTONE, recovered predominantly non-intact and moderately weathered throughout with locally heavily weathered sections'*. *'very weak to weak reddish brown thinly laminated MUDSTONE, heavily weathered and recovered predominantly non-intact'*; *'moderately strong whitish grey thickly bedded crystalline LIMESTONE, lightly to locally moderately weathered'*; *'moderately strong bluish grey thickly bedded crystalline LIMESTONE, lightly weathered'*; or *'hard white speckled orange SANDSTONE, fresh to locally lightly weathered'*. The conjectured depth to rockhead is indicated on Drawing No. P23/075/SI/R/F/08, based on the definitive depths encountered in the rotary boreholes.

4.3 Obstructions

4.3.1 A hard obstruction was encountered within BH01, BH01A, BH01B, BH02, BH02A, BH03, BH04A, BH04B, BH05 and BH06 at varying depths of between 0.30 mbgl and 4.00 mbgl. A concrete obstruction was also encountered within BH04 at 0.65 mbgl.

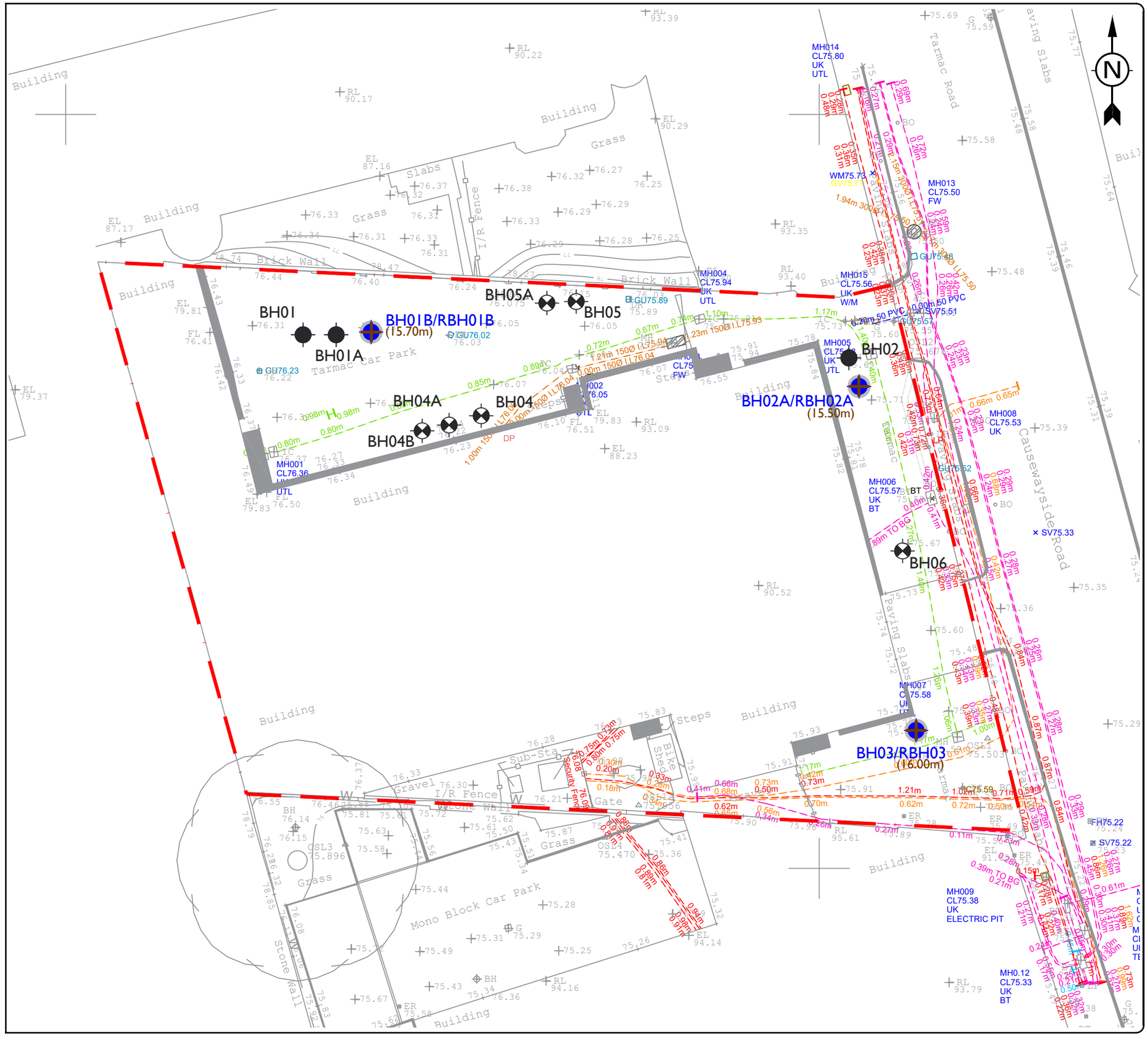
4.3.2 The obstructions encountered within BH01/BH01A/BH01B, BH02/BH02A and BH03 are considered likely to be representative of boulder obstructions, while the obstructions encountered within BH04B, BH05 and BH06 could potentially be representative of concrete slab obstructions. Note that given the previous development of the site, the possibility of encountered further obstructions at shallow depths cannot be discounted.

4.4 Groundwater

4.4.1 A groundwater strike (of an unspecified flow rate) was encountered within BH01, BH01A and BH01B at 2.00 mbgl. The remaining terrier and cable-percussive boreholes were recorded as dry during formation.

4.4.2 A groundwater strike was also recorded within the rock strata in rotary follow-on borehole BH03/RBH03 at 16.40 mbgl. No groundwater strikes were recorded within the rock strata during the formation of BH01B/RBH01B or BH02A/RBH02A.

4.4.3 Monitoring wells were installed into 3 No. of the cable-percussive boreholes (BH01B/RBH03B, BH02A/RBH02A and BH03/RBH03) to provide a more accurate assessment of the groundwater behaviour within the superficial soils The boreholes have been monitored on 5 No. occasions and a copy of the results are included I Appendix 7 for reference and summarized in the table below:



NOTES

- Approximate site boundary
- BH01 to BH03 — Cable-percussive boreholes sunk by Phoenix Drilling Ltd, on behalf of Mason Evans (April 2023)
  - Rotary follow-on-element, also carried out by Phoenix Drilling Ltd undertaken in boreholes BH01B, BH02A and BH03 (referenced as RBH01B, RBH02A and RBH03)
- BH04 to BH06 — Terrier boreholes sunk by Phoenix Drilling Ltd, on behalf of Mason Evans (April 2023)
- Gas/groundwater monitoring well installation
- (1.00m) — Recorded depth to definite rockhead (m)

REV	DATE	DETAILS

BALFOUR BEATTY INVESTMENTS Ltd
3rd FLOOR, FOCUS POINT
21 CALEDONIAN ROAD
LONDON, N1 9GB

PROJECT TITLE

CAUSEWAYSIDE EDINBURGH

DRAWING TITLE

RECORDED DEPTH TO DEFINITE ROCKHEAD

DRAWN BY	CHK'D BY	APP'D BY	DATE	SCALES
TR	LR	LR	21.07.23	1:250 @ A3

PROJECT No.	DRAWING No.	REVISION
P23/075	P23/075/SI/R/F/08	

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TABLE 10 - Summary of Groundwater Monitoring Results within the Superficial Soils

Location	BH01B/RBH01B	BH02A/RBH02A	BH03/RBH03*
Minimum Depth (mbgl)	1.00	1.00	1.35
*Minimum Level (mAOD)***	75.27	74.69	74.43
Maximum Depth (mbgl)	1.55	1.55	1.35
*Maximum Level (mAOD)***	74.72	74.14	74.43

** Note that the groundwater sampling bailer got stuck down to the borehole after the first round of monitoring in BH03/RBH03. As such, this borehole was only able to be monitored on 1 No occasion.*

*** Surface levels were extrapolated from the existing topographic survey (supplied to us by the Client's Engineer)*

4.4.4 The water depths results suggest that there is a shallow groundwater table within the made ground deposits beneath the site at depths of between 1.00 mbgl (75.27 mAOD) and 1.55 mbgl (74.72 mAOD), with the underlying cohesive glacial till deposits actively as a relatively impermeable barrier to downward migration. We have therefore classified this shallow groundwater beneath the site as a water body, as defined by SEPA Document WAT-PS-10-01 (and supporting guidance WAT-SG-53), as it could potentially sustain 10 m^3 per day extraction/supply long term. Note that SEPA indicate there to be no classified groundwater body within the superficial soils beneath the site.

4.4.5 Monitoring wells were not installed into the rock strata during our investigation and therefore the groundwater behaviour within the rock strata (i.e. bedrock aquifer) could not be determined. However, the Environment Report (Appendix 2) indicates the proposed development site to be underlain by a *'minor or moderately permeable aquifer – fractured or potentially fractured rocks which do not have a high primary permeability or other formations of variable permeability'* and therefore it is considered unlikely that any shallow groundwater (with the superficial soils) would interact with the any groundwater encountered within the bedrock aquifer. The proposed development site is also underlain by at least 12.00 m of stiff or stronger cohesive glacial till, which would act as relatively impermeable barrier (i.e. an aquitard) to the downward migration of groundwater within the superficial soils to the deeper groundwater. Consequently, the risk to the bedrock aquifer from the proposed development site was considered to be low. Note that SEPA indicate the deeper bedrock aquifer beneath the site to be named 'Edinburgh' (ID: 150591) and to have an overall status of *'Good'*, dated 2020.

4.5 Visual/Olfactory Evidence of Contamination

4.5.1 No visual or olfactory evidence of contamination was observed during our site investigation works, which is consistent with the lack of a significantly contaminative former site usage (i.e. a school and then more recently as the Scottish Law Commission).

5.0 CONTAMINATION RISK ASSESSMENT

5.1 Human Health and Groundwater Risk Assessment Screening Criteria

5.1.1 Consideration of analytical results against applicable, conservative risk-based screening criteria has been used to provide an assessment of risk. A tiered risk-based approach comprises:

- Preliminary Risk Assessment (e.g. establishing potential pollutant linkages);
- Generic Quantitative Risk Assessment (GQRA) (e.g. the comparison of contaminant concentrations against Soil Guideline Values (SGV) or other Generic Assessment Criteria (GAC)); and
- Detailed Quantitative Risk Assessment (DQRA) (e.g. the comparison of contaminant concentrations against site specific assessment criteria).

5.1.2 A GQRA has been carried out as part of this assessment. While any remedial recommendation may be refined by completing a DQRA, in this particular site assessment, the additional cost and time constraints associated with a DQRA are not considered appropriate or necessary. Soil chemical analysis data has been assessed in terms of risks to human health and vegetation while leachate data has been assessed in terms of risks to the water environment. The GACs utilised are the published Suitable 4 Use Levels (S4ULs) derived by LQM/CIEH, based on the exposure parameters, outlined in the DEFRA publication SP1010 (Category 4 Screening Levels (C4SLs) (March 2014). The S4ULs are derived in accordance with current UK legislation, and national policy using the most recent version of the CLEA software (v1.06). Normally the CLEA software utilises the default exposure pathways and land use assumptions outlined in SR3 (Environment Agency 2009b). In order to implement the revised exposure and land use assumptions introduced by DEFRA publication SP1010, a number of modifications were made to the land use and receptor databases of the CLEA model including the introduction of two additional land use scenarios: Public Open Space 'park' and Public Open Space, near residential housing. These changes are summarised in both DEFRA publication SP1010 (2014), and LQM/CIEH publication S4UL3203 (2015).

5.1.3 The derived S4ULs are based on the concept of minimal tolerable risk as described in SR2 (Environment Agency 2009a) which underpins all previous EA SGVs and other GACs. Please note that S4ULs do not incorporate any toxicological parameter changes to the CLEA base model, however recent toxicological data has been incorporated into the contaminant databases. Furthermore, S4UL GACs are considered to be equivalent to the previously published Environment agency SGVs, and previous iterations of LQM/CIEH GACs and as such are suitable for use in generic quantitative risk assessments under both planning and Part IIa regimes.

5.1.4 In this case, we have utilised S4UL values appropriate to a **residential with plant-uptake end-use,** to be conservative given the development proposals for student accommodation.

5.2 Statistical Analysis of Data

5.2.1 Where appropriate, chemical data for soils can be considered statistically in general accordance with the guidelines given in the Chartered Institute of Environmental Health Publication *Guidance on comparing Soil Contamination Data with a Critical Concentration* (May 2008). In this site assessment, statistical analysis had not been undertaken as the data set was considered to be relatively small and therefore detailed statistical analysis was deemed unnecessary.

Sample Depths

5.2.2 At the generic assessment stage, it should be assumed that all pathways contained within the generic model applied will be active. In reality, unless a contaminant is volatile (e.g. organic), exposure by direct contact will likely be mitigated by the depth of the contaminant or available surface cover. CLEA-SR3 states that *'Whether or not soil contamination at greater depth or beneath hardstanding poses a risk to health depends on the importance of the contact pathways (primary injection and dermal contact and the likelihood that such soils may be brought to the surface through activities such as gardening and building works'*. Generally, direct contact with contaminants at greater than 600 mm depth is considered unlikely to occur while direct contact with contaminants under hardstanding is highly unlikely to occur unless the ground is to be disturbed through removal of surfacing or earthworks.

5.3 Ground Gas Assessment

5.3.1 The potential presence of carbon dioxide and methane at the target site have been appraised in compliance with *'BS Standard 8576:2013.'* This document details site investigations methodologies and risk assessment procedures for assessing the results from such investigations. The risk assessment procedures are primarily based on those detailed by Wilson and Card (1999).

5.4 Building Materials Assessment

5.4.1 BRE Special Digest 1 'Concrete in Aggressive Ground' (3rd Edition, 2005) has been used to determine an appropriate concrete class for the development.

6.0 HUMAN HEALTH RISK ASSESSMENT

6.1 Contaminants In Soils

6.1.1 The results of analysis for a range of contaminants have been compared directly to their respective Generic Assessment Criteria. A summary table of all chemical results is included in Appendix 8. Made ground deposits of up to 2.50 m in thickness were encountered across the site. Therefore, this study was intended to confirm the nature of the soils encountered, along with the underlying natural soils, which were also identified.

TABLE 11– Exceedance of Guideline Levels (Residential with Plant Uptake)*

Contaminant	Effect	Measured Concentrations in Excess of SGV/GSV/SSTL (mg/kg)		Measured Exceedance Concentrations (mg/kg)		SGV/GSV/ SSV (mg/kg)	Source
		Natural	Made Ground	Natural	Made Ground		
Metals							
Arsenic	Toxic	0 out of 3	0 out of 8	-	-	38	LQM/CIEH S4ULs (2015)
Mercury (Inorganic)	Toxic	0 out of 3	0 out of 8	-	-	40	LQM/CIEH S4ULs (2015)
Boron (water soluble)	Toxic	0 out of 3	0 out of 8	-	-	290	LQM/CIEH S4ULs (2015)
Chromium III	Toxic	0 out of 3	0 out of 8	-	-	910	LQM/CIEH S4ULs (2015)
Chromium VI	Toxic	0 out of 3	0 out of 8	-	-	6.0	LQM/CIEH S4ULs (2015)
Lead***	Toxic	0 out of 3	0 out of 8	-	-	200	C4SL (DEFRA SP1010) (2014))
Cadmium	Toxic	0 out of 3	0 out of 8	-	-	11	LQM/CIEH S4ULs (2015)
Selenium	Toxic	0 out of 3	0 out of 8	-	-	250	LQM/CIEH S4ULs (2015)
Nickel	Toxic	0 out of 3	0 out of 8	-	-	130	LQM/CIEH S4ULs (2015)
Nickel**	Phytotoxic	0 out of 3	0 out of 8	-	-	110	BS:3882 (2015)
Copper	Toxic	0 out of 3	0 out of 8	-	-	2,400	LQM/CIEH S4ULs (2015)
Copper**	Phytotoxic	0 out of 3	0 out of 8	-	-	200	BS:3882 (2015)
Zinc	Toxic	0 out of 3	0 out of 8	-	-	3,700	LQM/CIEH S4ULs (2015)
Zinc**	Phytotoxic	0 out of 3	0 out of 8	-	-	300	BS:3882 (2015)
Inorganics							
Total Sulphate	Phytotoxic	0 out of 3	0 out of 8	-	-	10,000	ICRCL/SAC
Total Cyanide	Toxic	0 out of 3	0 out of 8	-	-	34	ATRISK (2015)
Phenols							
Phenol	Toxic	0 out of 3	0 out of 8	-	-	120	LQM/CIEH S4ULs (2015)
Total Petroleum Hydrocarbons (TPHs)							
Aliphatic C5-C6	Toxic	0 out of 3	0 out of 8	-	-	42	LQM/CIEH S4ULs (2015)
Aliphatic C6-C8	Toxic	0 out of 3	0 out of 8	-	-	100	LQM/CIEH S4ULs (2015)
Aliphatic C8-C10	Toxic	0 out of 3	0 out of 8	-	-	27	LQM/CIEH S4ULs (2015)
Aliphatic C10-C12	Toxic	0 out of 3	0 out of 8	-	-	130	LQM/CIEH S4ULs (2015)
Aliphatic C12-C16	Toxic	0 out of 3	0 out of 8	-	-	1,100	LQM/CIEH S4ULs (2015)
Aliphatic C16-C21	Toxic	0 out of 3	0 out of 8	-	-	65,000	LQM/CIEH S4ULs (2015)
Aliphatic C16-C35	Toxic	0 out of 3	0 out of 8	-	-	65,000	LQM/CIEH S4ULs (2015)
Aliphatic C21-C35	Toxic	0 out of 3	0 out of 8	-	-	65,000	LQM/CIEH S4ULs (2015)
Total Aliphatics C5-C35	Toxic	0 out of 3	0 out of 8	N/A	N/A	N/A	N/A
AromaticC5-C7	Toxic	0 out of 3	0 out of 8	-	-	70	LQM/CIEH S4ULs (2015)
Aromatic C7-C8	Toxic	0 out of 3	0 out of 8	-	-	130	LQM/CIEH S4ULs (2015)
Aromatic C8-C10	Toxic	0 out of 3	0 out of 8	-	-	34	LQM/CIEH S4ULs (2015)
Aromatic C10-C12	Toxic	0 out of 3	0 out of 8	-	-	74	LQM/CIEH S4ULs (2015)
Aromatic C12-C16	Toxic	0 out of 3	0 out of 8	-	-	140	LQM/CIEH S4ULs (2015)
Aromatic C16-C21	Toxic	0 out of 3	0 out of 8	-	-	260	LQM/CIEH S4ULs (2015)
Aromatic C21-C35	Toxic	0 out of 3	0 out of 8	-	-	1,100	LQM/CIEH S4ULs (2015)
Total Aromatics C5-C35	Toxic	0 out of 3	0 out of 8	N/A	N/A	N/A	N/A
Total Aliphatics and Aromatics C5-C35	Toxic	0 out of 3	0 out of 8	N/A	N/A	N/A	N/A

*Guidelines based on an SOM of 1.0%. Average SOM calculated from the soil samples tested to be 1.836%.
** Based on a pH value of >7.0. The average pH calculated from the soil samples tested is 8.9%
*** In the absence of a formalized guideline value for lead, we have been conservative and utilised the provision C4SL guideline value as outlined in the DEFRA SP1010 document.
Values below guideline value/laboratory limit of detection (LoD), unless individually listed.

TABLE 11 *CONT*– Exceedance of Guideline Levels (Residential with Plant Uptake)*

Contaminant	Effect	Measured Concentrations in Excess of SGV/GSV/SSTL (mg/kg)		Measured Exceedance Concentrations (mg/kg)		SGV/GSV/ SSV (mg/kg)	Source
		Natural	Made Ground	Natural	Made Ground		
Polycyclic Aromatic Hydrocarbons (PAHs)							
Acenaphthene	Toxic	0 out of 3	0 out of 8	-	-	210	LQM/CIEH S4ULs (2015)
Acenaphthylene	Toxic	0 out of 3	0 out of 8	-	-	170	LQM/CIEH S4ULs (2015)
Anthracene	Toxic	0 out of 3	0 out of 8	-	-	2,400	LQM/CIEH S4ULs (2015)
Benzo(a)anthracene	Toxic	0 out of 3	0 out of 8	-	-	7.2	LQM/CIEH S4ULs (2015)
Benzo(a)pyrene	Toxic	0 out of 3	0 out of 8	-	-	2.2	LQM/CIEH S4ULs (2015)
Benzo(b)fluoranthene	Toxic	0 out of 3	0 out of 8	-	-	2.6	LQM/CIEH S4ULs (2015)
Benzo(g,h,i)perylene	Toxic	0 out of 3	0 out of 8	-	-	320	LQM/CIEH S4ULs (2015)
Benzo(k)fluoranthene	Toxic	0 out of 3	0 out of 8	-	-	77	LQM/CIEH S4ULs (2015)
Chrysene	Toxic	0 out of 3	0 out of 8	-	-	15	LQM/CIEH S4ULs (2015)
Dibenzo(a,h)anthracene	Toxic	0 out of 3	0 out of 8	-	-	0.24	LQM/CIEH S4ULs (2015)
Fluoranthene	Toxic	0 out of 3	0 out of 8	-	-	280	LQM/CIEH S4ULs (2015)
Fluorene	Toxic	0 out of 3	0 out of 8	-	-	170	LQM/CIEH S4ULs (2015)
Indeno(1,2,3-c.d) Pyrene	Toxic	0 out of 3	0 out of 8	-	-	27	LQM/CIEH S4ULs (2015)
Naphthalene	Toxic	0 out of 3	0 out of 8	-	-	2.3	LQM/CIEH S4ULs (2015)
Phenanthrene	Toxic	0 out of 3	0 out of 8	-	-	95	LQM/CIEH S4ULs (2015)
Pyrene	Toxic	0 out of 3	0 out of 8	-	-	620	LQM/CIEH S4ULs (2015)
Total PAH's	Toxic	0 out of 3	0 out of 8	N/A	N/A	N/A	N/A
Other							
Asbestos	Toxic	0 out of 3	3 out of 8	-	Chrysotile (BH02) Amosite (BH03 at 0.50 m and 1.00 m depth)	Detection	HSE
Volatile Organic Compounds (VOCs)#							
Vinyl Chloride	Toxic	-	1 out of 1	-	<0.01	0.00064	LQM/CIEH S4ULs (2015)
1,2-dichlorethane	Toxic	-	1 out of 1	-	<0.01	0.071	LQM/CIEH S4ULs (2015)
Semi-Volatile Organic Compounds (SVOCs)#							
Bis(2-ethylhexyl)phalate	Toxic	0	1 out of 1		2.2	0.1	LoD

*Guidelines based on an SOM of 1.0%. Average SOM calculated from the soil samples tested to be 1.836%.
** Based on a pH value of >7.0. The average pH calculated from the soil samples tested is 8.9%
*** In the absence of a formalized guideline value for lead, we have been conservative and utilised the provision C4SL guideline value as outlined in the DEFRA SP1010 document.
Values below guideline value/laboratory limit of detection (LoD), unless individually listed.

6.1.2 The GRQA has identified no contaminants at concentrations about the recommended guideline values within the samples tested.

6.1.3 However, asbestos described as; *'chrysotile present as fibre bundles'*; *'amosite present in visible insulating board debris and as fibre bundles'* and *'amosite present in microscopic insulating board debris and present as fibres bundles'*, was encountered within the made ground samples obtained from BH02 (at 0.50 mbgl) and BH03 (at 0.50 mbgl and 1.00 mbgl). Quantification testing was undertaken on each of the samples with a positive identification for asbestos and the results of the phase contrast optical microscopy indicated the recorded level of asbestos to be between 0.002% vol and 0.020% vol.

6.1.4 The asbestos encountered within BH03 (0.50 mbgll) was identified to contain visible asbestos containing materials (ACMs), described as *'visible insulation board debris'*, and therefore these asbestos impacted soils must be removed from site prior to development by a licensed contractor and would need to be disposed of at a *'hazardous'* waste facility.

6.1.5 Conversely, the concentration of asbestos encountered within BH02 (at 0.50 mbgl) and BH03 (at 1.00 mbgl) were 0.002% vol. and 0.020% vol, respectively (i.e below the hazardous level of 0.1% vol. in the event that the asbestos containing soils were removed from site to a suitably licensed landfill). As such, the asbestos impacted soils, identified around BH02 and BH03 (at 1.00 mbgl), can be retained onsite beneath a suitable environmental capping layer or capped beneath a less sensitive area onsite (i.e. beneath areas of hardstanding or solumns).

6.1.6 Note that the asbestos identified in BH03 at 1.00 mbgl was described as containing *'insulation board debris'*. However, this was described as *'microscopic'* and therefore would not classify as an ACM, meaning that the asbestos impacted soils around BH03 (at 1.00 mbgl) can remain onsite beneath a suitable environmental capping layer, or be disposed offsite at a suitable licensed waste facility size.

6.1.7 Lastly, we would highlight that given the presence of made ground soils onsite and the fact that half of the site was occupied by building cover during our investigations, the possibility of encountering further asbestos during a future past-demolition phase of works cannot be discounted.

6.2 Volatile Organic Compounds (VOCs) and Semi-Volatile Organic Compounds (SVOCs) Testing

6.2.1 1 No. soil sample (BH03 at 1.00 m depth) was obtained from within the car park area at the front of the site, to confirm, or otherwise, the potential presence of spillages/leakages of fuels/oils associated with onsite vehicles. However, please note that this testing was only carried out as a precautionary measure as the car park only every has a few cars in it at any one time and there was no visual evidence or olfactory evidence of contamination within the site area during our site walkover survey (included in Appendix 1) or during our site investigation works.

6.2.2 Vinyl Chloride and 1,2-dichloroethane (VOCs) were highlighted as exceedances in the sample tested as the laboratory limit of detection ((LoD) 0.01 mg/kg) is greater than the current assessment criteria (0.00087 mg/kg and 0.0071 mg/kg, respectively). However, both vinyl chloride and 1,2-dichloroethane were recorded to be lower the LoD in the sample tested. Therefore, for the purpose of our risk assessment, we have assumed that non-detects are considered to pose a low risk to human health. Consequently, VOCs were not considered to pose a significant risk to the proposed development.

6.2.3 In addition, bis(2-ethylhexylphalate (an SVOC)) was recorded to be above the laboratory limit of detection (LoD (0.1 mg/kg)) in the sample tested, where the concentration was at 2.2 mg/kg. However, when we assess the concentration of bis(2-ethylhexylphalate recorded against the guideline values presented within the CL:AIRE *'Soil Generic Assessment Criteria for Human Health'* guidance document, dated January 2010 (280 mg/kg (using 1% SOM residential with plant-uptake guideline value)), the value recorded within our sample is orders of magnitude lower than the GAC guideline value. Note that we are that the toxicity values within the GAC guidance have been updated, however in the absence of formalised updated guideline value, the 2010 guideline values were considered appropriate to use, as opposed to just comparing the results against the LoD.

6.2.4 Therefore, SVOCs were also not considered to pose a significant risk to human health.

7.0 WATER ENVIRONMENT RISK ASSESSMENT

7.1 Water Environment Vulnerability

7.1.1 As previously discussed, the nearest natural classified surface water body to the proposed development site was considered to be the Pow Burn/Braid Burn, located approximately 1.00 km to the south-east of the proposed development site (at its closest point), flowing generally towards the north-east, towards the coast. Therefore, we have taken the Pow Burn/Braid Burn as the main Water Environment receptor with regard to the subject site.

7.1.2 Based on the results of the groundwater depth monitoring undertaken (5 No. rounds), it was also considered likely that a shallow persistent groundwater table exists beneath the site, within the made ground deposits at depths of between 1.00 mbgl (75.27 mAOD) and 1.55 mbgl (74.72 mAOD). The Scottish Environment Protection Agency (SEPA) provides guidance in document WAT-PS-10-01 'Assigning Groundwater Assessment Criteria for Pollutant Inputs' (August 2014) for assessing contamination risks to the groundwater and the water environment and it was considered possible that the groundwater within the superficial soils beneath the site could meet the minimum criteria to be classified as a water body i.e. an abstraction could achieve a sustained 10 m^3 per day.

7.1.3 As the Pow Burn/Braid Burn is located approximately 1.00 km to the south-east of the proposed development site (at its closest point) and a shallow groundwater table is present beneath the site, it was considered that both of these be regarded as sensitive receptors. Therefore, a groundwater assessment has been undertaken as well as a surface water assessment.

7.2 Surface Water and Groundwater Assessments

7.2.1 Following SEPA Position Statement WAT-PS-10-01, *'Assigning Groundwater Assessment Criteria for Pollutant Inputs'* (August 2014), the following assessment should be carried out for potential pollutant linkages to the water environment:

1) Assess which receptors (including surface / coastal waters, wetlands, potable water extractions, and future drinking water potential) may be affected by contamination sources.
2) For potential pollutant linkages, assess contaminant concentrations against relevant screening values at the recommended assessment point, taking into consideration mixing and upstream/upgradient concentrations, where appropriate.
3) Evaluate whether remedial measures would be either disproportionately costly, a risk to other receptors, or cause deterioration of the natural environment.

7.2.2 Leachate analysis was carried out on a selection of soil samples obtained from across the proposed development site to determine the risks posed by the on-site conditions (if any) to the Water Environment. Leachate testing of TPHs and PAHs was not undertaken as part of this assessment, as by their very nature, TPHs and PAHs are very leachable and would present themselves in samples of the groundwater.

7.2.3 Groundwater samples were obtained from BH01B/RBH01B on 3 No separate occasions and from BH02A/RBH02A on 2 No. separate occasions, with each of the samples obtained being tested for a comprehensive suite of chemical analysis, including TPHs and PAHs. Note that no groundwater samples could be obtained from BH03/RBH03 as the sampling bailer got stuck down the standpipe installation during the first round of monitoring.

7.2.4 In addition, note that on the third round of monitoring (12.06.23), a strong sulphurous odour was noted within borehole BH01B/RBH01B. Therefore, a grab sample was taken from this bore (i.e. prior to any purging being carried out), with the sample being tested for chemical analysis. A copy of the chemical analysis results, including tabulated summary tables, for both the leachate analysis and groundwater testing carried out are attached in Appendix 8 for reference.

7.2.5 The leachate and groundwater results have been compared to the appropriate guideline values (Tables 12 and 17). Where surface water is considered a primary risk, Environmental Quality Standards (EQSs) are used as obtained from SEPA document WAT-SG-53 (April 2020). In the absence of any SEPA published EQS we have reverted to the laboratory Limits of Detection (LoD), as recommended in SEPA position Statement WAT-PS-10-01.

7.2.6 Where groundwater is considered a primary risk, Minimum Reporting Values (MRV's) and Resource Protection Values (RPV's) have been utilised, obtained from SEPA position statement WAT-PS-10-01 (August 2014). In the absence of any SEPA published MRV's and RPV's we have reverted to the laboratory Limits of Detection (LoD). Note that we have utilised the guideline values from the World Health Organisation's (WHO) Guidelines for Drinking Water Quality (4th Edition, 2011) for both copper and zinc in the absence of any other formalised guideline values. However, we are aware that SEPA do not approve of the guideline values presented within the World Health Organisation's (WHO) Guidelines for Drinking Water Quality (4th Edition, 2011), CL:AIRE Petroleum Hydrocarbons in Groundwater (2017) or WHO: Petroleum Products in Drinking Water (2005) for groundwater risk assessments and cognisance of this has been noted within our assessment.

Surface Water Assessment

7.2.7 The nearest surface water body, the Pow Burn/Braid Burn, is considered a sensitive receptor and therefore the leachate results have been compared to the relevant assessment criteria as detailed above and as shown in Table 12 below.

TABLE 12: Analysis of Leachate Samples Compared with Marie Environmental Quality Standards

Potential Contaminant	Environmental Quality Standard (µg/l)	Laboratory Limits of Detection (µg/l)	No of Samples Above Assessment Levels	Range of Concentrations which Exceeded Relevant Guidelines ug/l
Metals				
Arsenic	50	-	0 out of 7	-
Cadmium*	<0.08	-	0 out of 7	-
Chromium III	4.7	-	1 out of 7	17
Copper	1.0	-	7 out of 7	2.1 – 6.0
Lead	1.2	-	1 out of 7	3.7
Mercury (inorganic (low level))	0.07	-	0 out of 7	-
Nickel (toxic)	4.0	-	0 out of 7	-
Selenium**	-	0.25	3 out of 7	0.29 – 0.68
Zinc (toxic)	10.9	-	0 out of 7	-
Inorganics				
Sulphate as SO4	400 (mg/l)	-	0 out of 7	-
Phenol	7.7	-	0 out of 7	-

* Based on an average water hardness, from all samples tested, of 24.075 mg/l.

** Most conservative laboratory Limit of Detection (LoD) value used as no reporting value was available in the WAT-SG-53 (April 2020) document.

7.2.8 Against current guideline values, the leachate testing identified exceedances of the contaminants; chromium III, copper, lead and selenium.

TABLE 13: Analysis of Groundwater Samples Compared with Environmental Quality Standards

Potential Contaminant	Environmental Quality Standard (µg/l)	Laboratory Limit of Detection (µg/l)**	No of Samples Above Assessment Levels	Range of Concentrations which Exceeded Relevant Guidelines ug/l
Metals				
Arsenic	50	-	0 out of 5	-
Boron (water soluble)	2000	-	0 out of 5	-
Cadmium*	0.15	-	0 out of 5	-
Chromium III	4.7	-	0 out of 5	-
Copper	1.0	-	3 out of 5	1.1 – 4.6
Lead	1.2	-	0 out of 5	-
Mercury (inorganic)	0.07	-	0 out of 5	-
Nickel	4.0	-	0 out of 5	-
Selenium**	-	0.25	4 out of 5	0.39 – 2.0
Zinc	10.	-	4 out of 5	14 - 99
Inorganics				
Sulphate as SO4	400 (mg/l)	-	0 out of 5	-
Total Cyanide	1.0	-	1 out of 5	1.7
Total Petroleum Hydrocarbons (TPHs)				
Aliphatic C5-C6**	-	0.1	0 out of 5	-
Aliphatic C6-C8**	-	0.1	0 out of 5	-
Aliphatic C8-C10**	-	0.1	0 out of 5	-
Aliphatic C10-C12**	-	1.0	1 out of 5	5.7
Aliphatic C12-C16**	-	1.0	2 out of 5	4.9 - 27
Aliphatic C16-C21**	-	1.0	2 out of 5	33 – 42
Aliphatic C21-C35**	-	1.0	2 out of 5	230 - 400
Total Aliphatic Hydrocarbons (C5-C35)	N/A	N/A	0 out of 5	N/A
Aromatic C5-C7**	-	0.1	0 out of 5	-
Aromatic C7-C8**	-	0.1	0 out of 5	-
Aromatic C8-C10**	-	0.1	0 out of 5	-
Aromatic C10-C12**	-	1.0	0 out of 5	-
Aromatic C12-C16**	-	1.0	0 out of 5	-
Aromatic C16-C21**	-	1.0	0 out of 5	-
Aromatic C21-C35**	-	1.0	1 out of 5	59
Total Aromatic Hydrocarbons (C5-C35)	N/A	N/A	0 out of 5	N/A
Total Petroleum Hydrocarbons (Aliphatics and Aromatics (C5-C35)	N/A	N/A	0 out of 5	N/A

* Based on an average water hardness, from all samples tested of 198 mg/l

** Most conservative laboratory Limit of Detection (LoD) value used as no reporting value was available in the WAT-SG-53 (April 2020) document.

TABLE 13 *CONT*: Analysis of Groundwater Samples Compared with Environmental Quality Standards

Potential Contaminant	Environmental Quality Standard (µg/l)	Laboratory Limit of Detection (µg/l)**	No of Samples Above Assessment Levels	Range of Concentrations which Exceeded Relevant Guidelines ug/l
Polycyclic Aromatic Hydrocarbons (PAHs)				
Naphthalene	2.0	-	0 out of 5	-
Acenaphthylene**	-	0.01/0.10	0 out of 5	-
Acenaphthene**	-	0.01/0.10	0 out of 5	-
Fluorene**	-	0.01/0.10	1 out of 5	0.02
Phenanthrene**	-	0.01/0.10	5 out of 5	0.03 – 0.48
Anthracene	0.1	-	0 out of 5	-
Fluoranthene	0.0063	0.01/0.10***	5 out of 5	0.01 – 0.30
Pyrene**	-	0.01/0.10	3 out of 5	0.09 – 0.29
Benzo[a]anthracene**	-	0.01/0.10	1 out of 5	0.14
Chrysene**	-	0.01/0.10	1 out of 5	0.11
Benzo(b)fluoranthene	0.00017	0.01/0.10***	5 out of 5	<0.01 – 0.20
Benzo(k)fluoranthene	0.00017	0.01/0.10***	5 out of 5	<0.01 – 0.02
Benzo(a)pyrene	0.00017	0.01/0.10***	5 out of 5	<0.01 – 0.04
Indeno(1,2,3-c,d)Pyrene	0.00017	0.01/0.10***	5 out of 5	<0.01 – 0.02
Dibenzo(a,h)anthracene**	-	0.01	0 out of 5	-
Benzo(g,h,i)perylene	0.00017	0.01/0.10***	5 out of 5	<0.01 – 0.12
Total PAHs	N/A	N/A	0 out of 5	N/A
Phenols				
Phenols – Monohydric	7.7	-	0 out of 5	-

* Based on an average water hardness, from all samples tested, of 198 mg/l

** Most conservative laboratory Limit of Detection (LoD) value used as no reporting value was available in the WAT-SG-53 (April 2020) document.

*** Laboratory Limit of Detection (LoD) is greater than the EQS guideline value.

7.2.9 Against current guideline values, the groundwater testing identified exceedances of the contaminants; copper, selenium, zinc, total cyanide, aliphatic C10-C12, aliphatic C12-C16, aliphatic C16-C21, aliphatic C21-C35, aromatic C5-C35, fluorene, phenanthrene, fluoranthene, pyrene, benzo(a)anthracene, pyrene, benzo(b)fluoranthene, benzo(k)fluoranthene, benzo(a)pyrene, indeno(1,2,3,c-d)pyrene and benzo(g,h,i)perylene.

7.2.10 However, it should be noted that; fluoranthene (1 No. out of 5 No. samples tested), benzo(b)fluoranthene (3 No. out of 5 No. samples tested), benzo(k)fluoranthene (4 No. out of 5 No. samples tested) benzo(a)pyrene (4 No. out of 5 No. samples tested), indeno(1,2,3,c-d)pyrene (4 No. out of 5 No. samples tested) and benzo(g,h,i)perylene (3 No. out of 5 No. samples tested) were highlighted as exceedances as the laboratory limit of detection (LoD) was greater than the current assessment criteria. Therefore, for the purpose of our risk assessment we have assumed that non-detects are considered to pose a low risk to the surface Water Environment. In addition, it is also worthy to note that as the EQS values for these PAHs are very low, laboratories are currently unable to achieve a limit of detection below the current EQS via a UKAS accredited test (LGC/R/2015/374).

Interpretation of Results

7.2.11 Although potentially leachable, chromium and lead were not recorded at elevated concentrations within the groundwater samples. This suggests that although these contaminants are potentially mobile, they do not appear to be impacting the groundwater and that there is no linkage between contamination in the soils and offsite water receptors. As such, chromium and lead are not considered to pose a significant risk to the surface Water Environment.

7.2.12 Groundwater modelling of; copper, selenium, zinc, total cyanide, aliphatic C10-C12, aliphatic C12-C16, aliphatic C16-C21, aliphatic C21-C35, aromatic C5-C35, fluorene, phenanthrene, fluoranthene, pyrene, benzo(a)anthracene, pyrene, benzo(b)fluoranthene, benzo(k)fluoranthene, benzo(a)pyrene, indeno(1,2,3-c,d)pyrene and benzo(g,h,i)perylene was undertaken to assess the potential risk of these contaminants to the surface Water Environment. We utilized RD P20 RTM (V3.2) published by the Environment Agency in our assessment and a copy of the models are included in Appendix 9. Based on average groundwater levels obtained from across the site (5 No. rounds of groundwater monitoring had been undertaken at the time of reporting), groundwater flow was indicated to be towards the east, towards the Pow Burn/Braid Burn (located approximately 1.00 km to the south-east (at its closest point), as anticipated. Therefore, we have chosen the Pow Burn/Braid Burn as the compliance point within our assessment. Furthermore, based on the results of our investigations, any groundwater flow beneath the site was anticipated to be largely through the made ground deposits, which were generally described as sand and gravel with extraneous materials or sandy gravelly clay. Therefore, to be conservative, we have utilised as gravelly sand as the main constituent of the aquifer material to represent a worst-case scenario. In addition, where a contaminant was encountered at elevated concentrated within more than one groundwater sample, we have utilised an average concentration of that contaminant from across all of the groundwater samples in our assessment due to the more widespread nature of the contamination. Where a contaminant was encountered at an elevated concentration within a single groundwater sample, that maximum concentration was utilised and the area of influence around that borehole was modelled. The model was run for each of the contaminants which were recorded to exceed EQS or LoD. The results of the modelling are shown in Table 14 below.

TABLE 14 – Results of P20 Groundwater Modelling (compliance point – Pow Burn/Braid Burn)

Contaminant	Recorded Exceedance (µg/l)	Target Value (µg/l)	Target Value Source	Final concentration (µg/l)
Copper	1.68**	1.0	EQS (AA)	0.0198
Selenium	0.746**	0.25	LoD	0.00881
Zinc	39.26	10.9	EQS (AA)	0.463
Total Cyanide	1.7*	1.0	EQS (AA)	0.00505
Aliphatic C10-C12	5.7*	1.0	LoD	0.00249
Aliphatic C12-C16	6.98**	1.0	LoD	0.0250
Aliphatic C16-C21	15.6**	1.0	LoD	0.0606
Aliphatic C21-C35	126.6**	1.0	LoD	0.550
Aromatic C21-C35	59*	1.0	LoD	0.128
Fluorene	0.02*	0.01	LoD	0.00000691
Phenanthrene	0.11**	0.01	LoD	0.000613
Fluoranthene	0.098**	0.0063	EQS (AA)	0.000448
Pyrene	0.11**	0.01	LoD	0.000613
Benzo(a)anthracene	0.14*	0.01	LoD	0.000216
Chrysene	0.11*	0.01	LoD	0.000179
Benzo(b)fluoranthene	0.038**	0.00017	EQS(AA)	0.000187
Benzo(k)fluoranthene	0.02*	0.00017	EQS (AA)	0.0000346
Benzo(a)pyrene	0.04*	0.00017	EQS (AA)	0.0000589
Indeno(1,2,3-c,d)pyrene	0.02*	0.00017	EQS (AA)	0.0000312
Benzo(g,h,i)perylene	0.058**	0.00017	EQS (AA)	0.000289

*Maximum concentration utilised as the contaminant was encountered within a singular borehole
**Average concentration from across the site utilised as the contaminant was encountered within a number of boreholes
EQS (AA) – Annual Average (SEPA document WAT-SG-53, April 2020)
LoD – Laboratory Limit of Detection

7.2.13 Contaminant half-lives have been utilised in our assessment, as outlined in CL:AIRE, 2017's guidance *'Petroleum Hydrocarbons in Groundwater: Guidance on assessing petroleum hydrocarbons using existing hydrogeological risk assessment methodologies' (p27),* dissolved oxygen concentrations of <1.0 mg/l are indicative of anaerobic conditions. Therefore, given that the dissolved oxygen content from the groundwater samples obtained from the across the site (5 No.) was recorded to range between 2.0 mg/l and 6.4 mg/l it was considered reasonable to assume that the groundwater beneath the site (within the superficial soils) is an aerobic environment, suggesting that biodegradation is occurring and factoring in this element (by way of half-lives) is applicable to the P20 modelling. Note that we are aware that by using the traditional bailer method to obtain the required groundwater samples, oxygen could potentially have been introduced into the samples. However, care was taken during sampling to try and minimize aeration of the samples during transportation to the chemical laboratory (i.e. the water was filled to the top of each sample container to eliminate any headspace within the container and each sample was collected in laboratory supplied glass and plastic bottles).

7.2.14 Modelling to a compliance point of 1000 m (i.e. the distance from the edge of the proposed development site to the Pow Burn/Braid Burn (at its closest point), the final concentrations of the contaminants; copper, selenium, zinc, total cyanide, aliphatic C10-C12, aliphatic C12-C16, aliphatic C16-C21, aliphatic C21-C35, aromatic C5-C35, fluorene, phenanthrene, fluoranthene, pyrene, benzo(a)anthracene, pyrene, benzo(k)fluoranthene, benzo(a)pyrene and indeno(1,2,3-c,d)pyrene were recorded to be lower than the target concentrations. Therefore, these contaminants are considered to pose a low risk to the nearest surface water body i.e. the Pow Burn/Braid Burn. Note that these results are not unexpected given the distance from the proposed development site to the nearest natural surface water body.

7.2.15 The model utilised above was created to represent a worst case scenario and benzo(b)fluoranthene and benzo(g,h,i)perylene were recorded to marginally exceed the EQS value at the chosen compliance point (i.e. the Powe Burn/Braid Burn). The EQS value utilised for risk assessing these PAH's was the annual average (0.00017 ug/l). However, when the maximum allowable concentration was considered (0.07 ug/l and 82 ug/l, respectively), the modelled PAH results were recorded to be below their respective values. It is also worthy of note that we have not allowed for any dilution within the receiving surface water body (i.e. the Pow Burn/Braid Burn and then ultimately the coast) and that as the EQS value for PAHs is very low, laboratories are currently unable to achieve a limit of detection below the current EQS via a UKAS accredited method (LGC/2015/374). It was therefore considered that benzo(b)fluoranthene and benzo(g,h,i)perylene pose a low risk to the surface Water Environment.

Groundwater Assessment

7.2.16 Based on the results of the groundwater monitoring undertaken (5 No. rounds), it was also considered likely that a shallow persistent groundwater table exists beneath the site, within the made ground deposits at depths of between 1.00 mbgl (75.27 mAOD) and 1.55 mbgl (74.72 mAOD). Therefore, we have also considered this water body as a sensitive receptor.

TABLE 15: Analysis of Leachate Samples Compared with RPV/MRV

Potential Contaminant	RPV/MRV (µg/l)	DWI PR14	WHO (4th Edition, 2011) (ug/l)	(MEP) Limit of Detection (µg/l)	No of Samples Above Assessment Levels	Range of Concentrations which Exceeded Relevant Guidelines (µg/l)
Metals						
Arsenic	10	-	-	-	0 out of 7	-
Cadmium	0.1	-	-	-	0 out of 7	-
Chromium	50	-	-	-	0 out of 7	-
Copper	-	-	2000	-	0 out of 7	-
Lead		10	-	-	0 out of 7	-
Mercury (inorganic)	0.01	-	-	-	3 out of 7	0.026 – 0.056-
Nickel	20	-	-	-	0 out of 7	-
Selenium	10	-	-	-	0 out of 7	-
Zinc	-	-	3000	-	0 out of 7	-
Inorganics						
Sulphate as SO4	250	-	-	-	0 out of 7	-
Phenols						
Phenol*	-	-	-	0.5	0 out of 7	-

* Most conservative Laboratory Limit of Detection (LoD) value used as no reporting value was available in the WAT-PS-10 (Aug 2010), or WHO Document (2011). Note that the WHO guideline values have only been utilised for metals.

7.2.17 Against current guideline values, the leachate testing identified exceedances of the contaminants; mercury (inorganic (low level)).

TABLE 16: Analysis of Groundwater Samples Compared with RPV/MRV

Potential Contaminant	RPV/MRV(µg/l)	DWI PR14 (ug/l)	WHO (4th Edition, 2011) (ug/l)	Laboratory Limit of Detection (µg/l)	No of Samples Above Assessment Levels	Range of Concentrations which Exceeded Relevant Guidelines ug/l
Metals						
Arsenic	10	-	-	-	0 out of 5	-
Boron	1000	-	-	-	0 out of 5	-
Cadmium	0.1	-	-	-	0 out of 5	-
Chromium III	50	-	-	-	0 out of 5	-
Copper		-	2000	-	0 out of 5	-
Lead	-	10	-	-	0 out of 5	-
Mercury (inorganic (low level))	0.01	-	-		0 out of 5	-
Nickel	20	-	-	-	0 out of 5	-
Selenium**	10	-	-	-	0 out of 5	-
Zinc			3000	-	0 out of 5	-
Inorganics						
Sulphate as SO4	250 (mg/l)	-	-	-	0 out of 5	-
Total Cyanide	50	-	-	-	0 out of 5	-
Phenols						
Phenol*	-	-	-	0.5/5.0	0 out of 5	-
Total Petroleum Hydrocarbons (TPHs)						
Aliphatic C5-C6*	-	-	-	0.1	0 out of 5	-
Aliphatic C6-C8*	-	-	-	0.1	0 out of 5	-
Aliphatic C8-C10*	-	-	-	0.1	0 out of 5	-
Aliphatic C10-C12*	-	-	-	1.0	1 out of 5	5.7
Aliphatic C12-C16*	-	-	-	1.0	2 out of 5	4.9 – 27
Aliphatic C16-C21**	-	-	-	1.0	2 out of 5	33 - 42
Aliphatic C21-C35**	-	-	-	1.0	2 out of 5	230 - 400
Aliphatic C5-C35	N/A	-	N/A	N/A	0 out of 5	N/A
Aromatic C5-C7+	1.0	-	-	-	0 out of 5	-
Aromatic C7-C8+	4.0	-	-	-	0 out of 5	-
Aromatic C8-C10++	3.0	-	-	-	0 out of 5	-
Aromatic C10-C12*	-	-	-	1.0	0 out of 5	-
Aromatic C12-C16*	-	-	-	1.0	0 out of 5	-
Aromatic C16-C21*	-	-	-	1.0	0 out of 5	-
Aromatic C21-C35*	-	-	-	1.0	1 out of 5	59
Aromatic C5-C35	N/A	-	N/A	N/A	0 out of 5	N/A

* Most conservative Laboratory Limit of Detection (LoD) value used as no reporting value was available in the WAT-PS-10 (Aug 2010) or WHO Document (2011). Note that the WHO guideline values have only been utilised for metals.
** Guideline value for Xylene utilised
*** Laboratory Limit of Detection (LoD) is greater than the current assessment criteria
+ Guideline values of 1.0 ug/l (benzene) and 4.0 ug/l (toluene) have been developed within this aromatic fraction range – the most conservative guideline value has been utilised.
++ Guideline values of 300 ug/l (ethylbenzene) and 3.0 ug/l (xylenes) has been developed within this aromatic fraction range – the most conservative guideline value shas been utilised.

TABLE 16 *CONT*: Analysis of Groundwater Samples Compared with RPV/MRV

Potential Contaminant	RPV/MRV(µg/l)	DWI PR14 (ug/l)	WHO (4th Edition, 2011) (ug/l)	Laboratory Limit of Detection (µg/l)	No of Samples Above Assessment Levels	Range of Concentrations which Exceeded Relevant Guidelines ug/l
Polycyclic Aromatic Hydrocarbons (PAHs)						
Naphthalene*	-	-	-	0.05	1 out of 5	0.28
Acenaphthylene*	-	-	-	0.01/0.10	0 out of 5	-
Acenaphthene*	-	-	-	0.01/0.10	0 out of 5	-
Fluorene*	-	-	-	0.01/0.10	1 out of 5	0.02
Phenanthrene*	-	-	-	0.01/0.10	5 out of 5	0.03 – 0.48
Anthracene*	-	-	-	0.01/0.10	0 out of 5	-
Fluoranthene	0.1	-	-	-	1 out of 5	0.3
Pyrene*	-	-	-	0.01/0.10	3 out of 5	0.09 – 0.29
Benzo[a]anthracene*	-	-	-	0.01/0.10	2 out of 5	0.05 – 0.14
Chrysene*	-	-	-	0.01/0.10	2 out of 5	0.03 – 0.11
Benzo(b)fluoranthene	0.1	-	-	-	1 out of 5	0.20
Benzo(k)fluoranthene	0.1		-	-	0 out of 5	-
Benzo(a)pyrene	0.01		-	0.01/0.10	3 out of 5	<0.10 - 0.04
Indeno(1,2,3-c,d)Pyrene	0.01		-	-	1 out of 5	0.02
Dibenzo(a,h)anthracene*	-		-	0.01/0.10	0 out of 5	-
Benzo(g,h,i)perylene	0.1		-	-	1 out of 5	0.12
Total PAHs	N/A		N/A	N/A	0 out of 5	N/A

* Most conservative Laboratory Limit of Detection (LoD) value used as no reporting value was available in the WAT-PS-10 (Aug 2010) or WHO Document (2011). Note that the WHO guideline values have only been utilised for metals.
** Guideline value for Xylene utilised
*** Laboratory Limit of Detection (LoD) is greater than the current assessment criteria
+ Guideline values of 1.0 ug/l (benzene) and 4.0 ug/l (toluene) have been developed within this aromatic fraction range – the most conservative guideline value has been utilised.
++ Guideline values of 300 ug/l (ethylbenzene) and 3.0 ug/l (xylenes) has been developed within this aromatic fraction range – the most conservative guideline value shas been utilised.

7.2.18 Against current guideline values, the groundwater testing identified exceedances of the contaminants; aliphatic C10-C12, aliphatic C12-C16, aliphatic C16-C21, aliphatic C21-C35, naphthalene, fluorene, phenanthrene, fluoranthene, pyrene, benzo(a)anthracene, chrysene, benzo(b)fluoranthene, benzo(a)pyrene, indeno(1,2,3-c.d)pyrene and benzo(g,h,i)perylene.

7.2.19 However, it should be noted that; benzo(a)pyrene (2 No. out of 5 No. samples tested) were highlighted as exceedances as the laboratory limit of detection (LoD) was greater than the current assessment criteria. Therefore, for the purpose of our risk assessment we have assumed that non-detects are considered to pose a low risk to the Groundwater Environment.

Interpretation of Results

7.2.20 Although potentially leachable, mercury (inorganic (low level)) was not recorded at elevated concentrations within the groundwater samples. This suggests that although this contaminant is potentially mobile, it does not appear to be impacting the groundwater and that there is no linkage between contamination in the soils and offsite water receptors. As such, mercury (inorganic (low level)) is not considered to pose a significant risk to the Groundwater Environment (within the superficial soils).

7.2.21 Groundwater modelling of; aliphatic C10-C12, aliphatic C12-C16, aliphatic C16-C21, aliphatic C21-C35, naphthalene, fluorene, phenanthrene, fluoranthene, pyrene, benzo(a)anthracene, chrysene, benzo(b)fluoranthene, benzo(a)pyrene (1 No. out of 5 No. samples tested), indeno(1,2,3-c.d)pyrene and benzo(g,h,i)perylene was undertaken to assess the potential risk of these contaminants to the groundwater environment. Again, we utilized RD P20 RTM (V3.2) published by the Environment Agency in our assessment and a copy of the models are included in Appendix 19. As previously discussed, groundwater flow was indicated to be towards the east, towards the Pow Burn/Braid Burn, and then ultimately the coast, as anticipated. We have also again assumed gravelly sand as the main constituent of the aquifer material to represent a worst-case scenario, given that groundwater flow beneath the site was largely through the made ground deposits, which were generally described as sand and gravel with extraneous materials or sandy gravelly clay. Furthermore, where a contaminant was encountered at elevated concentrated within more than one groundwater sample, we have utilised an average concentration of that contaminant from across all of the groundwater samples in our assessment due to the more widespread nature of the contamination. Where a contaminant was encountered at an elevated concentration within a single groundwater sample, that maximum concentration was utilised and the area of influence around that borehole was modelled. The results of the modelling are shown in Table 17 below.

TABLE 17 – Results of P20 Groundwater Modelling (compliance point – 250 m)

Contaminant	Recorded Exceedance (µg/l)	Target Value (µg/l)	Target Value Source	Final concentration (µg/l)
Aliphatic C10-C12	5.7*	1.0	LoD	0.154
Aliphatic C12-C16	6.98**	1.0	LoD	0.551
Aliphatic C16-C21	15.6**	1.0	LoD	1.26
Aliphatic C21-C35	126.6**	1.0	LoD	2.53
Aromatic C21-C35	59*	1.0	LoD	10.5
Naphthalene	0.28*	0.05	LoD	0.0104
Fluorene	0.02*	0.01	LoD	0.000362
Phenanthrene	0.174**	0.01	LoD	0.0136
Fluoranthene	0.3*	0.1	WAT-PS-10-01	0.00808
Pyrene	0.11**	0.01	LoD	0.00975
Benzo(a)anthracene	0.044&&	0.01	LoD	0.00389
Chrysene	0.052**	0.01	LoD	0.0467
Benz0(b)fluoranthene	0.2*	0.1	WAT-PS-10-01	0.00274
Benzo(a)pyrene	0.04*	0.01	WAT-PS-10-01	0.00109
Indeno(1,2,3-c,d)pyrene	0.02*	0.1	WAT-PS-10-01	0.000553
Benzo(g,h,i)perylene	0.12*	0.1	WAT-PS-10-01	0.00330

*Maximum concentration utilised as the contaminant was encountered within a singular borehole
**Average concentration from across the site utilised as the contaminant was encountered within a number of boreholes
WAT-PS-10-01 (August 2014)
LoD – Laboratory Limit of Detection

7.2.22 Note that contaminant half-lives have again between utilised in our assessment as the dissolved oxygen concentration recorded in the groundwater samples taken, suggest that the groundwater beneath the site is an aerobic environment.

7.2.23 Modelling to a compliance point of 250 m (as indicated in WAT-PS-10-01 (August 2014), assuming that the proposed development *'limits the exploitation of the groundwater resource for the foreseeable future'),* the final concentration of the contaminants: aliphatic C10-C12, aliphatic C12-C16, naphthalene, fluorene, fluoranthene, pyrene, benzo(a)anthracene, benzo(a)fluoranthene, benzo(a)pyrene, indeno(1,2,3-c,d)pyrene and benzo(g,h,i) perylene were recorded to be below the target values and were therefore not considered to pose a significant risk to the Groundwater Environment (within the superficial soils).

7.2.24 Whilst we recognise that aliphatic C16-C21, aliphatic C21-C35, aromatic C21-C35, phenanthrene and chrysene was recorded at marginally elevated concentrations at the compliance point (i.e. 250 m), it should be noted that we have assessed a theoretical risk only, as opposed to a known or proven risk, as the assessment criteria used for these contaminants was not a risk-based value, but rather the laboratory limit of detection (LoD), in the absence of a formalized guideline value in the SEPA WAT-PS-10-01 (August 2014) guidance document. Therefore, exceedances of the LoD do not necessarily indicate that there is risk to groundwater, only that said contaminants may be detected at the compliance point.

7.2.25 Furthermore, we would note that the total petroleum hydrocarbon (TPHs) contaminants were generally recorded at elevated concentrations on the first round of sampling, which was carried out not long after the boreholes were installed, and that on subsequent sampling rounds, the TPH concentrations were all below the laboratory Limit of Detection (LoD) Therefore, it is considered possible that the elevated concentrations of TPHs encountered within BH01B/RBH01B on the first round of sampling (particularly aliphatic C16 and above), could have been sourced from the mineral oils used to lubricate the drill rods during borehole formation.

7.2.26 In addition, the superficial soils beneath the site were confirmed to be made ground (of up to 2.50 m in thickness), underlain by stiff or stronger cohesive glacial till. As such, it is also considered possible that made ground could have been picked up by the drill rods during formation of the bore and then pushed into the glacial till, to the depth at which the standpipe was installed. Assuming that this was the case, this would account for the higher levels of contaminant encountered within the groundwater samples on the first round of sampling, in that all of the contaminants within the made ground would have been leached out at an early stage, and the lower concentration recorded on the second and third round of sampling are now just representative of background concentrations within the groundwater.

7.2.27 Therefore, in conclusion, given the lack of a significant source of contamination onsite, the risk to the Groundwater Environment (within the superficial soils) from the proposed development was considered to be generally low overall. In addition, given the nature of the proposed development, which is understood to be for multi-storey student accommodation, with limited areas of soft landscaping (i.e. primarily hardstanding and building cover), surface water infiltration will be considerably reduced by 40% or more following development of the site, which will further serve to reduce any potential loading of near to surface waters, and thus further reducing overall risk to the Groundwater Environment.

P20 Modelling Parameters

7.2.28 The tables below present justification for the input parameters used in our P20 modelling exercise undertaken for surface water and are considered to be relevant to the site conditions encountered during our investigation works.

TABLE 18 – P20 Groundwater Modelling Justifications

Criteria	Value Utilised	Justification
Width of plume in aquifer at source (perpendicular to flow)	36 m	If a contaminant was encountered in a number of groundwater samples then the width of the plume was considered to be width of the site from north to south i.e. ~36 m (measures perpendicular to flow, given the indicated groundwater flow direction to the east).
	17.5 m (BH01B/RBH01B) 17.5 m (BH02A/RBH02A)	If a contaminant was encountered in a singular groundwater sample, the area of influence around that borehole was modelled. The area of influence around BH01B/RBH01B was calculated to be 17.5 m (i.e. the half distance between BH01B/RBH01B and BH02A/RBH02A). The area of influence around BH02A/RBH02A was calculated to be 17.5 m (i.e. the half distance between BH01B/RBH01B and BH02A/RBH02A).
Plume thickness at source	6.75 m	Plume thickness has been estimated at 50% saturated aquifer thickness for metals (using maximum groundwater thickness encountered within site)
	3.375 m	Plume thickness has been estimated at 25% saturated aquifer thickness for hydrocarbons (using maximum groundwater thickness encountered within the site).
	3.375 m	Plume thickness has been estimated at 25% saturated aquifer thickness for hydrocarbons (BH01B/RBH01B)
	2.075 m	Plume thickness has been estimated at 25% saturated aquifer thickness for hydrocarbons (BH02A/RBH02A).
Saturated aquifer thickness	13.50 m	If a contaminant was encountered within a number of groundwater samples, then the maximum groundwater thickness recorded across the site during the groundwater monitoring phase of the investigations (5 No. rounds had been carried out at the time of reporting) was utilised (BH01B/RBH01B on 24.05.23)
	13.50 m (BH01B/RBH01B on 24.05.23)	If a contaminant was encountered in a singular groundwater sample, the maximum groundwater thickness observed within that borehole during the various groundwater monitoring phases of the investigation was utilised.
	8.30 m (BH02A/RBH0A on 15.05.23)	
Bulk density of aquifer materials	2.2505 g/cm^3	Based on the results of our site investigations, groundwater flow beneath the site was indicated to be largely through the made ground deposits, which were generally described as sand and gravel with extraneous materials or sandy gravelly CLAY. As such, a gravelly sand was assumed as the main constituent of the aquifer material to be conservative. Therefore, we have utilised a value of 2.2505 g/cm^3 in our assessment as the academic literature (StructX: Density Ranges for Different Soil Types: https://structx.com/Soil_Properties_002.html) indicates *'well-graded gravel/sand/silt and clay'* to have a bulk density range of between 2.002 g/cm^3 and 2.499 g/cm^3. As such, a reasonable midpoint value of 2.2505 g/cm^3 has been assumed within our assessment.
Effective porosity of aquifer	0.52 fraction	www.geotechdata.info – Soil Porosity A value of 30% was utilised in our assessment as the effective porosity in the literature for *'poorly graded sands, gravelly sands, with little or no fines'* was recorded to range between 30% and 75%. Therefore, we have utilised a mid-value between 30% and 75% of 52% within our assessment to be conservative.
Hydraulic gradient	0.0179 fraction	Calculated in the direction of groundwater flow (i.e. towards the east) between BH01B/RBH01B and BH02A/RBH02A.
Hydraulic conductivity of aquifer	86.4 m/d	Freeze and Cherry (1979) indicate clean sand and gravel to have a hydraulic conductivity of between 0.864 m/day to 86400 m/day. As such, a reasonable midpoint value of 86.4 m/day has been utilised within our assessment.

TABLE 18 *CONT* – P20 Groundwater Modelling Justifications

Criteria	Value Utilised	Justification
Fraction of Organic Carbon (FOC)	0.0362 fraction	Average dissolved organic carbon value from all groundwater sample results obtained from the site.
Distance to compliance point	1000 m	Value utilised was the distance from edge of the proposed development site to the Pow Burn/Braid Burn (~1000 m), at its closest point.
	Groundwater – 250 m	As stated in WAT-PS-10-01 (August 2014) Section 7.3 (p37), an assessment point of 250 m can be utilised, *'where present or planned future land-use limits the exploitation of the groundwater resource for the foreseeable future'* e.g. presence of sewered urban areas.

TABLE 19 – P20 Groundwater Modelling Justifications (Kd, KoC and Half Life)

	Substance	Kd (l/kg)	Reference	Half Life (days)	Reference
Metals	Copper	8175	EPA (2005)	Metal – assumed no decay	N/A
	Selenium	1995			
	Zinc	3135			
	Total Cyanide	1000			
	Substance	**KoC (l/kg)**	**Reference**	**Half Life (days)**	**Reference**
TPHs	Aliphatic C10-C12	316228	TPH CWG	102	Howard et al. (2019)
	Aliphatic C12-C16	5011872		440	
	Aliphatic C16-C21	398107171		530	
	Aliphatic C21-C35	398107171		730	
	Aromatic C21-C35	125893		730	
	Substance	**KoC (l/kg)**	**Reference**	**Half Life (days)**	**Reference**
PAHs	Chrysene	992.27	R&D Technical Report	2000	Howard et al. (2019)
	Naphthalene	49.90		258	
	Fluorene	237.46		120	
	Benzo(a)anthracene	1480.30		1361.45	
	Benzo(b)fluoranthene	441.42		1219	
	Benzo(k)fluoranthene	1093.63		4270.5	
	Indeno(1,2,3-c,d)pyrene	692.29		1460	
	Benzo(g,h,i)perylene	523.22		1314	
	Phenanthrene	454.86		401.5	
	Pyrene	671.83		3796	
	Fluoranthene	589.93		879.65	
	Benzo(a)pyrene	4023.87		1058.5	

EPA: Partition Coefficients for Metals in Surface Water, Soil and Waste, July 2005 (EPA/600/R-05/074)
TPH CWG: TPH Criteria Working Group Series, Volume 3, Selection of Representative TPH Fractions Based on Fate and Transport Consideration (1997)
Handbook for Environmental Degradation Rates, Howard et al, 2019
Draft R&D Technical Report P5-079/TP1, Table 2.4 pages 29 and 30. Note that the inverse log of 'log Koc' has been utilised

7.3 Conclusions

7.3.1 As demonstrated through chemical analysis and groundwater modelling, the proposed development is not considered to pose a significant risk to the Water Environment.

8.0 GROUND GAS EMISSIONS

8.1 General

8.1.1 A ground gas assessment has been undertaken to assess risks associated with carbon dioxide and methane to new buildings and their users. The identified potential sources of gas onsite were the made ground deposits, which were encountered across the site, extending to a maximum recorded depth of 2.50 mbgl (BH02). The Coal Authority indicate there to be no recorded mining beneath the site and unrecorded workings are considered unlikely. Consequently, mine gas was not considered a significant risk.

8.1.2 In accordance with the preliminary assessment procedures outlined in BS8576:2013, it was concluded that the generation potential for gas at the site was generally *'very low to low',* given that the made ground had a low degradable content and an average total organic content (TOC) of less than 6.0% (the average TOC concentration for the made ground encountered was 1.08%).

8.1.3 Therefore, in accordance with CIRIA C665 (2007), given the development proposals for flatted student accommodation (*'moderate'* end-use) and the *'low'* risk of generation potential for ground gases (assumed *'low'* risk as the worst-case scenario given the identified *'very low to 'low'* risk), 6 No. round of ground gas monitoring over a 3 No. month period were considered necessary.

8.1.4 The assessment of risk due to ground gases has been further discussed in publications for CIRIA and BRE, which have indicated a number of 'characteristic situations' depending on the concentrations and flow rates of gas. This classification system has been further developed by Wilson and Card (1999), and Boyle and Witherington (2006) and a revised industry guidance has been provided within CIRIA Report C665 (2007) and BS8485 (2015) & A1:2019.

8.1.5 The gas monitoring data was reviewed and a risk assessment prepared in line with British Standard BS8485 (2015) & A1:2019, whereby a scoring system is used to design suitable gas preclusion measures.

8.2 Ground Gas – Results

8.2.1 Ground gas standpipes were installed within each of the cable-percussive/rotary follow-on boreholes (BH01B/RBH01B, BH02A/RBH02A and BH03/RBH03) and have been monitored on 5 No occasions to date, using a portable gas meter. Confirmatory monitoring is continuing at the site and the results will be reported in a separate addendum letter report upon completion. The results of the monitoring carried out to date, along with a copy of the calibration certificates for the gas monitors used onsite, are attached in Appendix 7 for reference.

8.2.2 Measurements were taken over a variety of atmospheric conditions, including falling pressure conditions, with barometric pressure ranging from 997 mB to 1116 mB. Carbon dioxide concentrations ranged between 0.2% vol. and 1.7 % vol. and methane concentrations were recorded to always be 0.0 % vol. Oxygen concentrations ranged between 8.4% vol. and 18.0% vol. and the gas flow rate (steady state) was recorded to range between 0.0 l/hr and 2.5 l/hr (peak at 56.4 l/hr). Note that the above results were assessed after each of the readings reached steady state. The results of the gas monitoring undertaken are indicated on Drawing No. P23/075/SI/R/F/ 09.

8.3 Ground Gas – Assessment

8.3.1 Gas Screening Values have been calculated in line with CIRIA 665 and BS 8485 (2015) & A1:2019 guidance.

8.3.2 This is done by calculating a Q_{hg} for each monitoring point, for each monitoring event. Hazardous gas flow rate Q_{hg} (in lh^{-1}) is calculated using the following:

$$\mathbf{Q_{hg} = C_{hg}/100 \times q}$$

Where:

C_{hg} is the measured hazardous gas concentration (in percentage volume-by-volume)

q is the flow rate (in litres per hour) of combined gasses from the standpipe found by direct measurement.

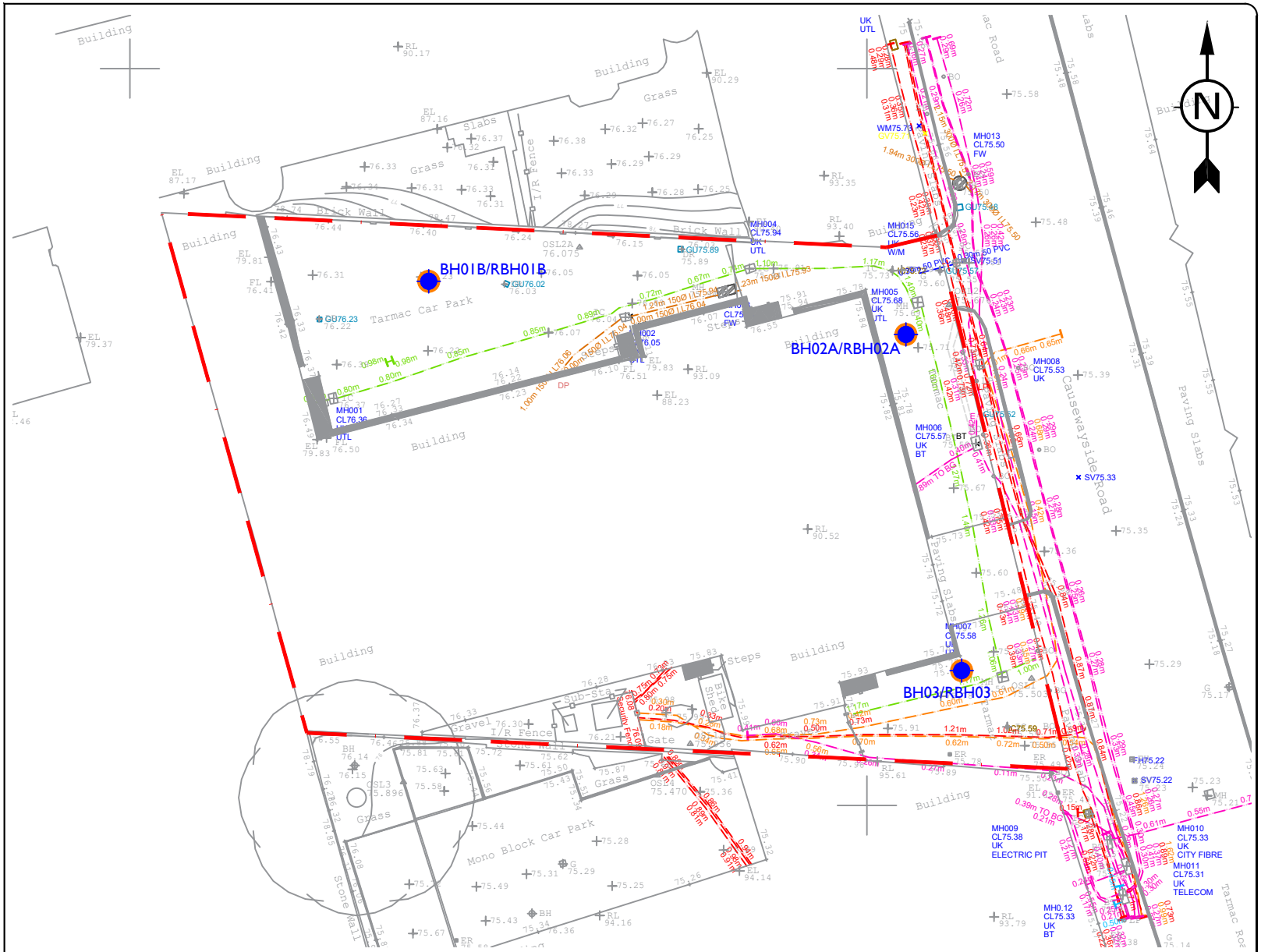
8.3.3 Hazardous gas flow rates were calculated for each monitoring point during each event (Appendix 7). Based on the monitoring undertaken to date, a worst-case scenario was realised on the 6th July 2023 in BH01B/RBH01B where CO_2 was 1.1% vol. and a gas flow rate of 2.5 l/hr was recorded. The resultant hazardous gas flow rate is therefore as follows:

$$\mathbf{Q_{hg} = 1.1/100 \times 2.4}$$

$$\mathbf{Q_{hg} = 0.0275}$$

The value derived above is the highest hazardous gas flow rate calculated over the duration of the gas monitoring. Given the low number of monitoring rounds, it was considered conservative to use this maximum value to proceed with the characterisation of the sites and evaluate the gas protection measures employed.

As a conservative measure, we have also calculated a theoretic worst-case scenario using the highest gas flow rate and the highest gas concentration recorded from any of the boreholes during the monitoring period. Note that the highest gas concentration (1.7% vol.) was recorded in BH01B/RBH01B on 21.06.23 and the highest flow rate (2.5 l/hr) was recorded in BH01B/RBH01B on 06.07.23.



Borehole BH01B/RBH01B

Date.	CH4 % Vol.	CO2 % Vol.	O2 % Vol.	Ave Flow (l/h)	Atmos. Pressure (mb)	G/Water Depth (m)
10.05.23	0.0	0.6	13.2	0.0	997	14.40
24.05.23	0.0	0.4	18.0	0.0	1015	14.60
12.06.23	0.0	0.6	17.8	0.0	1008	13.90
21.06.23	0.0	1.7	14.7	0.0	999	14.00
06.07.23	0.0	1.1	16.7	0.0	997	13.90

Borehole BH02A/RBH02A

Date.	CH4 % Vol.	CO2 % Vol.	O2 % Vol.	Ave Flow (l/h)	Atmos. Pressure (mb)	G/Water Depth (m)
10.05.23	-	-	-	-	997	-
24.05.23	0.0	0.2	16.9	0.0	1015	9.00
12.06.23	0.0	0.4	16.5	0.0	1008	8.80
21.06.23	0.0	0.4	17.7	0.0	999	8.80
06.07.23	0.0	0.5	17.8	0.0	997	8.95

Borehole BH03/RBH03

Date.	CH4 % Vol.	CO2 % Vol.	O2 % Vol.	Ave Flow (l/h)	Atmos. Pressure (mb)	G/Water Depth (m)
10.05.23	0.0	0.2	8.3	0.0	997	11.60
24.05.23	0.0	0.2	14.0	0.0	1015	N/A
12.06.23	0.0	0.2	15.7	0.0	1008	N/A
21.06.23	0.0	0.2	17.9	0.0	999	N/A
06.07.23	0.0	0.4	16.7	0.0	997	N/A

NOTES

- Approximate site boundary
- BH01 to BH03 — Cable-percussive boreholes sunk by Phoenix Drilling Ltd, on behalf of Mason Evans (April 2023)
  - Rotary follow-on-element, also carried out by Phoenix Drilling Ltd undertaken in boreholes BH01B, BH02A and BH03 (referenced as RBH01B, RBH02A and RBH03)
- Gas/groundwater monitoring well installation

REV	DATE	DETAILS

BALFOUR BEATTY INVESTMENTS Ltd
3rd FLOOR, FOCUS POINT
21 CALEDONIAN ROAD
LONDON, N1 9GB

PROJECT TITLE

CAUSEWAYSIDE EDINBURGH

DRAWING TITLE

GROUND GAS EMISSIONS SURVEY

DRAWN BY	CHK'D BY	APP'D BY	DATE	SCALES
TR	LR	LR	21.07.23	1:300 @ A4

PROJECT No.	DRAWING No.	REVISION
P23/075	P23/075/SI/R/F/09	

MASON EVANS
Part of the CTS Group
Geo-Environmental Consultants
t: 0141 420 2025 e: mail@masonevans.co.uk
The Piazza, 95 Morrison Street, Glasgow, G5 8BE

A4 border: 297 X 210

The resultant hazardous gas flow rate is therefore as follows:

Q_{hg} = 1.7/100 x 2.5

Q_{hg} = 0.0425

This worst-case hazardous gas flow rate will therefore be utilised in our following assessment on the ground gas regime at the proposed development site.

8.3.4 Based on the BS 8485 (2015) & A1:2019 guidance, the calculated GSV noted above corresponds generally to '*Characteristic Situation 1'* (Table 20), which does not require gas preclusion measures. This was considered appropriate given that no abandoned mineworkings were recorded beneath the site, no cemeteries or landfill sites were recorded within the immediate vicinity of the site and the made ground deposits encountered onsite recorded no significant organics or biodegradable materials. Confirmatory gas monitoring is continuing at the site and the results will be reported as a separate addendum letter report upon completion. However, it is assumed that gas protection measures will unlikely be required within the future development.

TABLE 20 – Assessment of Gas Characterisation

Characteristic Situation	Hazard Potential	Gas Screening Value (GSV) (l/hr))	Additional Limiting Factors
1	Very Low	<0.07	Methane not to exceed 1% by volume and carbon dioxide not to exceed 5% by volume.
2	Low	0.07 to <0.7	Borehole air flow not to exceed 70 l/hr, otherwise increase to CS3.
3	Moderate	0.7 to <3.5	None
4	Moderate to High	3.5 to <15	None
5	High	15 to <70	None
6	Very High	>70	None

8.3.5 The construction and use of the buildings, together with the control of future structural changes to the building and its maintenance (the building's management) should be assessed, since potential risks posed by ground gases are strongly influenced by these factors. The assessment should lead to the categorisation of the building as a whole, or each different part of the building, into one of four building types: Type A, Type B, Type C or Type D.

8.3.6 New buildings should be categorised in accordance with Table 21, as outlined in BS8485 (2015) & A1:2019. The proposed development is considered to be *'Type B'* (i.e. flatted student accommodation with central building management).

TABLE 21 – Building Types and Descriptions (BS 8485 – 2015 & A1:2019)

Building Type	Description
A	Private ownership with no building management controls on alterations to the internal structure, the use of rooms, the ventilation of rooms or the structural fabric of the building. Some small rooms present. Probably conventional building construction (rather than civil engineering). Examples include private housing and some retail premises.
B	Private or commercial property with central building management control of any alterations to the building or its uses but limited or no central building management control of the maintenance of the building, including the gas protection measures. Multiple occupancy. Small to medium size rooms with passive ventilation of rooms and other internal spaces throughout ground floor and basement areas. May be conventional building or civil engineering construction. Examples include managed apartments, multiple occupancy offices, some retail premises and parts of some public buildings (such as schools, hospitals, leisure centres) and parts of hotels.
C	Commercial building with central building management control of any alterations to the building or its uses and central building management control of the maintenance of the building, including the gas protection measures. Single occupancy of ground floor and basement areas. Small to large size rooms with active ventilation or good passive ventilation of all rooms and other internal spaces throughout ground floor and basement areas. Probably civil engineering construction. Examples include offices, some retail premises, and parts of some public buildings (such as schools, hospitals, leisure centres and parts of hotels).
D	Industrial style building having large volume internal space(s) that are well ventilated. Corporate ownership with building management controls on alterations to the ground floor and basement areas of the building and on maintenance of ground gas protective measures. Probably civil engineering construction. Examples are retail park sales buildings, factory shop floor areas, warehouses. (Small rooms within these style buildings should be separately categorized as Type B or Type C).

8.3.7 From the identified Characteristic Situation and type of building, the minimum gas protection score should be calculated, in accordance with Table 22 below.

TABLE 22: Gas Protection Score by Characteristic Situation and Type of Building

Characteristic Situation (CIRIA 149)	MINIMUM GAS PROTECTION SCORE (POINTS)			
	High Risk Type A Building	Type B Building	Medium Risk Type C Building	Low Risk Type D Building
1	0	0	0	0
2	3.5	3.5	2.5	1.5
3	4.5	4.0	3.0	2.5
4	6.5	5.5	4.5	3.5
5	-	6.5	5.5	4.5
6	-	-	7.5	6.5

8.3.8 Following the determination of the minimum gas protection score, gas protection measures are not considered necessary within the proposed development. Confirmatory gas monitoring is continuing at the site and the results will be reported as a separate addendum letter report upon completion. However, it is assumed that gas protection measures will not be required.

8.4 Radon

8.4.1 Researches of BGS information and the BRE Report BR211(2015): "Radon: Protective Measures for New Buildings" indicate that site is located in a Lower Probability Radon Area (less than 1% of homes are estimated to be at or above the Action Level) and that in accordance with the current guidelines indicated within the report, protection measures are not required.

8.5 Conclusions

8.5.1 Based on the results of the ground gas monitoring undertaken to date and subsequent risk assessment, gas protection measures are not considered necessary within the proposed development. Confirmatory gas monitoring at the site and the results will be reported as a separate addendum letter report upon completion.

8.5.2 Radon protection is not considered necessary within the proposed development.

9.0 RISKS TO CONSTRUCTED DEVELOPMENT

9.1 Sulphate Attack on Construction Materials

9.1.1 Laboratory testing was undertaken on selected soil samples recovered from the site to determine the sulphate content and acidity and hence the concrete class required for buried concrete (results included in Appendix 8). The results of chemical tests carried out are summarised below:

TABLE 23 -Sulphate and pH Summary

Determinant	Recorded Value	SD1 DS Class	SD1 ACEC Class
pH*	7.325	DS-1	AC-1
Total Sulphate as SO_4 (%)**	0.115		
Sulphate as SO_4 (mg/l)**	55.75		

*Concrete class was based on the mean of the lowest 20% test result as a characteristic value in accordance with guidance document BRE Special Digest (3rd Edition, 2005)

**Concrete class was based on the mean of the highest 20% test results as a characteristic value in accordance with guidance document BRE Special Digest (3rd Edition, 2005)

9.1.2 In accordance with BRE Special Digest 1:2005 *'Concrete in Aggressive Ground'*, our recommendations for concrete would be Aggressive Chemical Environment for Concrete (ACEC) Classification AC-1 with a Design Sulphate Class for the site of DS-1.

9.2 Water Supply Pipes

9.2.1 UK Water Industry Research (UKWIR) document, '*Guidance for the Selection of Water Supply Pipes to be Laid in Brownfield Sites*', ref 10/WM/03/21, states that on brownfield sites, MDPE/HDPE water supply pipes could be at risk from organic contaminants including mineral oils, VOC's and SVOC's, if the pipes are laid within 15 m of recorded contamination. Additionally, UKWIR states that where metallic pipes are being considered for use, conductivity, pH and redox state of the soil should be assessed to determine if the pipes are at risk of being corroded.

9.2.2 UKWIR testing was not undertaken as part of our site assessment as the proposed water pipe alignment and finalised levels were not available at the time of reporting. Therefore, we would recommend that UKWIR testing is undertaken along the proposed water supply route once the site layout and levels have been finalised.

9.3 Phytotoxicity

9.3.1 Guidance on the effects of metal contamination on plant growth is provided within BS3882:2015 Specification for Topsoil and similar guidance issued by the Scottish Agricultural College (SAC). A summary of test results, versus the recommended phytotoxic screening criteria is presented below:

TABLE 24: Summary of Soil Results vs Phytotoxic Screening Criteria

Contaminant	Screening Value (mg/kg)	Conc Range (mg/kg)	Max > MAFF Screening Value
Zinc	300	29 - 94	NO
Copper	200	21 - 63	NO
Nickel	110	17 - 34	NO

Note – screening value based on pH of >7.0 (average pH calculated from across the soil samples was calculated to be 8.9)

9.3.2 No exceedances of BS3882:2015 screening values have been noted in the samples tested. Therefore, the phytotoxic risk at the site was considered to be low. Note that the development proposals include for flatted student accommodation, therefore areas of soft landscaping within the future development will be minimal, if any.

10.0 REVISED CONCEPTUAL SITE MODEL

10.1 Contamination Sources

Human Health: 2 No. localised *'hotspots'* of asbestos were identified within the made ground in the areas surrounding BH02 (at 0.50 mbgl) and BH03 at (0.50 mbgl and 1.00 mbgl) Quantification testing was undertaken on each of the samples with a positive identification for asbestos and the results of the phase contrast optical microscopy indicated the recorded level of asbestos to be between 0.002% vol and 0.020% vol.

Therefore, the risk to human health was considered to be moderate to high around BH02 and BH03.

Elsewhere across the site, no other toxic contaminants were identified and the risk to human health was considered to be low.

Water Environment: Several contaminants were present above current guidelines values. However, modelling was undertaken and the overall risk to the Water Environment is considered to be low with the potential for off-site migration also considered to be low.

Ground Gas: Based on the results of the gas monitoring undertaken to date and our subsequent risk assessment, gas protection measures are not considered necessary within the future development. Confirmatory gas monitoring is continuing at the site and the results will be reported as a separate addendum letter report upon completion. However, it is currently assumed that gas protection measures will not be required.

The site was indicated to be located in an area where radon was not indicated to be present at significant levels.

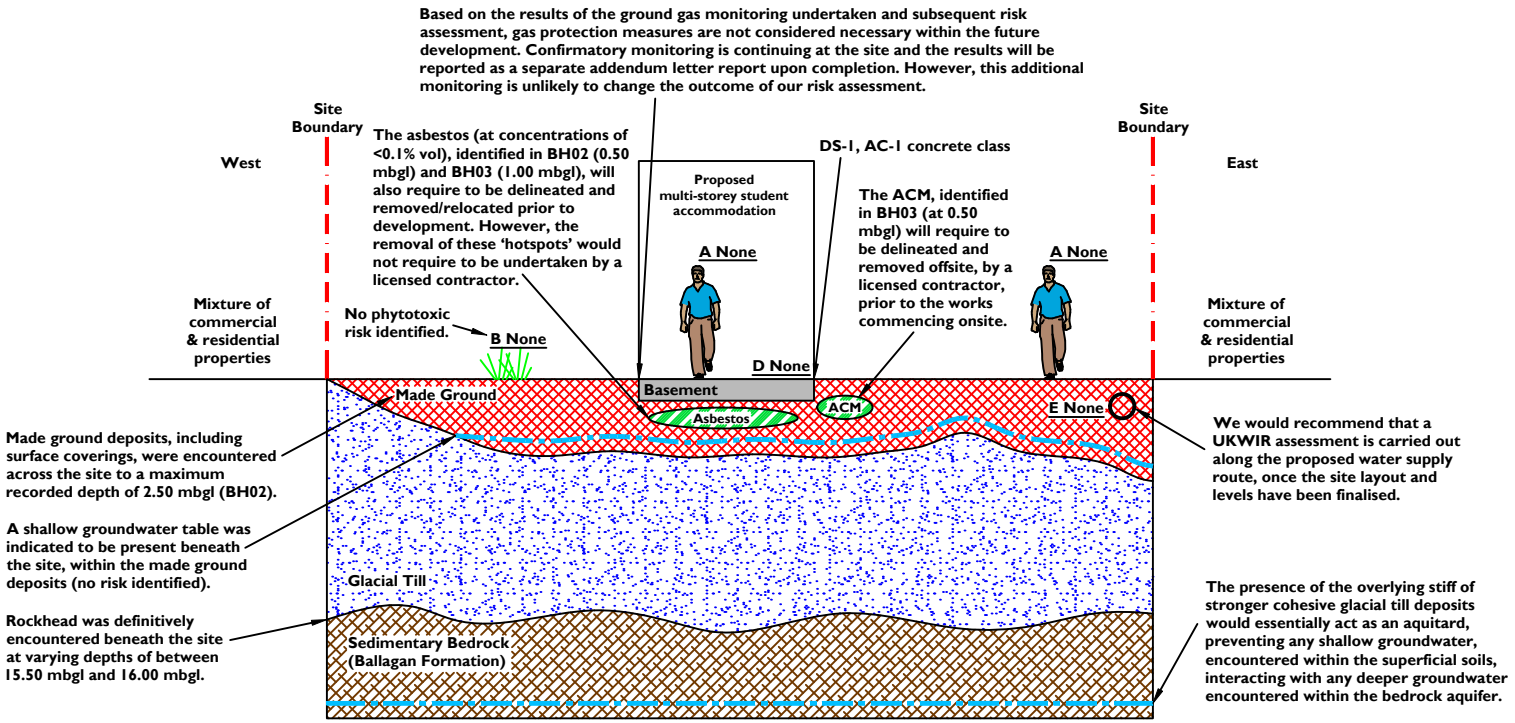
Built Environment: Samples analysed for pH and sulphate are below BRE thresholds.

UKWIR testing should be undertaken along the proposed water supply route the site layout and levels have been finalised.

Plant Life: Based on the soil chemical analysis results, the phytotoxic risk was considered to be low.

10.2 Pollutant Linkage Assessment

10.2.1 Based on the ground and groundwater contamination conditions noted at the site, a significant viable source-pathway-receptor pollutant linkages has been identified in relation to asbestos, and the conceptual site model has been revised as indicated on Drawing No. P23/075/SI/R/F/10.



Potential Source

- Localised 'hotspots' of asbestos and ACM's.

Potential Exposure Pathways

- None following remedial measures.

Potential Receptors

A. Site users / construction personnel.
B. Vegetation / fauna.
C. Groundwater.
D. Buried concrete (Service and foundations)
E. Plastic water supply pipes.

NOTES

REV	DATE	DETAILS

BALFOUR BEATTY INVESTMENTS Ltd
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LONDON, N1 9GB

PROJECT TITLE

CAUSEWAYSIDE
EDINBURGH

DRAWING TITLE

REVISED CONCEPTUAL SITE MODEL

DRAWN BY	CHK'D BY	APP'D BY	DATE	SCALES
TR	LR	LR	21.07.23	Not to Scale

PROJECT No.	DRAWING No.	REVISION
P23/075	P23/075/SI/R/F/10	

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10.3 Mitigation Measures

Contamination

10.3.1 The asbestos encountered in BH02 (at 0.50 mbgl) has been subject to analysis that has indicated the quantity of asbestos to be less than 0.1% vol. of the sample. Therefore, we would recommend that this *'hotspot'* is delineated and removed or relocated prior to development. It is anticipated that these works would comprise the removal of a limited volume of material around the affected area and would require to be supervised and validated by a suitably qualified geo-environmental scientist. The excavated soils would then need to be disposed of off-site, or alternatively, the soils could be retained on site and placed beneath less sensitive areas, such as beneath roads and/or areas of hardstanding/building cover. Note that we would not recommend that the asbestos contaminated soils are placed beneath roads due to the potential risk to future maintenance personnel.

10.3.2 Conversely, the asbestos identified in BH03 (at 0.50 mbgl) was recorded to contain visible asbestos, described as *'visible insulation board debris'* (at 0.50 mbgl). Therefore, this *'hotspot*' will require to be delineated in the first instance to confirm its size and extent and then removed by a suitably licensed contractor, with the excavated soils being removed from site so a specialist *'hazardous'* waste facility site.

10.3.3 The asbestos identified in BH03 at 1.00 mbgl was below the 0.1% vol. threshold and was recorded to contain no visible ACM's (asbestos containing materials). Therefore, once the ACM at 0.50 mbgl has been removed, the deeper lying asbestos (at 1.00 m depth) can be retained onsite beneath a less sensitive area or disposed offsite to a suitably licensed waste facility site, as per the *'hotspot'* identified in BH02.

10.3.4 However, please note that half of the site was occupied by building cover during our investigations, and therefore the possibility of encountered further asbestos during a future phase of post-demolition works cannot be discounted.

10.3.5 Construction personnel should be aware of the nature of the soils on the site and vigilance should be maintained for any soils at variance from those anticipated during construction works, particularly for the presence of further asbestos. Appropriate health and safety procedures should be adopted at the site in relation to these matters (including ACM's).

Groundwater

10.3.6 The shallow soils were considered to pose a low risk to the Water Environment.

Ground Gas

10.3.7 Based on the results of the ground gas monitoring undertaken to date and subsequent risk assessment, gas protection measures are not considered necessary within the proposed development of the site. Confirmatory gas monitoring is continuing at the site and the results will be reported as a separate addendum letter report upon completion. However, this additional monitoring is unlikely to change the outcome of our risk assessment.

10.3.8 Radon protection is also not considered necessary within the proposed development.

The Built Environment

10.3.9 We would recommend that a UKWIR assessment is undertaken along the proposed water supply route, once the site layout and levels have been finalised. Note that this assessment would inform the likely water supply pipe selection material.

10.3.10 Concrete class DS-1, AC-1 is considered sufficient to protect buried concrete from pH and sulphate levels in the soils and groundwater.

Fuel Tanks

10.3.11 Although it is considered unlikely that any underground tanks will be present onsite, especially given the history of the site as a school, vigilance should still be maintained during any future site works for the presence of tanks. If present, the tanks should be removed under our supervision and validation testing undertaken subsequent to removal, to ensure that there has been no localised impact on the soils from fuel leakage/spillage. If such impacts are identified, the soils would likely require to be removed or remediated.

Construction/Maintenance Workers

10.3.12 All site staff should remain vigilant to the possible risk of encountering isolated areas of unrecorded contaminated material, particularly the presence of further asbestos and ACM's. Should such materials be encountered, further testing may be required to assess the risk to health and safety of the site workers and the environment.

10.3.13 Good site working practices should be followed, including:

- Use of appropriately qualified personnel for the task;
- Use of appropriate PPE (including for working with asbestos, locally at concentrations >0.1% vol.)
- Provision of on-site washing facilities;
- Maintenance of a high standard of basic hygiene; and
- Implementation of a non-smoking and eating policy within the working area, with designated clean areas set aside for these activities.

10.4 Waste Management Legislation

10.4.1 Should materials be removed and disposed off-site, the developer has a statutory responsibility under the Duty of Care Regulations of the Environmental Protection Act 1990 to ensure that contaminated soil and water is disposed of off-site to a suitably licensed waste management facility in a safe and approved manner.

Waste Classification Summary

10.4.2 The HazWasteOnline toolkit was used to undertake a Hazard Assessment Screen, to establish whether the sampled soils should be considered as representative of either *'hazardous'* or *'non-hazardous'* waste. This classification process is in accordance with technical guidance document WM3, *Guidance on the classification and assessment of waste (v1.2 GB, 2021)*, published jointly by Natural Resources Wales / Cyfoeth Naturiol Cymru, the Scottish Environment Protection Agency and the Environment Agency.

10.4.3 It should be noted that in Scotland, *'hazardous waste'* is known as *'special waste*'. The Special Waste Regulations 1996 is the principal piece of legislation covering special waste arsing in Scotland and sets out the procedures to be followed when disposing of, carrying and receiving special waste.

10.4.4 The HazWasteOnline output sheets for each of the 11 No. samples collected, between the depths of 0.25 mbgl and 3.00 mbgl, give an initial waste classification of *'non-hazardous waste',* based on the chemical properties of the material. The output sheets are presented in Appendix 10 for reference.

Asbestos Assessment

10.4.5 Asbestos was detected in 3 No. of the made ground samples, in the form of *'chrysotile bundles'* in BH02 at 0.50 mbgl (at a quantification of 0.0022% vol) and amosite present as *'visible insulation board debris and fibres bundles'* in BH03 at 0.50 mbgl and *'amosite present in microscopic insulation board debris and fibres bundles'* in BH03 at 1.00 mbgl (at quantifications of 0.01% vol. and 0.02%, respectively).

10.4.6 Chapter 3 Section 1 of WM3 states:

"*The assessment of asbestos containing waste considers both the presence of asbestos as:*

- *fibres that are free and dispersed, and*
- *identifiable pieces of asbestos containing material*

If the waste contains fibres that are free and dispersed then the waste will be hazardous if the waste as a whole contains 0.1% or more asbestos. If the waste contains any identifiable pieces of suspected asbestos containing material they must be assessed as set out below. This would also apply to any dispersed fibres produced by deliberately breaking up such identifiable pieces.

Where the waste contains identifiable pieces of asbestos containing material; (i.e. any particle of a size that can be identified as potentially being asbestos by a competent person if examined by the naked eye); then these pieces must be assessed separately. The waste is hazardous if the concentration of asbestos in the piece of asbestos containing material is 0.1% or more."

10.4.7 As the asbestos recorded in soil sample BH03 at 0.50 mbgl was visible by the naked eye, it is regarded as asbestos containing material (ACM). Waste is hazardous if the concentration of asbestos in the piece of ACM is 0.1% or more. However, in reality, the majority of ACMs by their nature will contain >0.1% asbestos which means that if a single piece of ACM is identified in the material, the whole material will be classified as hazardous, unless the ACM can be removed.

10.4.8 In this case, the material in which sample BH03 at 0.50 mbgl was collected (i.e. a layer of made ground was recorded from 0.30 m to 1.20 m described as *'orange and brown angular cobbles of brick and igneous rock'*) should be classified as *'hazardous waste'*. Furthermore, as the asbestos has been identified as 'insulation board' the works to remove the soil in this area will be regarded as licenced works.

10.4.9 Therefore, a previously discussed, it is recommended that the *'hotspot'* of asbestos containing materials (ACMs), identified in BH05 at 0.50 mbgl, is delineated in the first instance to determine the extent of the asbestos impacted soils which will require to be removed – these delineation and removal works should be undertaken post-demolition.

List of Waste Code

10.4.10 Based on the results of the analysis, and the assessment described above, the List of Waste code for the sampled material is considered to be '**17 05 04 Soil and stones other than those mentioned in 17 05 03**, i.e. non-hazardous waste.

10.5 Waste Acceptance Criteria (WAC)

10.5.1 Materials classified as *'non-hazardous'* may be disposed of at an inert waste facility subject to meeting *'inert waste'* acceptance thresholds. Alternatively, if the samples do not meet *'inert'* thresholds then the waste may disposed of at a *'non-hazardous'* facility. WAC testing has been completed on 4 No. samples (BH01 at 1.00 mbgl, BH02 at 2.00 mbgl, BH03 at 2.00 mbgl, BH06 at 1.00 mbg)l. No exceedances of *'inert'* thresholds were recorded in the samples tested. The results of the WAC testing are included in Appendix 8 for reference.

10.5.2 It is ultimately the decision of the waste receiver as to whether they will accept the material. All wastes removed from site should be consigned, transported and disposed of in full accordance with all relevant UK legislation.

10.6 Rate of Scottish Landfill Tax

10.6.1 Any waste producer disposing of waste at a landfill site in Scotland is due to pay Scottish Landfill Tax (SLfT). Revenue Scotland is responsible for the collection of landfill tax in Scotland, and they have set two rates of tax; these are known as the "standard rate" and "lower rate" of SLfT. For material to be eligible for the lower rate of SLfT, it must meet the criteria of "qualifying material", as defined in the Scottish Landfill Tax (Qualifying Material) Order 2016. Qualifying material can include soils and stones.

10.6.2 In October 2016, Revenue Scotland published guidance note SLfT2006 clarifying the SLfT position on qualifying materials containing a small amount of non-qualifying material. This can include non-qualifying material in the form of contaminants in soils, or organic materials such as fragments of wood, rootlets or tarmac.

10.6.3 In order for the lower rate of SLfT to apply to a load of soil, Revenue Scotland has stated that it must not be classed as hazardous waste under WM3. The person disposing of the waste should be asked to provide evidence of the non-hazardous classification of the load under WM3.

10.6.4 At the time of this report the current landfill tax rates are £102.10/t for standard rate and £3.25/t for lower rate. The tax rate changes annually on 01 April. Note that a gate fee and haulage costs are in addition to the landfill tax rates.

10.6.5 Based on the classification described above and the List of Waste code assigned in Section 10.5, it is considered likely that the material will be eligible for the lower rate of SLfT.

10.7 Contingent Liabilities

10.7.1 Assessments of the site include the determination of contingent liabilities in respect of current and future ownerships subsequent to remedial measures. These consider the impact of the environmental conditions on the study area and surrounding areas on site users, properties and also the liability of the site owners.

10.7.2 With regard to site users, considerations in relation to liability are inherent in the development of a suitable remedial strategy. In the site-specific circumstances presented by the identified conditions, the risk levels suggest minimal liability on ownership due to the environmental conditions, subsequent to development.

10.7.3 The potential for liability arising from site conditions impacting on the surrounding environment largely considers the potential for migration of pollutants beyond the site boundary normally associated with groundwater. Within this report we have concluded that there is a limited contamination source within the boundaries of the proposed development site. However, hydrological modelling to the nearest surface water body and groundwater (within the superficial soils) has shown no significant migration potential. Therefore, overall offsite migration is considered to be low. Consequently, the potential for liability arising from the site conditions would be considered to be low following appropriate remediation of the identified development constraints.

10.7.4 In the event that more definitive advice is required, we would recommend that the Client seeks specific advice on the liabilities incumbent on ownership from their legal advisors.

11.0 GEOTECHNICAL ASSESSMENT

11.1 General

11.1.1 Geotechnical laboratory testing was undertaken on a selection of soil samples recovered from the boreholes (cable-percussive and rotary follow-on) sunk within the site. In addition, visual assessment of the soils during the sinking of the boreholes (terrier, cable-percussive and rotary follow-on) was supplemented by in-situ standard penetration tests (SPT's). It should be recognised that SPT testing of cohesive soils will only provide an indicative assessment of soils strength, although testing of granular soils will provide more reliable test data. Note that a number of SPT tests could not be carried out within the cable-percussive and terrier boreholes due to hard obstructions at shallow depths and/or due to the stiffness of the glacial soils encountered.

11.1.2 Moisture content analysis, Atterberg Limits and triaxial testing was also undertaken on a selection of natural samples obtained from the soils boreholes and point load and unconfined compressive strength (UCS) testing was undertaken on a selection of rock core samples obtained from each of the rotary follow-on boreholes (RBH01B, RBH02A and RBH03). The geotechnical results are included in Appendix 11.

11.1.3 The geotechnical properties of the soils tested at the site are indicated in the table below:

TABLE 25: Geotechnical Summary

Visual Description	Test Sample Depth Range (mbgl)	In-situ Strength/ Density	Uncorrected 'N' Value*	Moisture Content (%)	Cohesion (kPa)	Friction Angle (°)
Glacial Till (cohesive)	1.20 – 15.50	Stiff, becoming very stiff	19 - 50	0.8 – 1.5	325.3	24.7

** Does not include for the water correction factor as the groundwater table beneath the site was encountered within the overlying made ground soils, which would not be suitable as a potential foundation bearing horizon.*

11.1.4 The SPT testing carried out in the boreholes indicated the cohesive glacial till deposits to be stiff and very stiff, which is generally consistent with the visual appraisal of these soils in the boreholes.

11.2 Atterberg Limits

11.2.1 The sample tested from BH01B/RBH01B (at 4.50 mbgl), indicated the cohesive glacial till deposits to be *'clay with low plasticity'*.

11.3 Triaxial/Undrained Shear Strength Testing

11.3.1 A sample of the cohesive glacial till was obtained from BH01B/BH01B (at 4.60 mbgl) and tested for triaxial testing. The cohesion value in the sample tested was 325.3 kPa, which indicates an undrained shear strength of very stiff. This result is generally consistent with the visual appraisal of these soils and the results of the SPT's, which indicated the superficial soils to be stiff and very stiff.

11.4 **Sedimentary Bedrock**

11.4.1 Rockhead was definitively encountered in each of the rotary follow-on boreholes at varying depths of between 15.50 mbgl and 16.00 mbgl. Rockhead was described as; *'very weak, locally extremely weak light grey thin to thickly bedded micaceous MUDSTONE, recovered predominantly non-intact and moderately weathered throughout with locally heavily weathered sections'*, *'very weak to weak reddish brown thinly laminated MUDSTONE, heavily weathered and recovered predominantly non-intact'*; *'moderately strong whitish grey thickly bedded crystalline LIMESTONE, lightly to locally moderately weathered'*; *'moderately strong bluish grey thickly bedded crystalline LIMESTONE, lightly weathered'*; or *'hard white speckled orange SANDSTONE, fresh to locally lightly weathered'*. A copy of the rock core photographs are attached in Appendix 6 for reference.

11.5 **Point Load and Unconfined Compressive Strength (UCS) Testing**

11.5.1 Point Load and Unconfined Compressive Strength (UCS) tests were carried out on samples of the underlying rock strata obtained from the rotary follow-on boreholes (RBH01B, RBH02A and RBH03).

11.5.2 The results of the tests are indicated below:

TABLE 26: Geotechnical Summary (Unconfined Compressive Strength)

Borehole	Rock Type	Depth (mbgl)	UCS (MPa)
RBH02A	LIMESTONE	16.05	80.2
RBH02A	LIMESTONE	22.60	34.0
RBH03	MUDSTONE	21.60	29.4
RBH03	SANDSTONE	24.50	121.5

11.5.3 The unconfined compressive strength tests results from RBH02A and RBH03 were recorded to range between 29.4 MPa and 121.5 MPa, which corresponds to a moderately strong to very strong rock with a presumed bearing value of 10 MPa. Note that no rock core samples from RBH01A were suitable for UCS testing as the geotechnical laboratory deemed the scheduled samples (at 16.00 m to 18.00 m and 18.00 m to 21.00 m) to be *'badly fractured/no intact section'*.

11.5.4 The results of the point load tests are indicated below:

TABLE 27: Geotechnical Summary (Point Load Tests)

Borehole	Depth (mbgl)	Rock Type	Mean Is(50) – Axial	Equivalent UCS Value (MPa) – Axial*	Mean Is(50) – Diametrical	Equivalent UCS Value (MPa) – Diametrical*
RBH01B	16.50 – 20.65	MUDSTONE	0.80	16.0	0.18	3.6
RBH02A	15.65 – 18.80	MUDSTONE	6.88	137.6	4.82	96.4
RBBH03	21.90 – 23.70	LIMESTONE	3.30	66.0	1.12	22.4

* Is(50) multiplied by 20 to produce an approximate equivalent UCS value in MPa

11.5.5 The results of the point load tests recorded equivalent UCS values of between 3.6 MPa and 137.6 MPa which confirmed the underlying sedimentary strata to again be a moderately strong, to very strong rock, which is locally weak in the areas of mudstone.

11.5.6 Based on a review of the cored borehole logs, the underlying solid strata was indicated to be generally MUDSTONE with LIMESTONE bands and localised SANDSTONE. Therefore, assuming MUDSTONE as the most conservative, given the fissile nature of this rock type, we would recommend that a presumed bearing value of 3.0 MPa is assumed for the underlying solid strata.

12.0 FOUNDATION RECOMMENDATIONS

12.1 Details of the Development

12.1.1 The development proposals were not available at the time of reporting but were understood to be for multi-storey student accommodation with areas of soft landscaping and associated road infrastructure.

12.2 Foundations (relative to existing site levels)

12.2.1 Made ground deposits, including surface coverings, were encountered across the site extending to a maximum recorded depth of 2.50 mbgl (BH02). These deposits were recorded to be of a variable density and degree of compaction and are not considered as a suitable foundation bearing horizon in their current condition.

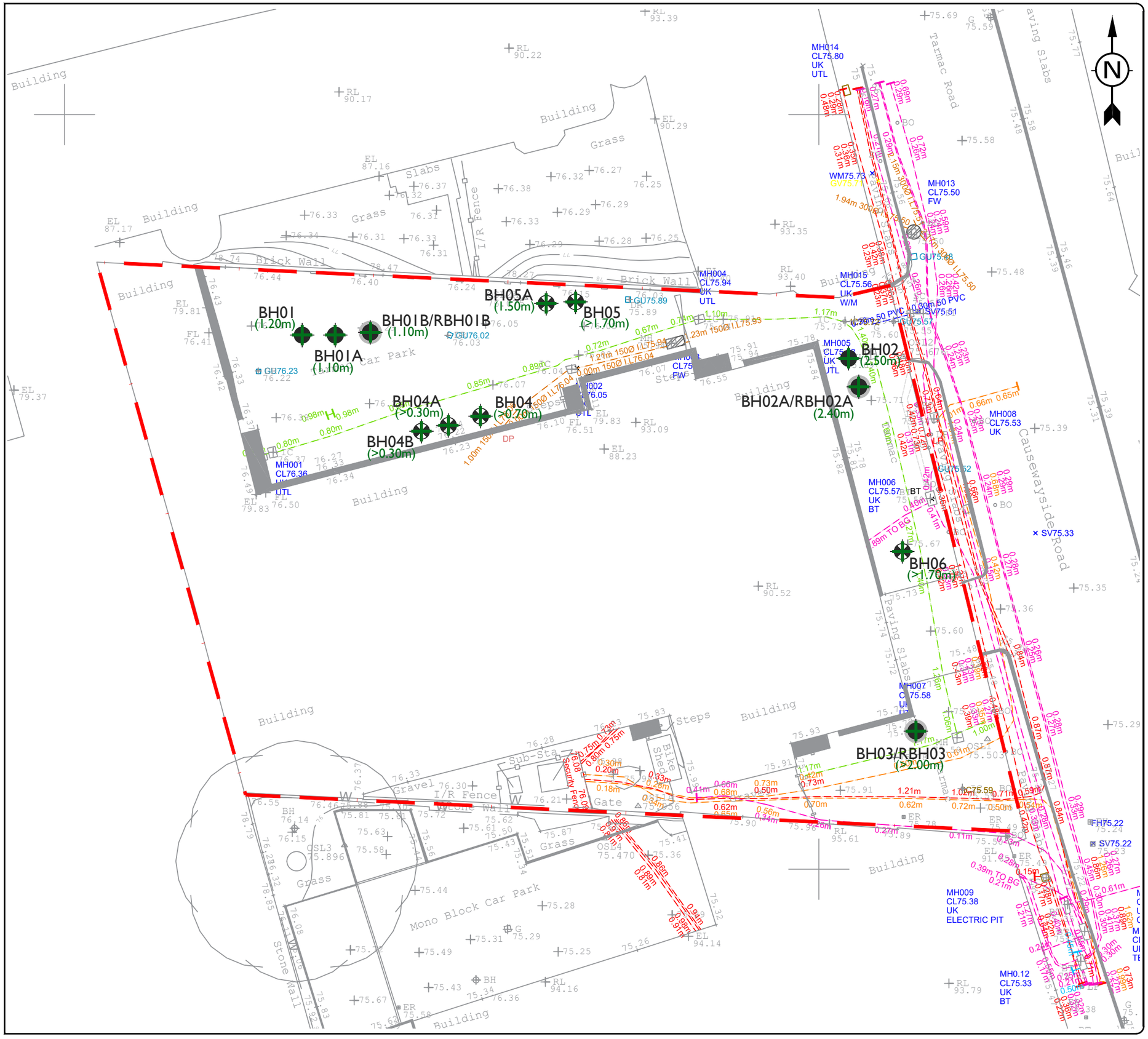
12.2.2 An allowable bearing capacity of approximately 125 kPa is estimated for the underlying stiff or stronger cohesive glacial deposits, which are present at depths of between approximately 1.10 m and 2.50 m, below current ground level. The conjectured depth to a potentially suitable foundation bearing horizon is indicated on Drawing No. P23/075/SI/R/F/11.

12.2.3 If, as anticipated given the nature of the proposed development, a greater bearing capacity is required, piling to the underlying competent rock strata will likely be required. Rockhead was definitively encountered beneath the site at varying depths of between 15.50 m and 16.00 m, below current ground level, and should provide a presumed bearing value of 3.0 MPa. However, the design of piles and method of piling should be developed in consultation with a specialist piling contractor, who will be able to devise the most economical and practical measures to be adopted. Once the development layout and site levels have been finalised, we would advise that discussions are held with us to ensure the pertinence of our recommendations.

12.3 Excavations

12.3.1 The exploratory holes indicate that the natural deposits are likely to be relatively stable in the short term in open excavations. However, all excavations below 2.00 m depth requiring manned entry should be battered well back or provided with close/continuous support. The exploratory holes also indicate that groundwater entry may occur at shallow depths, especially at the interface between the made ground and underlying cohesive soils.

12.3.2 If significant regrading of the site is to be undertaken, we should be consulted to ensure the continuing accuracy of our recommendations.



NOTES

- Approximate site boundary
- BH01 to BH03 — Cable-percussive boreholes sunk by Phoenix Drilling Ltd, on behalf of Mason Evans (April 2023)
  - Rotary follow-on-element, also carried out by Phoenix Drilling Ltd undertaken in boreholes BH01B, BH02A and BH03 (referenced as RBH01B, RBH02A and RBH03)
- BH04 to BH06 — Terrier boreholes sunk by Phoenix Drilling Ltd, on behalf of Mason Evans (April 2023)
- Gas/groundwater monitoring well installation
- (1.00m) — Conjectured depth to a potential foundation bearing horizon with an allowable bearing capacity of approximately 125 kPa (m)

REV	DATE	DETAILS

BALFOUR BEATTY INVESTMENTS Ltd
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PROJECT TITLE

CAUSEWAYSIDE EDINBURGH

DRAWING TITLE

CONJECTURED DEPTH TO A POTENTIAL FOUNDATION BEARING HORIZON

DRAWN BY	CHK'D BY	APP'D BY	DATE	SCALES
TR	LR	LR	21.07.23	1:250 @ A3

PROJECT No.	DRAWING No.	REVISION
P23/075	P23/075/SI/R/F/11	

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12.4 **Obstructions**

12.4.1 As previously discussed, a hard obstruction was encountered within BH01, BH01A, BH01B, BH02, BH02A, BH03, BH04A, BH04B, BH05 and BH06 at varying depths of between 0.30 mbgl and 4.00 mbgl and a concrete obstruction was encountered within BH04 at 0.65 mbgl. These obstructions are likely to impact upon the proposed foundation solution and will therefore require to be grubbed out/removed prior to development. Given the previous development of the site, the possibility that further obstructions could also be present beneath the site cannot be discounted.

13.0 ROAD CONSTRUCTION

13.1 General

13.1.1 The proposed development layout was not available at the time of reporting, but it was anticipated that there would be a new road access off of Causewayside, which extends parallel to the eastern site boundary.

13.2 Ground Conditions

13.2.1 Any proposed road within the proposed development site will be underlain by made ground deposits, which have not been subject to controlled compaction. Therefore, assuming that site levels are to be retained, any roads could be placed on the made ground soils and therefore a full capping layer will be required if roads are to be constructed to an adoptable standard.

13.2.2 Alternatively, if any roads are placed on the underlying stiff or stronger natural cohesive soils, it is considered likely that a lesser capping thickness would be required. However, this would require to be confirmed by California Bearing Ratio (CBR) testing.

13.3 Chemical Contamination

13.3.1 An ACM was identified within BH03 at 0.50 mbgl. Therefore, this *'hotspot'* of contaminated material must be removed from site prior to development by a licensed contractor and would need to be disposed of at a *'hazardous'* waste facility.

13.3.2 Following removal of the above *'hotspot'*, localised asbestos (at concentrations of <0.1% vol.) still remains with the shallow made ground soils and therefore appropriate health and safety measures are advised during construction works. Note that further detailed remedial recommendations in relation to the handling of asbestos will be provided within a detailed *'Remediation Method Statement*', which should be prepared prior to development.

13.4 Gas Emissions

13.4.1 No significant risk to road construction or maintenance personnel is envisaged due to ground gases. However, any trenches deeper than 1.20 m should be risk assessed by the contractor.

13.5 Mining and Mineral Stability

13.5.1 Based on the results of our researches, we have concluded that the site is not within area affected by shallow mineworkings and do not envisage any constraints to road construction in relation to the aspect of the ground conditions.

14.0 INVASIVE PLANT SURVEY

14.1 General

14.1.1 No obvious evidence of *'Commonly Known Species Controlled by Legislation'* (i.e. Giant Hogweed, Japanese Knotweed of Himalayan Balsam) or *'Problematic Non-Legislated Species'* (i.e. Horsetail) were identified during our recent 2022 site walkover survey (included in Appendix 1).

14.1.2 However, we would recommend that an invasive plat survey is undertaken at the site, preferably during the new growing season, by a specialist contractor. Kleerkut, or another invasive plant specialist, would be able to assist in the matter.

15.0 CONCLUSIONS AND RECOMMENDATIONS

15.1 General

15.1.1 Mason Evans have undertaken intrusive investigations in relation to the proposed redevelopment of an area of land located off of Causewayside, Edinburgh. The purpose of these investigations was to identify ground related risks that have the potential to impact on the proposed development at the site.

15.1.2 The ground conditions encountered during the investigation were generally consistent with the anticipated sequence of strata indicated by the desk study information.

15.2 Contamination and Gas Emissions

15.2.1 The asbestos encountered in BH02 (at 0.50 mbgl) has been subject to analysis that has indicated the quantity of asbestos to be less than 0.1% vol. of the sample. Therefore, we would recommend that this *'hotspot'* is delineated and removed or relocated prior to development. It is anticipated that these works would comprise the removal of a limited volume of material around the affected area and would require to be supervised and validated by a suitably qualified geo-environmental scientist. The excavated soils would then need to be disposed of off-site, or alternatively, the soils could be retained on site and placed beneath less sensitive areas, such as beneath roads and/or areas of hardstanding/building cover. Note that we would not recommend that the asbestos contaminated soils are placed beneath roads due to the potential risk to future maintenance personnel.

15.2.2 Conversely, the asbestos identified in BH03 (at 0.50 mbgl) was recorded to contain visible asbestos, described as *'visible insulation board debris'* (at 0.50 mbgl). Therefore, this *'hotspot*' will require to be delineated in the first instance to confirm its size and extent and then removed by a suitably licensed contractor, with the excavated soils being removed from site so a specialist *'hazardous'* waste facility site.

15.2.3 The asbestos identified in BH03 at 1.00 mbgl was below the 0.1% vol. threshold and was recorded to contain no visible ACM's (asbestos containing materials). Therefore, once the ACM at 0.50 mbgl has been removed, the deeper lying asbestos (at 1.00 m depth) can be retained onsite beneath a less sensitive area or disposed offsite to a suitably licensed waste facility site, as per the *'hotspot'* identified in BH02.

15.2.4 However, please note that half of the site was occupied by building cover during our investigations, and therefore the possibility of encountered further asbestos during a future phase of post-demolition works cannot be discounted.

15.2.5 Chemical analysis and groundwater modelling indicates that the proposed development site was not considered to pose a significant risk to the Water Environment (surface water and groundwater).

15.2.6 Based on the results of the ground gas monitoring undertaken to date and subsequent risk assessment, gas protection measures are not considered necessary within the proposed development. Confirmatory gas monitoring is continuing at 'the site and the results will be reported as a separate addendum letter report upon completion. However, it is assumed that gas protection measures will not be required.

15.2.7 Radon protection measures are also not considered necessary within the proposed development, given that the site is located within a Lower Probability Radon Area (less than 1% of homes are estimated to be at or above the Action Level).

15.3 The Built Environment

15.3.1 We would recommend that a UKWIR assessment is undertaken along the proposed water supply route, once the site layout and levels have been finalised. Note that this assessment would inform the likely water supply pipe selection material.

15.3.2 Concrete class DS-1, AC-1 is considered sufficient to protect buried concrete from pH and sulphate levels in the soils and groundwater.

15.4 Fuel Tanks

15.4.1 Although it is considered unlikely that any underground tanks will be present onsite, especially given the history of the site as a school, vigilance should still be maintained during any future site works for the presence of tanks. If present, the tanks should be removed under our supervision and validation testing undertaken subsequent to removal, to ensure that there has been no localised impact on the soils from fuel leakage/spillage. If such impacts are identified, the soils would likely require to be removed or remediated.

15.5 Foundation Recommendations (relative to existing site levels)

15.5.1 An allowable bearing capacity of approximately 125 kPa is estimated for the underlying stiff or stronger cohesive glacial till deposits, which are present at depths of between approximately 1.10 m and 2.50 m, below current ground level.

15.5.2 If, as anticipated given the nature of the proposed development, a greater bearing capacity is required, piling to the underlying competent rock strata will likely be required. Rockhead was definitively encountered beneath the site at varying depths of between 15.50 m and 16.00 m, below current ground level, and should provide a a presumed bearing value of 3.0 MPa. However, the design of piles and method of piling should be developed in consultation with a specialist piling contractor, who will be able to devise the most economical and practical measures to be adopted.

15.6 Mining

15.6.1 Mining instability constraints are not considered to pose a significant risk to the proposed development site.

15.7 Invasive Plants

15.7.1 No obvious evidence of *'Commonly Known Species Controlled by Legislation'* (i.e. Giant Hogweed, Japanese Knotweed of Himalayan Balsam) or *'Problematic Non-Legislated Species'* (i.e. Horsetail) were identified during our recent 2022 site walkover survey (included in Appendix 1).

15.7.2 However, we would recommend that an invasive plat survey is undertaken at the site, preferably during the new growing season, by a specialist contractor. Kleerkut, or another invasive plant specialist, would be able to assist in the matter.

15.8 **Additional Investigations**

15.8.1 As previously noted, at the time of our investigation, the Scottish Law Commission building (with associated outbuildings) occupied at least half of the site. Therefore, some further limited confirmatory site investigation works, including chemical testing, will be required following demolition of the onsite building.

15.9 **Consultations with Public Authorities**

15.9.1 It should be noted that various local authority departments may become involved in the review of the site conditions, including the issues of contaminated land. While measures proposed are consistent with conventional practice, we would advise that before design works are advanced to any considerable stage appropriate approvals are received from The City of Edinburgh Council. We would be pleased to liaise with the Council's representatives in this regard.

We trust that this will meet with your current requirements. However, should you require any further information, please do not hesitate to contact us.





Lindsay Reid
Associate

Patrick Barry
Managing Director

Appendix I

Site Walkover Survey (March 2023)

WALKOVER SURVEY RECORD

Project Name: Causewayside, Edinburgh **Project Number: P23-075**
Date of Survey: 29.03.23 **Surveyed By: LR**
Weather: Overcast with occasional showers

VICINITY OF THE SITE		DESCRIPTION
Are there any street/house/locality/pub names indicating current or former land use?		The site is currently occupied by the Scottish Law Commission building with associated car parking and garages at the rear.
What are the neighbouring land uses?	NORTH	Tenement residential properties.
	EAST	Causewayside Road with the National Library of Scotland building further beyond.
	SOUTH	Office buildings
	WEST	Residential properties.
Potential off-site receptors		The building onsite is occupied is partially occupied.

ACCESSES

Describe the site accesses - type, width and headroom.		Open access off of Causewayside (busy road). No headroom restrictions noted.
Describe any access difficulties for SI plant		The rigs may have to be offloaded on Causewayside as there is limited access at the front of the building between the building and the pavement. Access must be maintained to both of the fire escapes at all times, as well as to the garages in the north-west of the site.

SITE DESCRIPTION - GENERAL

What is the current land use?		The site is currently occupied by the Scottish Law Commission building with associated car parking and garages at the rear.
What is the topography?		Flat lying
What is the surface cover?		Mixture of building cover and tarmac hardstanding with an area in the north-west of the site surfaced in gravel.
Are there any waterlogged areas?	NO	None noted.
How are the boundaries formed?		***North:*** Brick wall ***South:*** Brick wall ***East:*** Metal posts and chains ***West***: the back of the building forms the boundary.
Does the topography suggest filling or platforming?	NO	None noted.
Are there any subsidence features?	NO	None noted.

EXISTING BUILDINGS

What proportion of the site do the buildings cover?		Approximately 75% of the site is surfaced with existing buildings.
Do the building(s) show any evidence of distress?		None obvious.
Indicate building usage on available site plan.		-
Indicate nature and location of materials in storage.		Could not get a good look around the inside of the building but it was understood to be office space for the Scottish Law Commission.
What processes are evident in the facility?		As above.

TANKS AND WASTE STORAGE

Are there any fuel or chemical storage tanks (surface and underground)? For each tank record whether it is above/under ground, nature of contents, whether full or empty, bunded/unbunded/leaking bund, presence of staining. Mark locations on plan.	NO	None noted.
Is there any evidence of waste storage or disposal?	NO	None noted.
Are there any chemical drums or other containers?	NO	None noted.
Are there any discharges to surface water?	NO	None noted.

HYDROLOGY

Describe any groundwater sources - including flow rate.		N/A
Record positions all springs, ponds and other water on site.		N/A

PUBLIC UTILITIES

Are there any overhead cables - indicate type and location?	NO	None noted.
Are there any manholes - describe?	YES	Various manhole covers noted, both square and rectangular (one identified as BT). A GPR survey had also ben carried out recently as a lot of the service locations had been marked out with spray paint.
Are there other indications of utilities?	YES	Lots of manhole covers for drainage were also noted.
Are there any electricity transformers	YES	An electrical substation was noted to be present within the central southern site area – a scar in the tarmac was also noted to extend from the pavement in the east up to the substation.

HAZARDS

Describe any obvious public health hazards.		The areas where we will be working at the front of the building is quite close to the pavement – consider heras fencing?.

SPILLAGES AND CONTAMINATION

Are there any indications of oil or other spillages?	NO	None noted.
Is their evidence of contaminated soils?	NO	None noted.
Is there evidence of distress to vegetation?	NO	None noted.
Describe constituents of any flytipping.		N/A
Is there surface evidence of asbestos contaminated soil?	NO	None noted.
Are there any noxious smells?	NO	None noted.

GEOLOGY

Soil and rock – record and describe any exposed soils or rocks that are present.		N/A

MINING AND QUARRYING

Are there any signs of mineral extraction in the area, such as old mine buildings, derelict or hummocky land, surface depressions, evidence of infilling or spoil heaps.	NO	None noted.
Is their evidence of any quarrying?	NO	None noted.

SLOPE STABILITY

Are there any risks of slope instability?	NO	None noted.
Is there evidence of previous land slipping?	NO	None noted.

INVASIVE PLANTS

Are there any obvious invasive plants?	NO	None obvious. However, a survey of the site should be undertaken by specialist invasive plant contractors.

Site Photographs



Photograph 1: View looking south-west across the southern half of the site (from the road)



Photograph 2: View looking north-north-west across the northern half of the site (from the road)



Photograph 3: View looking north-west along Causewayside (road)



Photograph 4: View looking north-west to the fire escape in the southern half of the site



Photograph 5: View looking east from the back of the site towards Causewayside



Photograph 6: View looking south-east towards the building onsite



Photograph 7: View looking west towards the garages at the back of the site



Photograph 8: Shows scarring within the tarmac within the southern site area (likely service for the substation)

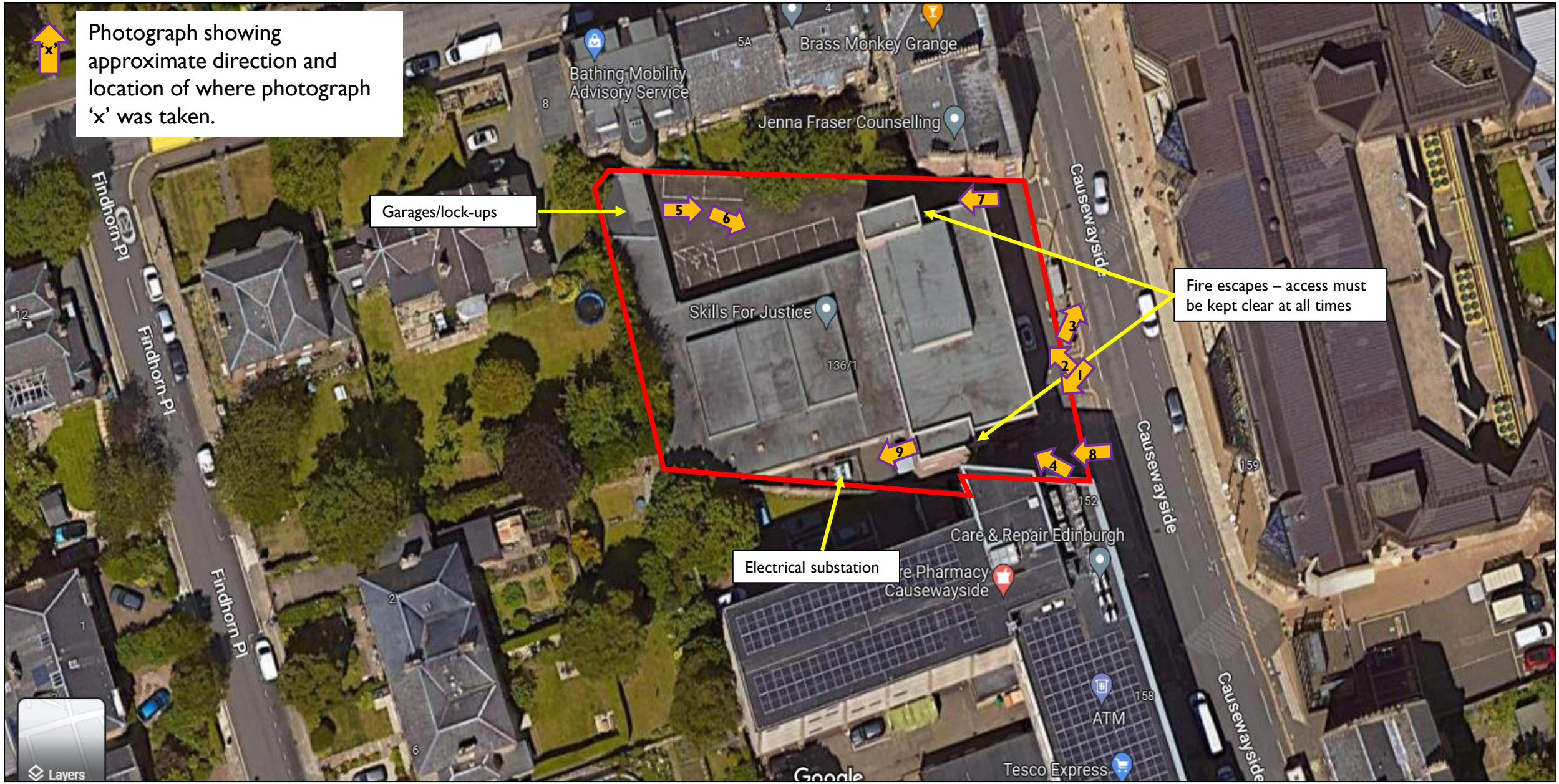


Photograph 9: View looking south-west towards the electrical substation within the south-western site area



Photograph 10: Examples of manhole covers/drainage noted onsite

Causewayside, Edinburgh – Site Walkover Survey Map



Appendix 2

Envirocheck Report

Envirocheck® Report:

Datasheet

Order Details:

Order Number:
309237239_1_1

Customer Reference:
P23-075

National Grid Reference:
326330, 672120

Slice:
A

Site Area (Ha):
0.21

Search Buffer (m):
1000

Site Details:

Scottish Law Commission, 140
Causewayside
EDINBURGH
EH9 1PR

Client Details:

Ms P Morton
Mason Evans Partnership
The Piazza
95 Morrison Street
(office side door on Dalentober St)
Glasgow
G5 8BE

Contents

Report Section	Page Number
Summary	-
Agency & Hydrological	1
Waste	7
Hazardous Substances	-
Geological	8
Industrial Land Use	10
Sensitive Land Use	20
Data Currency	21
Data Suppliers	25
Useful Contacts	26

Introduction

The Environment Act 1995 has made site sensitivity a key issue, as the legislation pays as much attention to the pathways by which contamination could spread, and to the vulnerable targets of contamination, as it does the potential sources of contamination.
For this reason, Landmark's Site Sensitivity maps and Datasheet(s) place great emphasis on statutory data provided by the Environment Agency/Natural Resources Wales and the Scottish Environment Protection Agency; it also incorporates data from Natural England (and the Scottish and Welsh equivalents) and Local Authorities; and highlights hydrogeological features required by environmental and geotechnical consultants. It does not include any information concerning past uses of land. The datasheet is produced by querying the Landmark database to a distance defined by the client from a site boundary provided by the client.
In this datasheet the National Grid References (NGRs) are rounded to the nearest 10m in accordance with Landmark's agreements with a number of Data Suppliers.

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Report Version v53.0

Summary

Data Type	Page Number	On Site	0 to 250m	251 to 500m	501 to 1000m (*up to 2000m)
Agency & Hydrological					
BGS Groundwater Flooding Susceptibility	pg 1	Yes	Yes	Yes	n/a
Contaminated Land Register Entries and Notices					
Discharge Consents					
Prosecutions Relating to Controlled Waters			n/a	n/a	n/a
Enforcement and Prohibition Notices					
Integrated Pollution Controls					
Integrated Pollution Prevention And Control					
Local Authority Integrated Pollution Prevention And Control					
Local Authority Pollution Prevention and Controls	pg 4		1		1
Local Authority Pollution Prevention and Control Enforcements					
Nearest Surface Water Feature	pg 4			Yes	
Pollution Incidents to Controlled Waters					
Prosecutions Relating to Authorised Processes					
Registered Radioactive Substances	pg 4				7
River Quality					
Substantiated Pollution Incident Register					
Water Abstractions					
Water Industry Act Referrals					
Groundwater Vulnerability	pg 5	Yes	n/a	n/a	n/a
Drift Deposits	pg 5	1	n/a	n/a	n/a
Source Protection Zones					
River Flood Data (Scotland)				n/a	n/a
OS Water Network Lines	pg 6				8
Waste					
BGS Recorded Landfill Sites					
Integrated Pollution Control Registered Waste Sites					
Licensed Waste Management Facilities (Landfill Boundaries)					
Licensed Waste Management Facilities (Locations)					
Local Authority Landfill Coverage	pg 7	1	n/a	n/a	n/a
Local Authority Recorded Landfill Sites					
Registered Landfill Sites					
Registered Waste Transfer Sites					
Registered Waste Treatment or Disposal Sites					

Summary

Data Type	Page Number	On Site	0 to 250m	251 to 500m	501 to 1000m (*up to 2000m)
Hazardous Substances					
Control of Major Accident Hazards Sites (COMAH)					
Explosive Sites					
Notification of Installations Handling Hazardous Substances (NIHHS)					
Planning Hazardous Substance Consents					
Planning Hazardous Substance Enforcements					
Geological					
BGS 1:625,000 Solid Geology	pg 8	Yes	n/a	n/a	n/a
BGS Recorded Mineral Sites	pg 8			1	6
CBSCB Compensation District			n/a	n/a	n/a
Coal Mining Affected Areas			n/a	n/a	n/a
Mining Instability			n/a	n/a	n/a
Man-Made Mining Cavities					
Natural Cavities					
Non Coal Mining Areas of Great Britain	pg 9	Yes	Yes	n/a	n/a
Potential for Collapsible Ground Stability Hazards	pg 9	Yes		n/a	n/a
Potential for Compressible Ground Stability Hazards				n/a	n/a
Potential for Ground Dissolution Stability Hazards				n/a	n/a
Potential for Landslide Ground Stability Hazards	pg 9	Yes		n/a	n/a
Potential for Running Sand Ground Stability Hazards	pg 9	Yes		n/a	n/a
Potential for Shrinking or Swelling Clay Ground Stability Hazards	pg 9	Yes		n/a	n/a
Radon Potential - Radon Affected Areas			n/a	n/a	n/a
Radon Potential - Radon Protection Measures			n/a	n/a	n/a
Industrial Land Use					
Contemporary Trade Directory Entries	pg 10		22	22	65
Fuel Station Entries	pg 19		2		
Gas Pipelines					

Summary

Data Type	Page Number	On Site	0 to 250m	251 to 500m	501 to 1000m (*up to 2000m)
Sensitive Land Use					
Ancient Woodland					
Areas of Adopted Green Belt	pg 20				1
Areas of Unadopted Green Belt					
Environmentally Sensitive Areas					
Forest Parks					
Local Nature Reserves					
Marine Nature Reserves					
National Nature Reserves					
National Parks					
National Scenic Areas					
Nitrate Sensitive Areas					
Nitrate Vulnerable Zones					
Ramsar Sites					
Sites of Special Scientific Interest	pg 20				1
Special Areas of Conservation					
Special Protection Areas					
World Heritage Sites	pg 20				1

Map ID	Details	Quadrant Reference (Compass Direction)	Estimated Distance From Site	Contact	NGR
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A13NE (SE)	0	1	326334 672122
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A13NW (NW)	10	1	326300 672150
	BGS Groundwater Flooding Susceptibility Flooding Type: Limited Potential for Groundwater Flooding to Occur	A13NE (NE)	44	1	326400 672150
	BGS Groundwater Flooding Susceptibility Flooding Type: Limited Potential for Groundwater Flooding to Occur	A13SW (SW)	55	1	326300 672050
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A13NW (NW)	79	1	326250 672200
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A13SE (SE)	107	1	326400 672000
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A13NW (N)	110	1	326300 672250
	BGS Groundwater Flooding Susceptibility Flooding Type: Limited Potential for Groundwater Flooding to Occur	A13NE (NE)	111	1	326450 672200
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A13NW (NW)	118	1	326250 672250
	BGS Groundwater Flooding Susceptibility Flooding Type: Limited Potential for Groundwater Flooding to Occur	A13SW (SW)	121	1	326250 672000
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A13SE (E)	135	1	326500 672100
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A13NW (W)	152	1	326150 672150
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A13NE (N)	160	1	326350 672300
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A13NE (N)	160	1	326334 672300
	BGS Groundwater Flooding Susceptibility Flooding Type: Limited Potential for Groundwater Flooding to Occur	A13NE (NE)	180	1	326500 672250
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A13SE (E)	185	1	326550 672100
	BGS Groundwater Flooding Susceptibility Flooding Type: Limited Potential for Groundwater Flooding to Occur	A13NE (E)	185	1	326550 672122
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding to Occur at Surface	A13NW (NW)	189	1	326200 672300
	BGS Groundwater Flooding Susceptibility Flooding Type: Limited Potential for Groundwater Flooding to Occur	A13NE (E)	191	1	326550 672150
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A13SE (S)	201	1	326350 671900
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A13SE (S)	202	1	326334 671900
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A13SE (S)	204	1	326400 671900

Map ID	Details	Quadrant Reference (Compass Direction)	Estimated Distance From Site	Contact	NGR
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A13NE (N)	210	1	326334 672350
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding to Occur at Surface	A13NW (NW)	220	1	326150 672300
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding to Occur at Surface	A13NW (NW)	229	1	326100 672250
	BGS Groundwater Flooding Susceptibility Flooding Type: Limited Potential for Groundwater Flooding to Occur	A13SE (E)	235	1	326600 672100
	BGS Groundwater Flooding Susceptibility Flooding Type: Limited Potential for Groundwater Flooding to Occur	A13NE (E)	235	1	326600 672122
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A13SE (SE)	239	1	326550 671950
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A13SE (E)	240	1	326600 672050
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A13SE (S)	242	1	326350 671850
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A13SE (S)	252	1	326334 671850
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A13NE (N)	260	1	326350 672400
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A13SW (S)	261	1	326250 671850
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding to Occur at Surface	A13NW (N)	265	1	326250 672400
	BGS Groundwater Flooding Susceptibility Flooding Type: Limited Potential for Groundwater Flooding to Occur	A13SW (SW)	281	1	326050 672000
	BGS Groundwater Flooding Susceptibility Flooding Type: Limited Potential for Groundwater Flooding to Occur	A13SE (E)	290	1	326650 672050
	BGS Groundwater Flooding Susceptibility Flooding Type: Limited Potential for Groundwater Flooding to Occur	A13SW (SW)	303	1	326050 671950
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A13SW (S)	304	1	326300 671800
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A13SW (S)	310	1	326250 671800
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding to Occur at Surface	A13NW (N)	310	1	326300 672450
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A13NE (N)	313	1	326400 672450
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding to Occur at Surface	A13SE (S)	313	1	326450 671800
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding to Occur at Surface	A13NW (N)	314	1	326250 672450
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A13SE (SE)	323	1	326650 671950

Map ID	Details	Quadrant Reference (Compass Direction)	Estimated Distance From Site	Contact	NGR
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A13SW (SW)	324	1	326200 671800
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A13NW (N)	326	1	326200 672450
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding to Occur at Surface	A13NW (NW)	344	1	326150 672450
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A14SW (E)	350	1	326700 672000
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding to Occur at Surface	A18SW (N)	360	1	326300 672500
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A18SE (N)	360	1	326350 672500
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding to Occur at Surface	A18SW (N)	363	1	326250 672500
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A14SW (SE)	367	1	326700 671950
	BGS Groundwater Flooding Susceptibility Flooding Type: Limited Potential for Groundwater Flooding to Occur	A13SW (SW)	370	1	326100 671800
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A8NW (S)	371	1	326200 671750
	BGS Groundwater Flooding Susceptibility Flooding Type: Limited Potential for Groundwater Flooding to Occur	A18SE (N)	372	1	326450 672500
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding to Occur at Surface	A13SE (SE)	380	1	326650 671850
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A18SW (NW)	390	1	326150 672500
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding to Occur at Surface	A13NW (NW)	399	1	326050 672450
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding to Occur at Surface	A18SW (N)	410	1	326300 672550
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A8NW (SW)	412	1	326100 671750
	BGS Groundwater Flooding Susceptibility Flooding Type: Limited Potential for Groundwater Flooding to Occur	A18SE (N)	421	1	326450 672550
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding to Occur at Surface	A18SW (N)	422	1	326200 672550
	BGS Groundwater Flooding Susceptibility Flooding Type: Limited Potential for Groundwater Flooding to Occur	A14SW (E)	438	1	326800 672050
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A8NW (SW)	440	1	326050 671750
	BGS Groundwater Flooding Susceptibility Flooding Type: Limited Potential for Groundwater Flooding to Occur	A14SW (E)	447	1	326800 672000
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding to Occur at Surface	A18SW (N)	460	1	326300 672600

Map ID	Details	Quadrant Reference (Compass Direction)	Estimated Distance From Site	Contact	NGR
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A18SE (N)	460	1	326350 672600
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A14SW (E)	460	1	326800 671950
	BGS Groundwater Flooding Susceptibility Flooding Type: Limited Potential for Groundwater Flooding to Occur	A18SE (N)	462	1	326400 672600
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A12NE (NW)	478	1	325900 672400
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A14SW (SE)	479	1	326800 671900
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A8NW (SW)	481	1	326050 671700
	BGS Groundwater Flooding Susceptibility Flooding Type: Potential for Groundwater Flooding of Property Situated Below Ground Level	A12NE (NW)	498	1	325850 672350
1	**Local Authority Pollution Prevention and Controls** Name: Kennet Ltd Location: 2-40 Ratcliffe Terrace, EDINBURGH, Midlothian, EH9 1SS Authority: Scottish Environment Protection Agency, East Region Permit Reference: Apc/E/20173 Dated: 27th August 1998 Process Type: Local Authority Air Pollution Control Description: PG1/14 Petrol filling station **Status: Not Supplied** Positional Accuracy: Manually positioned to the road within the address or location	A13SE (S)	204	2	326400 671900
2	**Local Authority Pollution Prevention and Controls** Name: Truetest Motors Ltd Location: 3a St. Albans Road, EDINBURGH, Midlothian, EH9 2PA Authority: Scottish Environment Protection Agency, East Region Permit Reference: Apc/E/20107 Dated: 4th March 1998 Process Type: Local Authority Air Pollution Control Description: PG1/1Waste oil burners, less than 0.4MW net rated thermal input **Status: Not Supplied** Positional Accuracy: Manually positioned to the address or location	A8NE (S)	504	2	326508 671618
	Nearest Surface Water Feature	A14NW (NE)	467	-	326709 672447
3	**Registered Radioactive Substances** Name: Lothian Health Board Location: Royal Hospital For Sick Children, Sciennes Road, EDINBURGH, Midlothian, EH9 1LF Authority: Scottish Environment Protection Agency, Head Office Permit Reference: IPB/3/4/11/6/1 Dated: 31st October 1995 Process Type: Authorisation under S13 or S14 RSA for the accumulation or disposal of Radioactive waste (was RSA60 S6/S7) Description: Authorisation under RSA **Status: Not Given** Positional Accuracy: Unknown	A12NE (NW)	510	3	325834 672345
3	**Registered Radioactive Substances** Name: University Of Edinburgh Location: 20 Sylvan Place, EDINBURGH, Midlothian, EH9 1UW Authority: Scottish Environment Protection Agency, Head Office Permit Reference: IPB/3/5/3/022 Dated: 27th December 1994 Process Type: Registration under S7 of RSA to keep or accumulate radioactive material Description: Use of one or more open source(s) **Status: Not Given** Positional Accuracy: Manually positioned to the address or location	A12NE (NW)	510	3	325837 672351

Map ID	Details	Quadrant Reference (Compass Direction)	Estimated Distance From Site	Contact	NGR
3	**Registered Radioactive Substances** Name: Lothian Health Board Location: Royal Hospital For Sick Children, Sciennes Road, EDINBURGH, Midlothian, EH9 1LF Authority: Scottish Environment Protection Agency, East Region Permit Reference: IPB/3/4/11/6/1 Dated: 29th August 1995 Process Type: Registration under S7 RSA for the keeping and use of Radioactive materials (was RSA60 S1) Description: Registration under S7 RSA of 1 or more open sources **Status: Not Given** Positional Accuracy: Unknown	A12NE (NW)	512	2	325834 672350
3	**Registered Radioactive Substances** Name: University Of Edinburgh Location: Dept Of Child Life & Health, University Of Edinburgh, Sylvan Place, EDINBURGH, Midlothian, EH9 1UW Authority: Scottish Environment Protection Agency, Head Office Permit Reference: IPB/3/5/3/022 Dated: 28th April 1995 Process Type: Authorisation under RSA (no specific reference) Description: Authorisation under RSA **Status: Not Given** Positional Accuracy: Manually positioned to the address or location	A12NE (NW)	515	3	325833 672354
4	**Registered Radioactive Substances** Name: University Of Edinburgh Location: Dept Of Pharmacology & The Mrc, Brain Metabolism Unit, 1 George Square, EDINBURGH, Midlothian, EH8 9JZ Authority: Scottish Environment Protection Agency, Head Office Permit Reference: IPB/3/5/3/034 Dated: 14th March 1995 Process Type: Registration under S7 RSA for the keeping and use of Radioactive materials (was RSA60 S1) Description: Registration under S7 RSA of 1 or more open sources **Status: Not Given** Positional Accuracy: Unknown	A17NE (NW)	977	3	325895 673029
4	**Registered Radioactive Substances** Name: University Of Edinburgh Location: Dept Of Pharmacology & The Mrc, Brain Metabolism Unit, 1 George Square, EDINBURGH, Midlothian, EH8 9JZ Authority: Scottish Environment Protection Agency, East Region Permit Reference: IPB/3/5/3/034 Dated: 8th June 1994 Process Type: Authorisation under S14 RSA for the accumulation of Radioactive waste (was RSA60 S6) Description: Authorisation under RSA **Status: Not Given** Positional Accuracy: Unknown	A17NE (NW)	979	2	325890 673029
4	**Registered Radioactive Substances** Name: University Of Edinburgh Location: Dept Of Pharmacology & The Mrc, Brain Metabolism Unit, 1 George Square, EDINBURGH, Midlothian, EH8 9JZ Authority: Scottish Environment Protection Agency, East Region Permit Reference: IPB/3/5/3/034 Dated: 8th June 1994 Process Type: Authorisation under S13 or S14 RSA for the accumulation or disposal of Radioactive waste (was RSA60 S6/S7) Description: Authorisation under RSA **Status: Not Given** Positional Accuracy: Unknown	A17NE (NW)	984	2	325890 673034
	Groundwater Vulnerability Geological Classification: Minor or Moderately Permeable Aquifer - Fractured or potentially fractured rocks which do not have a high primary permeability or other formations of variable permeability Soil Classification: Not classified Map Sheet: Map of Scotland Scale: 1:625,000	A13NE (SE)	0	3	326334 672122
	Drift Deposits Drift Deposit: Low permeability drift deposits which include till, head, peat, lacustrine deposits, clay-with-flints and brick earths Map Sheet: Map of Scotland Scale: 1:625,000	A13NE (SE)	0	3	326334 672122
	River Flood Data (Scotland) None				

Map ID	Details	Quadrant Reference (Compass Direction)	Estimated Distance From Site	Contact	NGR
5	**OS Water Network Lines** Watercourse Form: Inland river Watercourse Length: 517.7 Watercourse Level: Underground Permanent: True Watercourse Name: Pow Burn Catchment Name: Edinburgh Coastal Primacy: 1	A8SE (S)	667	4	326429 671438
6	**OS Water Network Lines** Watercourse Form: Inland river Watercourse Length: 21.9 Watercourse Level: On ground surface Permanent: True Watercourse Name: Pow Burn Catchment Name: Edinburgh Coastal Primacy: 1	A8SE (S)	696	4	326506 671420
7	**OS Water Network Lines** Watercourse Form: Inland river Watercourse Length: 57.6 Watercourse Level: Underground Permanent: True Watercourse Name: Pow Burn Catchment Name: Edinburgh Coastal Primacy: 1	A8SE (S)	706	4	326528 671415
8	**OS Water Network Lines** Watercourse Form: Inland river Watercourse Length: 576.1 Watercourse Level: Underground Permanent: True Watercourse Name: Pow Burn Catchment Name: Edinburgh Coastal Primacy: 1	A8SE (SE)	730	4	326656 671431
9	**OS Water Network Lines** Watercourse Form: Inland river Watercourse Length: 5.7 Watercourse Level: On ground surface Permanent: True Watercourse Name: Pow Burn Catchment Name: Edinburgh Coastal Primacy: 1	A8SE (S)	733	4	326584 671402
10	**OS Water Network Lines** Watercourse Form: Inland river Watercourse Length: 278.5 Watercourse Level: Underground Permanent: True Watercourse Name: Not Supplied Catchment Name: Edinburgh Coastal Primacy: 1	A19SE (NE)	855	4	327113 672538
11	**OS Water Network Lines** Watercourse Form: Inland river Watercourse Length: 520.5 Watercourse Level: On ground surface Permanent: True Watercourse Name: Pow Burn Catchment Name: Edinburgh Coastal Primacy: 1	A9NE (SE)	936	4	327136 671569
12	**OS Water Network Lines** Watercourse Form: Inland river Watercourse Length: 101.8 Watercourse Level: On ground surface Permanent: True Watercourse Name: Pow Burn Catchment Name: Edinburgh Coastal Primacy: 1	A8SW (S)	962	4	326103 671165

Map ID	Details	Quadrant Reference (Compass Direction)	Estimated Distance From Site	Contact	NGR
	Local Authority Landfill Coverage Name: City of Edinburgh Council - Has supplied landfill data		0	5	326334 672122

Map ID	Details	Quadrant Reference (Compass Direction)	Estimated Distance From Site	Contact	NGR
	BGS 1:625,000 Solid Geology Description: Inverclyde Group	A13NE (SE)	0	1	326334 672122
13	**BGS Recorded Mineral Sites** Site Name: Grange Quarry Location: Sciennes, Edinburgh, Midlothian Source: British Geological Survey, National Geoscience Information Service Reference: 62844 Type: Opencast **Status: Ceased** Operator: Unknown Operator Operator Location: Not Supplied Periodic Type: Devonian and Carboniferous Geology: Kinnesswood Formation Commodity: Sandstone Positional Accuracy: Located by supplier to within 10m	A12SE (SW)	400	1	325940 671955
14	**BGS Recorded Mineral Sites** Site Name: Beaufort Road Quarry Location: Sciennes, Edinburgh, Midlothian Source: British Geological Survey, National Geoscience Information Service Reference: 127678 Type: Opencast **Status: Ceased** Operator: Unknown Operator Operator Location: Not Supplied Periodic Type: Devonian and Carboniferous Geology: Kinnesswood Formation Commodity: Sandstone Positional Accuracy: Located by supplier to within 10m	A12SE (W)	655	1	325690 671897
15	**BGS Recorded Mineral Sites** Site Name: St Leonard'S Location: St Leonard'S, Edinburgh, Midlothian Source: British Geological Survey, National Geoscience Information Service Reference: 231298 Type: Opencast **Status: Ceased** Operator: Unknown Operator Operator Location: Not Supplied Periodic Type: Carboniferous Geology: Ballagan Formation Commodity: Sandstone Positional Accuracy: Located by supplier to within 10m	A18NE (N)	852	1	326492 672981
15	**BGS Recorded Mineral Sites** Site Name: St Leonard'S Location: St Leonard'S, Edinburgh, Midlothian Source: British Geological Survey, National Geoscience Information Service Reference: 231297 Type: Opencast **Status: Ceased** Operator: Unknown Operator Operator Location: Not Supplied Periodic Type: Carboniferous Geology: Ballagan Formation Commodity: Sandstone Positional Accuracy: Located by supplier to within 10m	A18NE (N)	890	1	326496 673019
16	**BGS Recorded Mineral Sites** Site Name: Quarry Close Location: South Side, Edinburgh, Midlothian Source: British Geological Survey, National Geoscience Information Service Reference: 6918 Type: Opencast **Status: Ceased** Operator: Unknown Operator Operator Location: Not Supplied Periodic Type: Carboniferous Geology: Ballagan Formation (Cementstone Group) Commodity: Sandstone Positional Accuracy: Located by supplier to within 100m	A18NW (N)	883	1	326100 673000

Map ID	Details	Quadrant Reference (Compass Direction)	Estimated Distance From Site	Contact	NGR
17	**BGS Recorded Mineral Sites** Site Name: Blackford Location: Morningside, Edinburgh, Midlothian Source: British Geological Survey, National Geoscience Information Service Reference: 231311 Type: Opencast **Status: Ceased** Operator: Unknown Operator Operator Location: Not Supplied Periodic Type: Silurian Geology: Blackford Hill Member Commodity: Igneous and Metamorphic Rock Positional Accuracy: Located by supplier to within 10m	A8SW (S)	888	1	326142 671232
18	**BGS Recorded Mineral Sites** Site Name: Burgh Muir Location: Bruntsfield, Edinburgh, Midlothian Source: British Geological Survey, National Geoscience Information Service Reference: 127680 Type: Opencast **Status: Ceased** Operator: Unknown Operator Operator Location: Not Supplied Periodic Type: Devonian and Carboniferous Geology: Kinnesswood Formation Commodity: Sandstone Positional Accuracy: Located by supplier to within 10m	A12NW (W)	933	1	325409 672414
	Coal Mining Affected Areas In an area that might not be affected by coal mining				
	Non Coal Mining Areas of Great Britain Risk: Highly Unlikely Source: British Geological Survey, National Geoscience Information Service	A13NE (SE)	0	1	326334 672122
	Non Coal Mining Areas of Great Britain Risk: Rare Source: British Geological Survey, National Geoscience Information Service	A13SW (SW)	25	1	326289 672091
	Potential for Collapsible Ground Stability Hazards Hazard Potential: Very Low Source: British Geological Survey, National Geoscience Information Service	A13NE (SE)	0	1	326334 672122
	Potential for Compressible Ground Stability Hazards Hazard Potential: No Hazard Source: British Geological Survey, National Geoscience Information Service	A13NE (SE)	0	1	326334 672122
	Potential for Ground Dissolution Stability Hazards Hazard Potential: No Hazard Source: British Geological Survey, National Geoscience Information Service	A13NE (SE)	0	1	326334 672122
	Potential for Landslide Ground Stability Hazards Hazard Potential: Very Low Source: British Geological Survey, National Geoscience Information Service	A13NE (SE)	0	1	326334 672122
	Potential for Running Sand Ground Stability Hazards Hazard Potential: Very Low Source: British Geological Survey, National Geoscience Information Service	A13NE (SE)	0	1	326334 672122
	Potential for Shrinking or Swelling Clay Ground Stability Hazards Hazard Potential: Very Low Source: British Geological Survey, National Geoscience Information Service	A13NE (SE)	0	1	326334 672122
	Radon Potential - Radon Affected Areas Affected Area: The property is in a Lower probability radon area (less than 1% of homes are estimated to be at or above the Action Level). Source: British Geological Survey, National Geoscience Information Service	A13NE (SE)	0	1	326334 672122
	Radon Potential - Radon Protection Measures Protection Measure: No radon protective measures are necessary in the construction of new dwellings or extensions Source: British Geological Survey, National Geoscience Information Service	A13NE (SE)	0	1	326334 672122

Map ID	Details	Quadrant Reference (Compass Direction)	Estimated Distance From Site	Contact	NGR
19	**Contemporary Trade Directory Entries** Name: Bath Lifts At Home Location: Grange Rd, Edinburgh, Midlothian, EH9 1UH Classification: Stairlifts - Manufacturers & Installers **Status: Inactive** Positional Accuracy: Manually positioned to the road within the address or location	A13NW (W)	48	-	326254 672149
20	**Contemporary Trade Directory Entries** Name: Caledonian Paint Co Location: 173, Causewayside, Edinburgh, EH9 1PH Classification: Painting & Decorating Supplies **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A13SE (SE)	54	-	326401 672061
20	**Contemporary Trade Directory Entries** Name: Foam Centre Group Location: 176, Causewayside, Edinburgh, EH9 1PN Classification: Bed & Mattress Manufacturers **Status: Active** Positional Accuracy: Automatically positioned to the address	A13SE (SE)	65	-	326379 672038
20	**Contemporary Trade Directory Entries** Name: Foam Centre Location: 176, Causewayside, Edinburgh, EH9 1PN Classification: Foam Products - Rubber & Plastics **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A13SE (SE)	65	-	326379 672038
21	**Contemporary Trade Directory Entries** Name: Frank Mcluskie Fine Art Restoration Location: 108c, Causewayside, Edinburgh, EH9 1PU Classification: Art Restoration & Picture Cleaning **Status: Active** Positional Accuracy: Automatically positioned to the address	A13NW (N)	98	-	326322 672238
21	**Contemporary Trade Directory Entries** Name: Carson Furniture Restorers Location: C, 108, Causewayside, Edinburgh, EH9 1PU Classification: Furniture - Repairing & Restoring **Status: Active** Positional Accuracy: Automatically positioned to the address	A13NW (N)	99	-	326322 672238
21	**Contemporary Trade Directory Entries** Name: South Side Centre Location: 86-92 Causewayside, Edinburgh, Midlothian, EH9 1PY Classification: Catering Equipment **Status: Active** Positional Accuracy: Automatically positioned to the address	A13NW (N)	120	-	326307 672260
21	**Contemporary Trade Directory Entries** Name: Mark Turnbull Jewellery Location: 93, Causewayside, Edinburgh, EH9 1QG Classification: Jewellery Manufacturers & Repairers **Status: Active** Positional Accuracy: Automatically positioned to the address	A13NE (N)	120	-	326344 672259
22	**Contemporary Trade Directory Entries** Name: Interprint Ltd Location: 210, Causewayside, Edinburgh, EH9 1PN Classification: Printers **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A13SE (SE)	124	-	326401 671983
22	**Contemporary Trade Directory Entries** Name: Home Sound Location: 246, Causewayside, Edinburgh, EH9 1UU Classification: Cinema Equipment **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A13SE (S)	156	-	326374 671946
22	**Contemporary Trade Directory Entries** Name: Tynecastle Printing Co Ltd Location: 246, Causewayside, Edinburgh, EH9 1UU Classification: Printers **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A13SE (S)	156	-	326374 671946
23	**Contemporary Trade Directory Entries** Name: Art Source Location: 64, Newington Road, Edinburgh, EH9 1QN Classification: Picture & Picture Frame Renovating & Restoring **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A13NE (NE)	174	-	326467 672276

Map ID	Details	Quadrant Reference (Compass Direction)	Estimated Distance From Site	Contact	NGR
23	**Contemporary Trade Directory Entries** Name: Johnson Cleaners (Uk) Ltd Location: 65, Newington Road, Edinburgh, EH9 1QW Classification: Dry Cleaners **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A13NE (NE)	208	-	326493 672298
24	**Contemporary Trade Directory Entries** Name: Hendry & Macdonald Location: Edinburgh, EH9 1SR Classification: Garage Services **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A13SE (SE)	198	-	326480 671940
24	**Contemporary Trade Directory Entries** Name: M R H Service Station Location: 2-40, Ratcliffe Terrace, Edinburgh, EH9 1SS Classification: Petrol Filling Stations **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A13SE (SE)	207	-	326426 671904
24	**Contemporary Trade Directory Entries** Name: Kennet Ltd Location: 2-40, Ratcliffe Terrace, Edinburgh, EH9 1SS Classification: Petrol Filling Stations - 24 Hour **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A13SE (SE)	207	-	326426 671904
24	**Contemporary Trade Directory Entries** Name: National Tyres & Autocare Edinburgh (Ratcliffe Terrace Eh9) Location: Ratcliffe Terrace, Edinburgh, Midlothian, EH9 1SS Classification: Tyre Dealers **Status: Active** Positional Accuracy: Manually positioned within the geographical locality	A13SE (SE)	207	-	326426 671904
24	**Contemporary Trade Directory Entries** Name: Ian Standing Location: 2-40, Ratcliffe Terrace, Edinburgh, EH9 1SS Classification: Petrol Filling Stations - 24 Hour **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A13SE (SE)	207	-	326426 671904
24	**Contemporary Trade Directory Entries** Name: B P Service Station Location: Ratcliffe Terrace, Edinburgh, Midlothian, EH9 1SS Classification: Petrol Filling Stations **Status: Active** Positional Accuracy: Automatically positioned to the address	A13SE (SE)	207	-	326426 671904
24	**Contemporary Trade Directory Entries** Name: Caledonian Paints Location: 9, Ratcliffe Terrace, Edinburgh, EH9 1SX Classification: Wallpapers & Wall Coverings **Status: Active** Positional Accuracy: Automatically positioned to the address	A13SE (SE)	212	-	326463 671913
25	**Contemporary Trade Directory Entries** Name: C Birss & Sons Location: 47, Upper Gray Street, Edinburgh, EH9 1SN Classification: Joinery Manufacturers **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A13SE (SE)	205	-	326534 671985
26	**Contemporary Trade Directory Entries** Name: Macwind Location: 42, Findhorn Place, Edinburgh, EH9 2NT Classification: Electricity Generating & Distributing Equipment **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A13SE (S)	212	-	326357 671890
27	**Contemporary Trade Directory Entries** Name: Euroclean Location: 17, Newington Road, Edinburgh, EH9 1QR Classification: Dry Cleaners **Status: Active** Positional Accuracy: Automatically positioned to the address	A13NE (NE)	256	-	326452 672378
28	**Contemporary Trade Directory Entries** Name: Causewayside Garage Location: 31, Ratcliffe Terrace, Edinburgh, Midlothian, EH9 1SX Classification: Garage Services **Status: Active** Positional Accuracy: Manually positioned to the address or location	A13SE (SE)	262	-	326515 671886

Map ID	Details	Quadrant Reference (Compass Direction)	Estimated Distance From Site	Contact	NGR
29	**Contemporary Trade Directory Entries** Name: The Chinese Laundry Co Ltd Location: 24, Grange Loan, Edinburgh, EH9 2NR Classification: Dry Cleaners **Status: Active** Positional Accuracy: Automatically positioned to the address	A13SE (S)	277	-	326431 671832
29	**Contemporary Trade Directory Entries** Name: Grange Dry Cleaners Location: 26, Grange Loan, Edinburgh, EH9 2NR Classification: Dry Cleaners **Status: Active** Positional Accuracy: Automatically positioned to the address	A13SE (S)	278	-	326425 671830
30	**Contemporary Trade Directory Entries** Name: Welgo Location: 51, Ratcliffe Terrace, Edinburgh, EH9 1SU Classification: Office Furniture & Equipment **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A13SE (SE)	311	-	326510 671826
31	**Contemporary Trade Directory Entries** Name: Heaven Location: 92, South Clerk Street, Edinburgh, EH8 9PT Classification: Cosmetic Manufacturers **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A13NE (N)	311	-	326387 672449
31	**Contemporary Trade Directory Entries** Name: Minuteman Press Location: 88, South Clerk Street, Edinburgh, EH8 9PT Classification: Printers **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A13NE (N)	311	-	326387 672449
32	**Contemporary Trade Directory Entries** Name: Eurotradeprint.Com Location: 21, Sciennes Road, Edinburgh, EH9 1NX Classification: Printers **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A13NW (NW)	323	-	326023 672304
33	**Contemporary Trade Directory Entries** Name: Christopher Ness Jewellery Location: 1 Summerhall, Edinburgh, Midlothian, EH9 1PL Classification: Jewellery Manufacturers & Repairers **Status: Active** Positional Accuracy: Automatically positioned to the address	A18SW (N)	350	-	326255 672487
33	**Contemporary Trade Directory Entries** Name: Pickering'S Gin Location: Summerhall, Edinburgh, Midlothian, EH9 1PL Classification: Distilleries **Status: Active** Positional Accuracy: Manually positioned within the geographical locality	A18SW (N)	373	-	326248 672510
34	**Contemporary Trade Directory Entries** Name: Maiden Cleaning Ltd Location: 6, Melville Terrace, Edinburgh, EH9 1ND Classification: Cleaning Services - Domestic **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A18SW (N)	362	-	326193 672486
35	**Contemporary Trade Directory Entries** Name: Poppies Of Edinburgh Location: 23, Lauder Road, EDINBURGH, EH9 2JG Classification: Cleaning Services - Domestic **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A13SW (SW)	377	-	326006 671883
36	**Contemporary Trade Directory Entries** Name: Chrysalis Ltd Location: 50 South Clerk St, Edinburgh, Midlothian, EH8 9PP Classification: Recycling Services **Status: Inactive** Positional Accuracy: Manually positioned to the address or location	A18SE (N)	383	-	326348 672523
37	**Contemporary Trade Directory Entries** Name: 3 D Creative Workshops Location: 2, Salisbury Road, Edinburgh, EH16 5AB Classification: Printers **Status: Active** Positional Accuracy: Automatically positioned to the address	A14NW (NE)	407	-	326730 672302

Map ID	Details	Quadrant Reference (Compass Direction)	Estimated Distance From Site	Contact	NGR
37	**Contemporary Trade Directory Entries** Name: Firking Brewery Location: 58, Dalkeith Road, Edinburgh, EH16 5AD Classification: Brewers **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A14NW (NE)	440	-	326764 672307
38	**Contemporary Trade Directory Entries** Name: Absolute Chem-Dry Location: 1, 26, Fountainhall Road, Edinburgh, Midlothian, EH9 2LW Classification: Carpet, Curtain & Upholstery Cleaners **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A8NE (S)	413	-	326380 671689
38	**Contemporary Trade Directory Entries** Name: 1st Call Chemdry Location: 1, 26, Fountainhall Road, Edinburgh, Midlothian, EH9 2LW Classification: Cleaning Services - Commercial **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A8NE (S)	413	-	326380 671689
39	**Contemporary Trade Directory Entries** Name: Momac Cleaning Services Limited Location: 4/2, Grantully Place, Edinburgh, EH9 1SQ Classification: Cleaning Services - Domestic **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A14SW (SE)	425	-	326759 671940
40	**Contemporary Trade Directory Entries** Name: Homestyle Blinds Location: 31e, Minto Street, Edinburgh, EH9 2BT Classification: Blinds, Awnings & Canopies **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A14SW (SE)	426	-	326706 671845
41	**Contemporary Trade Directory Entries** Name: Abroadbean Location: 1, 17, Lutton Place, Edinburgh, Midlothian, EH8 9PD Classification: Car Dealers **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A18SE (N)	435	-	326455 672563
42	**Contemporary Trade Directory Entries** Name: Lutton Press Ltd Location: Unit 7-8 Lutton Court, Lutton Place, Edinburgh, Midlothian, EH8 9PD Classification: Printers **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A18SE (N)	499	-	326494 672620
42	**Contemporary Trade Directory Entries** Name: Lutton Press Location: Unit 7-8 Lutton Court, Lutton Place, Edinburgh, EH8 9PD Classification: Printers **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A18SE (N)	499	-	326494 672620
43	**Contemporary Trade Directory Entries** Name: Brakesafe Location: 3, St. Albans Road, Edinburgh, EH9 2PA Classification: Garage Services **Status: Active** Positional Accuracy: Automatically positioned to the address	A8NE (S)	508	-	326505 671613
43	**Contemporary Trade Directory Entries** Name: Truetest Motors Ltd Location: 3a, St. Albans Road, Edinburgh, EH9 2PA Classification: Garage Services **Status: Active** Positional Accuracy: Automatically positioned to the address	A8NE (S)	514	-	326501 671606
44	**Contemporary Trade Directory Entries** Name: Robertsons Electrical Ltd Location: 21, South Clerk Street, Edinburgh, EH8 9JD Classification: Electrical Goods Sales, Manufacturers & Wholesalers **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A18SE (N)	510	-	326338 672650
44	**Contemporary Trade Directory Entries** Name: Pace Print Edinburgh Location: 19, South Clerk Street, Edinburgh, EH8 9JD Classification: Printers **Status: Active** Positional Accuracy: Automatically positioned to the address	A18SW (N)	529	-	326330 672668

Map ID	Details	Quadrant Reference (Compass Direction)	Estimated Distance From Site	Contact	NGR
44	**Contemporary Trade Directory Entries** Name: Sundial Properties Location: 13, South Clerk Street, Edinburgh, EH8 9JD Classification: Laundries & Launderettes **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A18SW (N)	529	-	326330 672668
44	**Contemporary Trade Directory Entries** Name: Ace Cleaning Centre Location: 13, South Clerk Street, Edinburgh, EH8 9JD Classification: Dry Cleaners **Status: Active** Positional Accuracy: Automatically positioned to the address	A18SW (N)	529	-	326330 672668
45	**Contemporary Trade Directory Entries** Name: Royal Hospital For Sick Children Location: Royal Hospital for Sick Children, 9, Sciennes Road, Edinburgh, EH9 1LF Classification: Hospitals **Status: Active** Positional Accuracy: Automatically positioned to the address	A12NE (NW)	513	-	325834 672350
46	**Contemporary Trade Directory Entries** Name: Gallery Workshop Location: 73, Clerk Street, Edinburgh, Midlothian, EH8 9JG Classification: Jewellery Manufacturers & Repairers **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A18SW (N)	552	-	326274 672692
47	**Contemporary Trade Directory Entries** Name: Saville Motor Co Location: 64, Mayfield Road, Edinburgh, EH9 2NH Classification: Car Dealers - Used **Status: Inactive** Positional Accuracy: Automatically positioned in the proximity of the address	A8NE (S)	582	-	326549 671550
48	**Contemporary Trade Directory Entries** Name: Macmillan Hunter Sundials Location: 3, Peel Terrace, Edinburgh, EH9 2AY Classification: Garden Ornament Manufacturers **Status: Active** Positional Accuracy: Automatically positioned to the address	A9NW (SE)	587	-	326847 671766
49	**Contemporary Trade Directory Entries** Name: Alba Heating Location: 11/3, Parkside Street, Edinburgh, EH8 9RJ Classification: Boilers - Servicing, Replacements & Repairs **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A18SE (N)	593	-	326545 672702
50	**Contemporary Trade Directory Entries** Name: Treehouse Print Solutions Location: 20a Rankeillor St, Edinburgh, Midlothian, EH8 9HZ Classification: Printers **Status: Inactive** Positional Accuracy: Manually positioned to the road within the address or location	A18SW (N)	652	-	326314 672792
51	**Contemporary Trade Directory Entries** Name: Marchmont Hardware Location: 8, Warrender Park Road, EDINBURGH, EH9 1JQ Classification: Hardware **Status: Active** Positional Accuracy: Automatically positioned to the address	A12NE (W)	662	-	325659 672300
52	**Contemporary Trade Directory Entries** Name: Trumps Location: 11-13 Clerk St, Edinburgh, Midlothian, EH8 9JH Classification: Photographic Processors **Status: Inactive** Positional Accuracy: Manually positioned to the address or location	A18NW (N)	672	-	326210 672806
53	**Contemporary Trade Directory Entries** Name: John Forbes Location: 7, Meadow Lane, Edinburgh, EH8 9NR Classification: Garage Services **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A18SW (N)	683	-	326081 672787
54	**Contemporary Trade Directory Entries** Name: Mediamate Web Solutions Location: 63, Mayfield Road, Edinburgh, EH9 3AA Classification: Printers **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A8SE (SE)	707	-	326625 671444

Map ID	Details	Quadrant Reference (Compass Direction)	Estimated Distance From Site	Contact	NGR
55	**Contemporary Trade Directory Entries** Name: Strachan Research Location: 6, Marchhall Crescent, EDINBURGH, EH16 5HN Classification: Laboratories **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A14SE (E)	710	-	327075 672080
56	**Contemporary Trade Directory Entries** Name: Marchhall House Location: 3, Marchhall Crescent, Edinburgh, EH16 5HP Classification: Hospitals **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A14NE (E)	712	-	327073 672176
57	**Contemporary Trade Directory Entries** Name: Mackerel Media Ltd Location: 31, Argyle Place, Edinburgh, EH9 1JT Classification: Gate Manufacturers **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A12NW (NW)	714	-	325657 672447
58	**Contemporary Trade Directory Entries** Name: Greenprint Ltd Location: 49-53, St. Leonards Street, Edinburgh, EH8 9QN Classification: Printers **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A18NE (N)	726	-	326407 672864
58	**Contemporary Trade Directory Entries** Name: Print Pixels Location: 49-53, ST. LEONARDS STREET, ST LEONARDS, EDINBURGH, EH8 9QN Classification: Printers **Status: Active** Positional Accuracy: Automatically positioned to the address	A18NE (N)	726	-	326407 672864
58	**Contemporary Trade Directory Entries** Name: Greenprint Ltd Location: 49-53, St. Leonards Street, Edinburgh, EH8 9QN Classification: Printers **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A18NE (N)	726	-	326407 672864
58	**Contemporary Trade Directory Entries** Name: Edingburgh Distributors Location: 39 St. Leonards St, Edinburgh, Midlothian, EH8 9QN Classification: Distribution Services **Status: Inactive** Positional Accuracy: Manually positioned to the address or location	A18NE (N)	740	-	326397 672879
59	**Contemporary Trade Directory Entries** Name: Butler Services Location: 2/2, Mentone Gardens, EDINBURGH, EH9 2DJ Classification: Cleaning Services - Domestic **Status: Active** Positional Accuracy: Automatically positioned to the address	A9NW (SE)	729	-	326850 671556
60	**Contemporary Trade Directory Entries** Name: The Copy & Print Shop Location: 37, Buccleuch Street, Edinburgh, EH8 9LT Classification: Copying & Duplicating Services **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A18NW (N)	743	-	326125 672862
60	**Contemporary Trade Directory Entries** Name: Cash Registers (Buccleuch) Ltd Location: 16-26, Buccleuch Street, Edinburgh, EH8 9JR Classification: Cash Registers & Check-Out Equipment **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A18NW (N)	776	-	326083 672885
61	**Contemporary Trade Directory Entries** Name: Sundial Launderettes Location: 17, Roseneath Street, Edinburgh, EH9 1JH Classification: Laundries & Launderettes **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A12NW (W)	744	-	325584 672337
62	**Contemporary Trade Directory Entries** Name: Centre For Research On Families & Relationships Location: 23, Buccleuch Place, Edinburgh, EH8 9LN Classification: Laboratories **Status: Active** Positional Accuracy: Automatically positioned to the address	A17SE (NW)	765	-	325905 672795

Map ID	Details	Quadrant Reference (Compass Direction)	Estimated Distance From Site	Contact	NGR
63	**Contemporary Trade Directory Entries** Name: Holyrood Distillery Location: 71-73, St Leonard's Hill, Edinburgh, EH8 9SB Classification: Distilleries **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A18NE (N)	766	-	326517 672889
63	**Contemporary Trade Directory Entries** Name: Crystal Clean Location: 71, St. Leonards Hill, Edinburgh, Midlothian, EH8 9SB Classification: Dry Cleaners **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A18NE (N)	767	-	326518 672890
64	**Contemporary Trade Directory Entries** Name: Blinds Location: 1, St. Patrick Square, Edinburgh, EH8 9EZ Classification: Blinds, Awnings & Canopies **Status: Active** Positional Accuracy: Automatically positioned to the address	A18NW (N)	774	-	326202 672908
64	**Contemporary Trade Directory Entries** Name: The Cottage Location: 136, Nicolson Street, Edinburgh, EH8 9EH Classification: Photocopiers **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A18NW (N)	823	-	326196 672957
64	**Contemporary Trade Directory Entries** Name: Cottage Copies Location: 136, Nicolson Street, Edinburgh, EH8 9EH Classification: Photocopiers **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A18NW (N)	823	-	326196 672957
65	**Contemporary Trade Directory Entries** Name: Osler & Robertson Location: 6, Beaufort Road, Edinburgh, EH9 1AG Classification: Cash Registers & Check-Out Equipment **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A12SW (W)	788	-	325532 671974
66	**Contemporary Trade Directory Entries** Name: Aegis Property Care Location: 53, East Crosscauseway, Edinburgh, EH8 9HG Classification: Damp & Dry Rot Control **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A18NW (N)	793	-	326279 672934
66	**Contemporary Trade Directory Entries** Name: Aegis Property Care Location: 53, East Crosscauseway, Edinburgh, EH8 9HG Classification: Timber Preservation Services **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A18NW (N)	793	-	326279 672934
66	**Contemporary Trade Directory Entries** Name: Aegis Location: 53, East Crosscauseway, Edinburgh, EH8 9HG Classification: Woodworm Control **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A18NW (N)	793	-	326279 672934
67	**Contemporary Trade Directory Entries** Name: Absolute Upholstery Location: 50, Relugas Place, Edinburgh, EH9 2PY Classification: Upholstery Manufacturers **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A8SW (S)	813	-	326203 671298
68	**Contemporary Trade Directory Entries** Name: Down The Kitchen Sink Location: 157, Dalkeith Road, Edinburgh, EH16 5HQ Classification: Waste Disposal Services **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A14SE (E)	815	-	327157 671906
69	**Contemporary Trade Directory Entries** Name: Move - It Mobile Skip Hire Location: 12, Marchmont Crescent, Edinburgh, EH9 1HN Classification: Waste Disposal Services **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A12NW (W)	820	-	325525 672404

Map ID	Details	Quadrant Reference (Compass Direction)	Estimated Distance From Site	Contact	NGR
70	**Contemporary Trade Directory Entries** Name: The Pregnancy & Parents Centre Location: 192, PLEASANCE, NEWINGTON, EDINBURGH, EH8 9RT Classification: Hospitals **Status: Active** Positional Accuracy: Automatically positioned to the address	A18NE (N)	843	-	326361 672983
70	**Contemporary Trade Directory Entries** Name: Shri Ram Hospitality Ltd Location: 201, Pleasance, Edinburgh, EH8 9RU Classification: Food Products - Manufacturers **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A18NW (N)	861	-	326321 673002
70	**Contemporary Trade Directory Entries** Name: Donaldson Cleaning Location: 193a-195, Pleasance, Edinburgh, EH8 9RU Classification: Carpet, Curtain & Upholstery Cleaners **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A18NW (N)	868	-	326316 673009
70	**Contemporary Trade Directory Entries** Name: Holyrood Distillery Location: 181 Pleasance, Edinburgh, Midlothian, EH8 9RU Classification: Distilleries **Status: Active** Positional Accuracy: Automatically positioned to the address	A18NW (N)	882	-	326309 673022
71	**Contemporary Trade Directory Entries** Name: J M Domestic Appliance Repairs Location: 100, Marchmont Road, Edinburgh, EH9 1HR Classification: Domestic Appliances - Servicing, Repairs & Parts **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A12NW (W)	848	-	325454 672140
72	**Contemporary Trade Directory Entries** Name: Marchmont Garage Location: 1, Thirlestane Lane, Edinburgh, EH9 1AJ Classification: Garage Services **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A12SW (W)	854	-	325464 671975
73	**Contemporary Trade Directory Entries** Name: Source Electronics Location: 114, Nicolson Street, Edinburgh, EH8 9EJ Classification: Electrical Goods Sales, Manufacturers & Wholesalers **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A18NW (N)	858	-	326189 672991
74	**Contemporary Trade Directory Entries** Name: Kwik-Fit Location: 17-19, Strathearn Road, Edinburgh, EH9 2AA Classification: Exhaust & Shock Absorber Centres **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A12SW (W)	862	-	325474 671898
74	**Contemporary Trade Directory Entries** Name: Farmer Autocare Location: 17-19, Strathearn Road, Edinburgh, EH9 2AA Classification: Tyre Dealers **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A12SW (W)	862	-	325474 671898
75	**Contemporary Trade Directory Entries** Name: Home Agency Location: 36, McLaren Road, EDINBURGH, EH9 2BN Classification: Cleaning Services - Domestic **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A9NE (SE)	871	-	327091 671620
76	**Contemporary Trade Directory Entries** Name: Capital Camera Repairs Location: 38, Marchmont Road, Edinburgh, EH9 1HX Classification: Photographic Equipment Repairs **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A12NW (W)	879	-	325440 672316
77	**Contemporary Trade Directory Entries** Name: Omni Electronics Location: 174, Dalkeith Road, Edinburgh, EH16 5DX Classification: Electrical Goods Sales, Manufacturers & Wholesalers **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A14SE (E)	883	-	327191 671788

Map ID	Details	Quadrant Reference (Compass Direction)	Estimated Distance From Site	Contact	NGR
77	**Contemporary Trade Directory Entries** Name: William Regan & Sons Location: 170, Dalkeith Road, Edinburgh, EH16 5DX Classification: Washing Machines - Servicing & Repairs **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A14SE (E)	883	-	327191 671788
77	**Contemporary Trade Directory Entries** Name: Arduus Distribution Location: 3, Dalkeith Road Mews, Edinburgh, EH16 5GA Classification: Distribution Services **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A9NE (SE)	885	-	327180 671754
77	**Contemporary Trade Directory Entries** Name: Leith Metal Polishing Co Location: 196b, Dalkeith Road, Edinburgh, Midlothian, EH16 5DU Classification: Engineers - General **Status: Inactive** Positional Accuracy: Manually positioned to the address or location	A9NE (SE)	924	-	327219 671749
77	**Contemporary Trade Directory Entries** Name: A 3 P Location: 196b, Dalkeith Road, Edinburgh, Midlothian, EH16 5DU Classification: Garage Services **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A9NE (SE)	924	-	327219 671749
77	**Contemporary Trade Directory Entries** Name: Mcdonald Minis Location: 196b, Dalkeith Road, Edinburgh, Midlothian, EH16 5DU Classification: Garage Services **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A9NE (SE)	924	-	327219 671749
78	**Contemporary Trade Directory Entries** Name: Arnold Seftor & Son Location: 1-1a, Marchmont Crescent, Edinburgh, EH9 1HN Classification: Furriers **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A17SW (W)	894	-	325467 672462
79	**Contemporary Trade Directory Entries** Name: Scoopz Location: 25-27 West Nicolson St, Edinburgh, Midlothian, EH8 9DD Classification: Ice Cream Manufacturers & Suppliers **Status: Inactive** Positional Accuracy: Manually positioned within the geographical locality	A18NW (N)	906	-	326080 673019
80	**Contemporary Trade Directory Entries** Name: Munro Klick Location: 90, Nicolson Street, Edinburgh, EH8 9EW Classification: Photographic Processors **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A18NW (N)	933	-	326176 673065
80	**Contemporary Trade Directory Entries** Name: Munro Cleaners Location: 90, Nicolson Street, Edinburgh, EH8 9EW Classification: Dry Cleaners **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A18NW (N)	933	-	326176 673065
81	**Contemporary Trade Directory Entries** Name: Capital Launderette & Dry Cleaners Location: 208, Dalkeith Road, Edinburgh, EH16 5DT Classification: Laundries & Launderettes **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A9NE (SE)	947	-	327237 671731
81	**Contemporary Trade Directory Entries** Name: Wet Wash New York Location: 206, Dalkeith Road, Edinburgh, EH16 5DT Classification: Ironing & Home Laundry Services **Status: Active** Positional Accuracy: Automatically positioned to the address	A9NE (SE)	947	-	327236 671730
82	**Contemporary Trade Directory Entries** Name: Beauty Concessions Ltd Location: 85, Nicolson Street, Edinburgh, EH8 9BZ Classification: Toiletries **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A18NW (N)	971	-	326114 673094

Map ID	Details	Quadrant Reference (Compass Direction)	Estimated Distance From Site	Contact	NGR
82	**Contemporary Trade Directory Entries** Name: M S Irvin Location: 67-71, Nicolson Street, Edinburgh, EH8 9BZ Classification: Electrical Engineers **Status: Inactive** Positional Accuracy: Automatically positioned to the address	A18NW (N)	1000	-	326103 673120
83	**Fuel Station Entries** Name: Causewayside Service Station Location: 35, Causewayside , , Edinburgh, Edinburgh, City Of, EH9 1QF Brand: OBSOLETE Premises Type: Not Applicable **Status: Obsolete** Positional Accuracy: Automatically positioned to the address	A13NW (N)	207	-	326307 672347
84	**Fuel Station Entries** Name: Mfg Ratcliffe Location: 2-4, Ratcliffe Terrace , , Edinburgh, Edinburgh, City Of, EH9 1SS Brand: Bp Premises Type: Petrol Station **Status: Open** Positional Accuracy: Automatically positioned to the address	A13SE (SE)	207	-	326426 671904

Map ID	Details	Quadrant Reference (Compass Direction)	Estimated Distance From Site	Contact	NGR
85	**Areas of Adopted Green Belt** Authority: City of Edinburgh Council Plan Name: Edinburgh Local Development Plan **Status: Adopted** Plan Date: 24th November 2016	A19SW (NE)	730	5	326703 672782
86	**Sites of Special Scientific Interest** Name: Arthurs Seat Volcano Multiple Areas: Y Total Area (m2): 2281277.63 Source: NatureScot Reference: 91 Designation Details: Mixed Designation Date: 24th April 1986 Date Type: Designated	A19SW (NE)	728	6	326705 672780
87	**World Heritage Sites** Name: Edinburgh World Heritage Site Boundary Reference: Edinburgh World Heritage Site Boundary Area(m^2): 4427976.99 Source: Historic Environment Scotland Designation Date: 20th September 2001	A17NE (NW)	917	7	325682 672817

Data Currency

Agency & Hydrological	Version	Update Cycle
Contaminated Land Register Entries and Notices		
Scottish Environment Protection Agency - Head Office	June 2020	Annually
Midlothian Council	October 2017	Annual Rolling Update
City of Edinburgh Council	September 2017	Annual Rolling Update
East Lothian Council	September 2017	Annual Rolling Update
Discharge Consents		
Scottish Environment Protection Agency - East Region	April 2002	
Enforcement and Prohibition Notices		
Scottish Environment Protection Agency - East Region	March 2013	
Integrated Pollution Controls		
Scottish Environment Protection Agency - Head Office	February 1998	
Scottish Environment Protection Agency - East Region	March 2002	
Local Authority Pollution Prevention and Controls		
Scottish Environment Protection Agency - East Region	March 2002	Not Applicable
Local Authority Pollution Prevention and Control Enforcements		
Scottish Environment Protection Agency - East Region	June 2001	Variable
Nearest Surface Water Feature		
Ordnance Survey	January 2023	
Prosecutions Relating to Authorised Processes		
Scottish Environment Protection Agency - East Region	March 2013	
Prosecutions Relating to Controlled Waters		
Scottish Environment Protection Agency - East Region	March 2013	
Registered Radioactive Substances		
Scottish Environment Protection Agency - East Region	April 1996	
Scottish Environment Protection Agency - Head Office	January 1998	
River Quality		
Scottish Environment Protection Agency - East Region	December 1990	Not Applicable
Scottish Environment Protection Agency - Head Office	December 1990	Not Applicable
Water Abstractions		
Scottish Government - Agriculture, Environment and Fisheries Department	February 2004	
Water Industry Act Referrals		
Scottish Environment Protection Agency - East Region	April 1996	
Groundwater Vulnerability		
Scottish Environment Protection Agency - East Region	December 1995	Not Applicable
Scottish Environment Protection Agency - Head Office	December 1995	
Drift Deposits		
Scottish Environment Protection Agency - East Region	December 1995	Not Applicable
Scottish Environment Protection Agency - Head Office	December 1995	Not Applicable
OS Water Network Lines		
Ordnance Survey	January 2023	Quarterly
BGS Groundwater Flooding Susceptibility		
British Geological Survey - National Geoscience Information Service	May 2013	As notified

Data Currency

Waste	Version	Update Cycle
BGS Recorded Landfill Sites		
British Geological Survey - National Geoscience Information Service	November 2002	As notified
Integrated Pollution Control Registered Waste Sites		
Scottish Environment Protection Agency - East Region	March 2002	Not Applicable
Scottish Environment Protection Agency - Head Office	March 2002	Not Applicable
Local Authority Landfill Coverage		
City of Edinburgh Council	February 2003	Not Applicable
East Lothian Council	February 2003	Not Applicable
Midlothian Council - Waste Management Department	February 2003	Not Applicable
Local Authority Recorded Landfill Sites		
City of Edinburgh Council	October 2018	
East Lothian Council	October 2018	
Midlothian Council - Waste Management Department	October 2018	
Registered Landfill Sites		
Scottish Environment Protection Agency - East Region	March 2006	Not Applicable
Scottish Environment Protection Agency - East Region - Alloa Office	March 2006	Not Applicable
Scottish Environment Protection Agency - East Region - Arboath Office	March 2006	Not Applicable
Scottish Environment Protection Agency - East Region - Galashiels Office	March 2006	Not Applicable
Scottish Environment Protection Agency - East Region - Glenrothes Office	March 2006	Not Applicable
Scottish Environment Protection Agency - East Region - Perth Office	March 2006	Not Applicable
Scottish Environment Protection Agency - East Region - Stirling Office	March 2006	Not Applicable
Scottish Environment Protection Agency - Head Office	March 2006	Not Applicable
Registered Waste Transfer Sites		
Scottish Environment Protection Agency - East Region	April 2018	
Scottish Environment Protection Agency - East Region - Alloa Office	April 2018	
Scottish Environment Protection Agency - East Region - Arboath Office	April 2018	
Scottish Environment Protection Agency - East Region - Galashiels Office	April 2018	
Scottish Environment Protection Agency - East Region - Glenrothes Office	April 2018	
Scottish Environment Protection Agency - East Region - Perth Office	April 2018	
Scottish Environment Protection Agency - East Region - Stirling Office	April 2018	
Scottish Environment Protection Agency - Head Office	April 2018	
Registered Waste Treatment or Disposal Sites		
Scottish Environment Protection Agency - East Region	June 2015	
Scottish Environment Protection Agency - East Region - Alloa Office	June 2015	
Scottish Environment Protection Agency - East Region - Arboath Office	June 2015	
Scottish Environment Protection Agency - East Region - Galashiels Office	June 2015	
Scottish Environment Protection Agency - East Region - Glenrothes Office	June 2015	
Scottish Environment Protection Agency - East Region - Perth Office	June 2015	
Scottish Environment Protection Agency - East Region - Stirling Office	June 2015	
Scottish Environment Protection Agency - Head Office	June 2015	

Hazardous Substances	Version	Update Cycle
Control of Major Accident Hazards Sites (COMAH)		
Health and Safety Executive	March 2023	Bi-Annually
Explosive Sites		
Health and Safety Executive	March 2017	Annually
Notification of Installations Handling Hazardous Substances (NIHHS)		
Health and Safety Executive	August 2001	
Planning Hazardous Substance Enforcements		
City of Edinburgh Council - City Development Department	February 2016	Variable
East Lothian Council - Planning Department	February 2016	Variable
Midlothian Council - Planning Department	February 2016	Variable
Planning Hazardous Substance Consents		
City of Edinburgh Council - City Development Department	February 2016	Variable
East Lothian Council - Planning Department	February 2016	Variable
Midlothian Council - Planning Department	February 2016	Variable
Geological	**Version**	**Update Cycle**
BGS 1:625,000 Solid Geology		
British Geological Survey - National Geoscience Information Service	January 2009	As notified
BGS Recorded Mineral Sites		
British Geological Survey - National Geoscience Information Service	November 2022	Bi-Annually
CBSCB Compensation District		
Cheshire Brine Subsidence Compensation Board (CBSCB)	August 2011	
Cheshire Brine Subsidence Compensation Board (CBSCB)	November 2020	As notified
Coal Mining Affected Areas		
The Coal Authority - Property Searches	February 2023	Annual Rolling Update
Mining Instability		
Ove Arup & Partners	June 1998	Not Applicable
Non Coal Mining Areas of Great Britain		
British Geological Survey - National Geoscience Information Service	May 2015	Not Applicable
Potential for Collapsible Ground Stability Hazards		
British Geological Survey - National Geoscience Information Service	April 2020	As notified
Potential for Compressible Ground Stability Hazards		
British Geological Survey - National Geoscience Information Service	January 2019	As notified
Potential for Ground Dissolution Stability Hazards		
British Geological Survey - National Geoscience Information Service	January 2019	As notified
Potential for Landslide Ground Stability Hazards		
British Geological Survey - National Geoscience Information Service	January 2019	As notified
Potential for Running Sand Ground Stability Hazards		
British Geological Survey - National Geoscience Information Service	January 2019	As notified
Potential for Shrinking or Swelling Clay Ground Stability Hazards		
British Geological Survey - National Geoscience Information Service	January 2019	As notified
Radon Potential - Radon Affected Areas		
British Geological Survey - National Geoscience Information Service	September 2022	Annually
Radon Potential - Radon Protection Measures		
British Geological Survey - National Geoscience Information Service	September 2022	Annually

Industrial Land Use	Version	Update Cycle
Contemporary Trade Directory Entries		
Thomson Directories	January 2023	Quarterly
Fuel Station Entries		
Catalist Ltd - Experian	February 2023	Quarterly
Gas Pipelines		
National Grid	October 2021	Bi-Annually
Sensitive Land Use	**Version**	**Update Cycle**
Ancient Woodland		
NatureScot	September 2017	Bi-Annually
Areas of Adopted Green Belt		
City of Edinburgh Council	July 2022	Quarterly
East Lothian Council	July 2022	Quarterly
Midlothian Council	July 2022	Quarterly
Areas of Unadopted Green Belt		
City of Edinburgh Council	July 2022	Quarterly
East Lothian Council	July 2022	Quarterly
Midlothian Council	July 2022	Quarterly
Environmentally Sensitive Areas		
Scottish Government	January 2017	
Forest Parks		
Forestry Commission	April 1997	Not Applicable
Local Nature Reserves		
City of Edinburgh Council	February 2018	Bi-Annually
East Lothian Council	February 2018	Bi-Annually
Midlothian Council	February 2018	Bi-Annually
Marine Nature Reserves		
NatureScot	July 2019	Bi-Annually
National Nature Reserves		
NatureScot	February 2023	Bi-Annually
National Parks		
Scottish Government	February 2018	Bi-Annually
National Scenic Areas		
Scottish Government	February 2018	Bi-Annually
Nitrate Vulnerable Zones		
Scottish Government	March 2023	Annually
Ramsar Sites		
NatureScot	March 2023	Bi-Annually
Sites of Special Scientific Interest		
NatureScot	March 2019	Bi-Annually
Special Areas of Conservation		
NatureScot	August 2020	Bi-Annually
Special Protection Areas		
NatureScot	February 2021	Bi-Annually
World Heritage Sites		
Historic Environment Scotland	July 2022	Bi-Annually

Data Suppliers

A selection of organisations who provide data within this report

Data Supplier	Data Supplier Logo
Ordnance Survey	
Environment Agency	
Scottish Environment Protection Agency	
The Coal Authority	
British Geological Survey	
Centre for Ecology and Hydrology	
Natural Resources Wales	
Scottish Natural Heritage	
Natural England	
Public Health England	
Ove Arup	
Stantec UK Ltd	

Useful Contacts

Contact	Name and Address	Contact Details
1	**British Geological Survey - Enquiry Service** British Geological Survey, Environmental Science Centre, Keyworth, Nottingham, Nottinghamshire, NG12 5GG	Telephone: 0115 936 3143 Fax: 0115 936 3276 Email: enquiries@bgs.ac.uk Website: www.bgs.ac.uk
2	**Scottish Environment Protection Agency - East Region** Clearwater House, Heriot Watt Research Park, Avenue North, Riccarton, Edinburgh, Midlothian, EH14 4AP	Telephone: 0131 449 7296 Fax: 0131 449 7277
3	**Scottish Environment Protection Agency - Head Office** Erskine Court, The Castle Business Park, Stirling, Stirlingshire, FK9 4TR	Telephone: 01786 457700 Fax: 01786 446885
4	**Ordnance Survey** Adanac Drive, Southampton, Hampshire, SO16 0AS	Telephone: 03456 05 05 05 Email: customerservices@ordnancesurvey.co.uk Website: www.ordnancesurvey.gov.uk
5	**City of Edinburgh Council** Council Headquarters, Wellington Court, 10 Waterloo Place, Edinburgh, EH1 3EG	Telephone: 0131 200 2000 Email: council.info@edinburgh.gov.uk Website: www.edinburgh.gov.uk
6	**NatureScot** Great Glen House, Leachkin Road, Inverness, IV3 8NW	Telephone: 01463 725000 Email: enquiries@nature.scot Website: www.nature.scot
7	**Landmark Information Group Limited** Imperium, Imperial Way, Reading, Berkshire, RG2 0TD	Telephone: 0844 844 9966 Fax: 0844 844 9951 Email: helpdesk@landmark.co.uk Website: www.landmark.co.uk
-	**Public Health England - Radon Survey, Centre for Radiation, Chemical and Environmental Hazards** Chilton, Didcot, Oxfordshire, OX11 0RQ	Telephone: 01235 822622 Fax: 01235 833891 Email: radon@phe.gov.uk Website: www.ukradon.org
-	**Landmark Information Group Limited** Imperium, Imperial Way, Reading, Berkshire, RG2 0TD	Telephone: 0844 844 9952 Fax: 0844 844 9951 Email: customerservices@landmarkinfo.co.uk Website: www.landmarkinfo.co.uk

Please note that the Environment Agency / Natural Resources Wales / SEPA have a charging policy in place for enquiries.